

Daily Briefing - October 29, 2026

Your Daily Tech & Programming Digest

Thursday, October 29, 2026

1000	122699	1174	35
ARTICLES	WORDS	MIN READ	SOURCES

📰 Today's Top Stories

Easily download files from the Open Science Framework with Papercheck

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-07-22 📖 3 min 📝 765 words

TWENTY PERCENT STATISTICIAN

> **Summary:** <p>Researchers increasingly use the Open Science Framework (OSF) to share files, such as data and code underlying scientific publications, or presentations and materials for scientific workshops. The OSF is an amazing service that has contributed immensely to a changed ...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/07/easily-download-files-from-open-science.html

How I use AI in my work

noreply@blogger.com (Daniel Lakens)
2026-10-11 21 min 4292 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Tomorrow we will have a discussion in our lab about how to use AI as a scientist. I have increasingly started to use AI, having moved from Copilot Chat that our university provides last year, to some free credit on GitHub Copilot under their educational program, to Claude Pro integrated in Visual...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/how-i-use-ai-in-my-work.html

A novel function of neuroligin 3: Beyond sociality, Regulating lifespan in Drosophila

Jurbaj, J. M., Ataei, R., Simon, A. F.
2026-10-28 1 min 153 words

BIORXIV NEUROSCIENCE

Summary: Neuroligins are conserved postsynaptic adhesion proteins involved in synaptic function and behaviour. In Drosophila melanogaster, Neuroligin 3 (Nlg3) regulates social behaviour, sleep and synaptic architecture. We now report, through analysis of publicly available transcriptomics data, confirmed wit...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.762327v1?rss=1

Reward changes the use and structure of predictive spatial maps

Oza, S., Cohen, A., Wager, G., Derdikman, D., Recanatesi, S.

2026-10-29 1 min 178 words

BIORXIV NEUROSCIENCE

Summary: Reward-related changes in hippocampal place-cell activity may reflect either different use of an existing spatial map or plasticity that reshapes the map. We separated these contributions in a two-pathway model: a recurrent network learned to predict sensory input and developed a spatial representat...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.28.762439v1?rss=1

Environmental risk reshapes high-confidence schizophrenia risk genes during a developmental window of vulnerability

Brack, A., Phadke, R. A., Fournier, L., Boberg, M. J., Salgado, M., Cruz-Martin, A.

2026-10-28

1 min 170 words

BIORXIV NEUROSCIENCE

Summary: Rare coding variants and environmental exposures such as early-life stress contribute to schizophrenia risk, but whether environmental adversity alters the expression of genetically implicated genes remains unclear. Combining the limited bedding and nesting (LBN) model of early-life stress (ELS) wit...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.762379v1?rss=1

Cholinergic and adrenergic modulation sculpts parallel inhibition of SST-INs in visual cortex

Gupta, J., Chiu, C., Veneto, F., Gjorgjieve, J., Higley, M.

2026-10-28 1 min 200 words

BIORXIV NEUROSCIENCE

Summary: GABAergic interneurons are fundamental to the normal function of neuronal circuits in the neocortex. A growing literature has highlighted the role of somatostatin-expressing interneurons (SST-INs) in regulating cortical activity, and inhibition of SST-INs by other interneurons has emerged as a circu...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.28.762553v1?rss=1

Cortical asymmetry and its developmental change have distinct relationships with brain connectivity

Venkadesh, S., Yeh, F.-c., Hof, P. R., Korenberg, J. R.

2026-10-28 1 min 222 words

BIORXIV NEUROSCIENCE

Summary: The two cerebral hemispheres specialize while remaining interconnected. Does greater structural asymmetry imply less connectivity, and does connectivity also relate to how asymmetry changes during development? We addressed these questions in three independent cohorts spanning birth to 22 years, meas...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.28.762579v1?rss=1

A proof-of-concept study on molecular fluorescence spectroscopy (EEM) combined with chemometrics applied to screening for autism spectrum disorder

📅 2026-10-29 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-78234-6

Baseline cognitive impairment predicts cerebrovascular events in stroke-free chinese adults

📅 2026-10-29 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-78298-4

4-methylumbelliferone attenuates amyloid pathology and learning deficits in the APP/PS1 mouse model

📅 2026-10-29 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-78028-w

Trusting the tag: electrophysiological signatures of source credibility in fact-checking

📅 2026-10-29 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-77776-z

Sustained antidepressant-like effects of psychedelics through 5-HT_{2A}R–β-arrestin signaling

👤 Sheng Wang 📅 2026-10-23 📖 1 min 📝 40 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 23 October 2026; doi:10.1038/s41593-026-02497-5 This study shows that the sustained antidepressant-like effect of psychedelics is driven by 5-HT2A receptor–β-arrestin signaling. Repeated low...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02497-5

Identification of candidate markers for the isolation of neuron-specific extracellular vesicles

David R. Walt
2026-10-23 1 min 41 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 23 October 2026; doi:10.1038/s41593-026-02466-y The authors developed methods to identify extracellular vesicles (EVs) in the blood that come from the brain by capturing EV surface proteins....

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02466-y

Neuropharmacology of language: ERP evidence for catecholaminergic modulation of syntactic processing

Yingying TanAshley G. LewisIris C. SchmitsRobert-Jan VerkesRoshan CoolsPeter Hagoortahttps://ror.org/01bn89z48Shanghai International Studies University, Institute of Language Sciences, Shanghai 201620, Chinabhttps://ror.org/00671me87Max Planck Institute for Psycholinguistics, Neurobiology of Language Department, Nijmegen 6525 XD, the Netherlandschttps://ror.org/03q8dnn23City University of Hong Kong, Department of Linguistics and Translation, Kowloon, Hong Kong Special Administrative Region, Chinadhttps://ror.org/016xsfp80Radboud University, Donders Institute for Brain, Cognition and Behaviour, Center for Cognitive Neuroimaging, Nijmegen 6525 EN, the Netherlandsehttps://ror.org/05wg1m734Radboud University Medical Center, Department of Psychiatry, Nijmegen 6525 GA, the Netherlands

📅 2026-10-22 📖 1 min 📝 42 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceHow the brain’s neuromodulatory systems influence language, a defining human capacity, remains poorly understood. Here we show that catecholaminergic neurotransmission—primarily involving dopamine ...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2606806123?af=R

Safety of lecanemab in real-world patients with Alzheimer’s disease: a prospective, observational case series supported by FAERS signal analysis

Erhao Bao 2026-10-02 1 min 354 words FRONTIERS NEUROINFORMATICS

Summary: ObjectiveLecanemab, a humanized immunoglobulin monoclonal antibody, has been approved for the treatment of Alzheimer’s disease (AD). Although clinical trials have demonstrated its favorable safety profile, real-world studies are crucial for identifying rare adverse events (AEs) and evaluating post-m...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1977226

Correction: The PREP pipeline: standardized preprocessing for large-scale EEG analysis

Kay A. Robbins 2026-10-12 1 min 0 words FRONTIERS NEUROINFORMATICS

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.2015651

EEG emotion-recognition datasets for neuroarchitecture: an appraisal of openness, AI-readiness and ontology maturity

Han Jong Jun 2026-10-20 1 min 341 words FRONTIERS NEUROINFORMATICS

Summary: IntroductionNeuroarchitecture studies how designed space shapes emotion, often with electroencephalography (EEG), and computational models that decode emotion from EEG must be trained on labelled recordings. Yet none of the widely shared EEG emotion datasets was recorded while people moved through d...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1971441

A systematic review of machine learning approaches for MRI-based brain tumor analysis

Almira Melsovna Kustubayeva 2026-10-12 1 min 213 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Machine learning is now widely applied to magnetic resonance imaging-based brain tumor analysis, with task-specific performance metrics: the Dice similarity coefficient (DSC) quantifies segmentation overlap, while the area under the curve (AUC) and accuracy measure classification performance. A PRIS...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1937002

Are cortical microcircuits optimized for information flux?—A simulation-based reverse engineering study

Patrick Krauss 2026-10-20 1 min 271 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionA sufficiently large information flux in recurrent neural networks, quantified by the mutual information between successive network states, has been proposed as a marker of rich information-processing capabilities. This raises the question of whether biological neural networks, such as c...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1886601

Efficient neuroanalytic local–global learning with dual-branch CNN–lightweight Swin Transformer for Alzheimer's disease analysis

Arshad Ali 2026-10-29 1 min 287 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Alzheimer's disease (AD) is one of the main causes of dementia, where early detection is critical for postponing the progression of the disease and timely treatment. Recent computer-aided diagnosis using deep learning with structural MRI has shown optimal performance; however, the rising number of m...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1972923

Virtual brain models: a roadmap towards effective brain digital twins

👤 Viktor Jirsa 📅 2026-10-29 📖 1 min 📝 205 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Virtual Brain Models (VBMs) are emerging computational tools promising to revolutionize personalised and precision medicine. VBMs reproduce brain functional dynamics through a generative model that can be constructed and optimized against single subject data, thereby generating what is called a brai...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1973956

Simulation-based neural dynamics of stress-related negative cognitive bias in depressive symptoms: a dynamical EEG/ERP framework

👤 Jianping Liu 📅 2026-10-29 📖 1 min 📝 192 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionThis simulation-based computational study develops a non-linear state-space framework for stress-related negative cognitive bias and multivariate EEG/ERP observations.MethodsA three-state model represents negative bias, cognitive control, and arousal, while symptom load and normalized mo...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1972914

Editorial: Emerging talents in human neuroscience: psychophysiology and cognitive neuroscience 2025

Gaia Risso 2026-10-29 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.2016840

A unified fixed-wavelet transformer decodes eight EEG classes across motor imagery, P300, and SSVEP paradigms

Hong-Gi Yeom 2026-10-29 1 min 314 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionMotor imagery (MI) can support self-initiated control, whereas brain–computer interfaces (BCIs) based on the P300 or the steady-state visual evoked potential (SSVEP) rely on ongoing visual stimulation. Repetitive visual stimulation can cause visual fatigue; when the stimuli are suspended...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1973838

Resolving the apparent paradox of multisensory expertise: coupling priors, common-cause priors, and how to distinguish among them

Pier Luigi Sacco 2026-10-29 1 min 348 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Perceptual expertise is widely assumed to refine multisensory perception, yet the empirical literature appears inconsistent. Trained musicians show narrower audiovisual temporal integration windows and quicker segregation, whereas experienced wine tasters seem more influenced by a wine's color in th...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1960361

Effect of robotic assistance on upper-limb sensorimotor recovery: a systematic review and meta-analysis

Sivakumar Balasubramanian 2026-10-20 1 min 359 words FRONTIERS NEUROROBOTICS

Summary: IntroductionRobot-assisted upper-limb neurorehabilitation enables several beneficial therapy components, including repetitive task-specific practice, high training intensity, real-time feedback, gamification, and continuous physical interaction with the patient’s limb. However, conventional comparis...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1860365

Dolphin-inspired adaptive coordination for online multi-robot path planning in dynamic environments

Lijuan Chen 2026-10-29 1 min 200 words FRONTIERS NEUROROBOTICS

Summary: IntroductionOnline multi-robot path planning requires repeated joint motion decisions amid static obstacles, moving obstacles, and other robots.MethodsThis study introduces the Dolphin Cooperative Foraging Optimizer (DCFO), a population-based method that adapts its search strategy within each replan...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1993109

Case Report: Probable paroxysmal sympathetic hyperactivity during vascular and postoperative critical illness with bilateral white matter abnormalities

Lei Wang 2026-10-29 1 min 194 words FRONTIERS NEUROSCIENCE

Summary: BackgroundRecognition of paroxysmal sympathetic hyperactivity (PSH) during vascular and postoperative critical illness is complicated by overlapping infection and seizure features.Case presentationA 37-year-old man developed encephalopathy and rigidity after extensive arterial thrombosis, thrombecto...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1935805

A systematic review and meta-analysis of quantitative pain outcomes in degenerative thoracic myelopathy

Oliver D. Mowforth
2026-10-29 1 min 322 words

FRONTIERS NEUROSCIENCE

Summary: IntroductionDegenerative thoracic myelopathy (DTM) refers to symptomatic compression of the thoracic spinal cord secondary to degenerative pathology, including thoracic disc herniation (TDH), ossification of the ligamentum flavum (OLF) and ossification of the posterior longitudinal ligament (OPLL). ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1937393

From confidence to caution: conformal gating for cross-session motor imagery brain–computer interfaces

Sixue Xing and Aarthy Nagarajan
2026-10-22 1 min 278 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Erroneous motor imagery brain–computer interface (MI-BCI) commands can cause unintended device actions, yet standard classifiers provide no mechanism to withhold uncertain predictions. We evaluate whether conformal prediction (CP) can serve as a command gate under cross-session EEG shift,...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aead26

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261029040750&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39870674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261029040750&v=2.20.1

Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 2025-06-26 1 min 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261029040750&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261029040750&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261029040750&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41358599/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261029040750&v=2.20.1

Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41676644/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261029040750&v=2.20.1

Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41678848/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261029040750&v=2.20.1

The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41703344/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261029040750&v=2.20.1

EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42538927/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261029040750&v=2.20.1

Communication patterns in home medication review for individuals with vision impairment: A qualitative study

👤 Woojin Park
📅 2026-10-27 📖 1 min 📝 37 words LOW VISION

Summary: CONCLUSIONS: In these consultations, professional dialogue supported verification and reassurance, while direct regimen review revealed unverified beliefs and gaps in medication knowledge. The findings suggest a role for accessible counseling that combines information provision with professional ver...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945707/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261029040746&v=2.20.1

Global, regional, and national burden of unintentional poisoning, 1990-2023: results from the Global Burden of Disease Study 2023

👤 GBD 2023 Unintentional Poisoning Collaborators

📅 2026-10-27 📖 1 min 📝 65 words

LOW VISION

Summary: BACKGROUND: Unintentional poisoning is a preventable cause of injury-related morbidity and mortality, yet global patterns, trends, and inequalities remain incompletely characterised. Within the Global Burden of Disease (GBD) framework, unintentional poisoning includes poisoning by carbon monoxide an...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945982/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261029040746&v=2.20.1

Directing Ru-based acidic oxygen evolution catalysts toward a universal dual-site pathway

👤 Chuangang Hu 📅 2026-10-28 📖 1 min 📝 62 words LOW VISION

Summary: The oxide path mechanism offers vision possibility from kinetics for the anodic oxygen evolution reaction, yet its universal activation remains elusive due to the difficulty of constructing globally appropriate interatomic distances. Here, we address this challenge in the integrated-screened Y(0.1)I...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42946705/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261029040746&v=2.20.1

Biomimetic computational modelling of retinal ganglion cell responses to contact lens designs with refractive and opaque non-refractive features

👤 Praveen K Bandela 📅 2026-10-28 📖 1 min 📝 67 words LOW VISION

Summary: CONCLUSIONS: In this computational model, both BF and AR lens designs produced measurable changes in simulated RGC temporal activity relative to SV, with AR inducing greater and more configuration-independent modulation. These findings must be considered hypothesis-generating only; they are consiste...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947083/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261029040746&v=2.20.1

Growth, nutritional status, and physical fitness trajectories among primary and secondary school students and their associations with poor vision: a school-based study in Chongqing, China

Yong Zhao 2026-10-28 1 min 42 words LOW VISION

Summary: CONCLUSION: From 2022 to 2026, growth and development indicators among primary and secondary school students in Yuzhong District improved, while poor vision remained prevalent. BMI and VCI trajectories were associated with poor vision, highlighting the importance of integrated monitoring and targete...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947915/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261029040746&v=2.20.1

From Awareness to Action : Fall Prevention in Practice

Jola Jimoh 2026-10-28 1 min 73 words LOW VISION

Summary: Falls represent a public health and economic risk for older adults, serving as the leading cause of fatal and nonfatal injuries nationally and within Delaware. This article describes the efforts of a multidisciplinary team of subject matter experts from area health care systems and academic institut...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948503/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261029040746&v=2.20.1

Leveraging Artificial Intelligence in Healthcare in Cambodia: Enhancing Access and Patient Outcomes

👤 Virak Sorn 📅 2026-10-28 📖 1 min 📝 58 words LOW VISION

Summary: Artificial intelligence (AI) is increasingly reshaping healthcare globally and may offer opportunities to strengthen health systems in low- and middle-income countries (LMICs), including Cambodia. AI-enabled technologies, including machine learning, deep learning, predictive analytics, computer visi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948795/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261029040746&v=2.20.1

Marine Carbonate System Observing and mCDR: New Processes, Needs, and Systems

👤 Zhaohui Aleck Wang 📅 2026-10-28 📖 1 min 📝 70 words LOW VISION

Summary: Marine carbon dioxide removal (mCDR) techniques have the potential to remove legacy emitted carbon dioxide from the atmosphere and store it durably in ocean reservoirs. While the ocean's vast surface area and buffering capacity offer benefits for scaling removal, monitoring remains challenging. This...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949060/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261029040746&v=2.20.1

Thirty-year follow-up of retinal vasculitis at a tertiary eye care center in India

👤 Parthopratim Dutta Majumder 📅 2026-10-28 📖 1 min 📝 47 words LOW VISION

Summary: CONCLUSION: Retinal vasculitis has a chronic and unpredictable course, warranting long-term follow-up to detect late bilateral involvement and recurrence. While recurrence becomes less frequent over time, it may still occur after many years. Most patients continue to have idiopathic disease without ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949077/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261029040746&v=2.20.1

Visual and Patient-Reported Outcomes of a Bilaterally Implanted Enhanced Monofocal IOL: A Prospective, Multicenter Study

👤 Eric Donnenfeld 📅 2026-10-28 📖 1 min 📝 40 words LOW VISION

Summary: CONCLUSION: Bilateral implantation of enVista Aspire partial-depth-of-field enhanced monofocal IOL demonstrated very good uncorrected visual acuity at far and intermediate and can be a good option for patients who prioritize good distance and intermediate vision and a favorable dysphotopsia profile.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949788/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261029040746&v=2.20.1

Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux 📅 2026-07-18 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42470115/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261029040744&v=2.20.1

A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42532383/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261029040744&v=2.20.1

Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42590984/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261029040744&v=2.20.1

Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42596634/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261029040744&v=2.20.1

Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42614895/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261029040744&v=2.20.1

A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42787568/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261029040744&v=2.20.1

Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42788316/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261029040744&v=2.20.1

Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42817759/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261029040744&v=2.20.1

Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42923236/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261029040744&v=2.20.1

Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42926384/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261029040744&v=2.20.1

Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42711906/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261029040741&v=2.20.1

Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42768901/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261029040741&v=2.20.1

Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42775591/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261029040741&v=2.20.1

Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42789611/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261029040741&v=2.20.1

Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42845092/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261029040741&v=2.20.1

Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42848198/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261029040741&v=2.20.1

Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42881664/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261029040741&v=2.20.1

Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885551/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261029040741&v=2.20.1

An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885828/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261029040741&v=2.20.1

Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42941752/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261029040741&v=2.20.1

Non-Invasive Brain Stimulation for Central Neuropathic Pain: An Exploratory Randomized Sham-Controlled Crossover Trial

Eunhee Sohn 2026-10-27 1 min 64 words TDCS TACS TRNS

Summary: Background/Objectives: Central neuropathic pain (CNP) is difficult to manage, and comparative evidence for non-invasive brain stimulation is limited. We explored the short-term efficacy and safety of repetitive transcranial magnetic stimulation (rTMS) and transcranial direct current stimulation (tDC...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939331/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261029040738&v=2.20.1

Noninvasive Neuromodulation for Cognitive Dysfunction in Parkinson's Disease: A Scoping Review

Reginald B O'Hara 2026-10-27 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: Evidence specific to PD-MCI is limited and mixed. Several studies pairing stimulation with cognitive training reported favorable outcomes, but this pattern was inconsistent and does not establish superiority over stimulation alone. No VNS trial enrolled participants with confirmed PD-MC...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939339/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261029040738&v=2.20.1

Comparative Plastid Phylogenomics of 29 Newly Sequenced Bougainvillea Cultivars Resolves the Phylogenetic Placement of Double-Bracted Cultivars

👤 Huiqi Zhao
📅 2026-10-27 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: Bougainvillea (Nyctaginaceae) cultivars are horticulturally prized worldwide for their vivid bracts; however, repeated interspecific hybridization and somatic mutations have obscured their maternal origins, particularly for double-bracted forms. Here, we sequenced and assembled the complete plastome...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939474/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261029040738&v=2.20.1

Anodal Transcranial Direct Current Stimulation Reduces Neuroinflammation and Cholinergic Dysregulation After Young Adult Traumatic Brain Injury

👤 Guven Akcay
📅 2026-10-27 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Traumatic brain injury (TBI) during childhood and adolescence can result in persistent motor and cognitive dysfunction due to secondary injury cascades involving neuroinflammation and disruption of neurotransmitter homeostasis. Although transcranial Direct Current Stimulation (tDCS) is a promising n...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939682/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261029040738&v=2.20.1

Assessing brain-behaviour coupling in non-invasive brain stimulation: Insights from reliable change indices and Monte Carlo simulations

👤 Alexander D Tang
📅 2026-10-27 📖 1 min 📝 64 words

TDCS TACS TRNS

Summary: Non-invasive brain stimulation (NiBS) studies frequently report exploratory brain-behaviour correlations following the application of NiBS. However, these analyses are typically underpowered and rely on ratio-based change scores with known statistical limitations. We addressed these limitations by p...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945749/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261029040738&v=2.20.1

The application of brain-computer interface combined with neuromodulation for stroke upper-limb rehabilitation: a systematic review and network meta-analysis

👤 Zhongjian Luo
📅 2026-10-28 📖 1 min 📝 42 words

TDCS TACS TRNS

Summary: CONCLUSION: Some non-invasive MI-BCI combined interventions like neuromodulation, ROB, or FES might improve upper-limb function and daily activities in stroke patients. However, rankings don't imply clinical importance. Due to small samples, limited evidence, and differences, these findings need con...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947035/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261029040738&v=2.20.1

Comparison of the efficacy of neuromodulation techniques based on corticospinal excitability in improving exercise-induced fatigue in athletes: a systematic review and network meta-analysis

👤 Li Qing 📅 2026-10-28 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSION: TEAS combined with acupuncture may be associated with the most favorable SUCRA ranking, but this should be interpreted as exploratory for improving EIF in athletes. However, the overall quality of current evidence is low, and SUCRA rankings should be interpreted as exploratory. Future hi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947696/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261029040738&v=2.20.1

Across established non-invasive neuromodulation and emerging deep-targeting strategies: a comparative bibliometric analysis of TI, TMS, tDCS, and tFUS/TUS/LIFU

👤 Shuang Wu
📅 2026-10-28 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Non-invasive brain stimulation (NIBS) is expanding from conventional cortical modulation toward approaches capable of engaging deeper brain targets, but the literature structures of major modalities have rarely been compared within a common framework. We conducted a comparative bibliometric mapping ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947882/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261029040738&v=2.20.1

mPFC-Dependent Valuation Shapes Information Sharing: Causal Evidence from tDCS and a Dyadic Social Media Paradigm

👤 Jinjin Wang 📅 2026-10-28 📖 1 min 📝 72 words TDCS TACS TRNS

Summary: Information sharing drives the spread of both accurate and misleading content, yet its psychological and neural mechanisms remain unclear. We investigated whether perceived self- and social relevance influence sharing through downstream content evaluations, and whether this process depends on medial...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949098/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261029040738&v=2.20.1

Joint a priori kinetic modeling with neural ODE for dual-tracer dynamic PET separation

👤 Su Ruan 📅 2026-10-28 📖 1 min 📝 66 words TDCS TACS TRNS

Summary: Simultaneous dual-tracer dynamic PET imaging acquires a time series of images and could streamline oncology workflows by enabling the quantification of complementary biological processes in a single session. However, separating the superimposed tracer signals from the resulting mixed image sequence ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42950478/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261029040738&v=2.20.1

Self-Selected Versus Assigned Emotion Regulation in Pain Expectancy: Behavioral and fNIRS Evidence

👤 Ping Shi 📅 2026-10-27 📖 1 min 📝 63 words FNIRS

Summary: Background: Expectation is a powerful cognitive modulator of pain, but how different types of emotion regulation approaches interact with expectancy to influence pain processing remains unclear. The present study compared self-selected habitual strategies (cognitive reappraisal or deep breathing) wi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939283/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261029040736&v=2.20.1

Comparative Effects of Acute HIIE and MICE on Executive Function, Prefrontal Oxyhaemoglobin Responses, and Related Biomarkers in Sedentary Young Adults: A Randomized Crossover Trial

👤 Xinyue Yang
📅 2026-10-27 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSIONS: Both exercise sessions were followed by faster reaction times on selected tasks. The small sample, multiple analyses, and potential systemic contributions to fNIRS signals preclude firm conclusions about exercise-mode superiority or biological mediation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939307/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261029040736&v=2.20.1

Cerebral Hemodynamic Patterns in People with Subjective Cognitive Decline: A Systematic Review Based on Functional Near-Infrared Spectroscopy

👤 Israel Contador
📅 2026-10-27 📖 1 min 📝 62 words FNIRS

Summary: Background/Objectives: Subjective cognitive decline (SCD) has been proposed as an early symptomatic manifestation of dementia/Alzheimer's disease. However, its associated brain patterns are not yet fully characterized. In recent years, functional near-infrared spectroscopy (fNIRS) has emerged as a c...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939380/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261029040736&v=2.20.1

Resting-State Functional Brain Connectivity and Neurodevelopmental Profiles in 3-Year-Olds with Congenital Heart Disease

👤 Anne Gallagher
📅 2026-10-27 📖 1 min 📝 67 words FNIRS

Summary: Background/Objectives: Congenital heart disease (CHD) is associated with altered brain development and an increased risk of neurodevelopmental difficulties, particularly in more severe forms of the condition. Although alterations in functional brain connectivity have been reported before and after t...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42940032/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261029040736&v=2.20.1

Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan
📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944288/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261029040736&v=2.20.1

Multimodal Sensing and Digital Biomarkers in Virtual Reality-Based Cognitive Rehabilitation: Clinical Validation, Adaptive Systems, and Real-World Translation

👤 Raffaele Ferri
📅 2026-10-27 📖 1 min 📝 58 words FNIRS

Summary: Virtual reality (VR) is evolving from a platform for cognitive exercises into an instrumented rehabilitation environment capable of simultaneously delivering treatment and quantifying behavior. This critical narrative review examines how behavioral, ocular, kinematic, physiological, and neurophysiol...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944342/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261029040736&v=2.20.1

Unpredictable human screams elicit anticipatory and phasic cortical responses that affect subsequent interpersonal space regulation

👤 Alessandro Grecucci
📅 2026-10-27 📖 1 min 📝 59 words

FNIRS

Summary: Unpredictable threats can sustain vigilance states that bias perception, cognition, and behaviour beyond immediate stimulus processing. Human screams, as evolutionarily salient alarm signals, are especially effective in eliciting defensive responses, yet their cortical dynamics and consequences for ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42946138/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261029040736&v=2.20.1

Brain activation changes during working memory training: Individual capacity effects on functional near-infrared spectroscopy measures

👤 Tali Bitan 📅 2026-10-28 📖 1 min 📝 66 words FNIRS

Summary: Working memory (WM) training has shown promise for cognitive enhancement, but the neural mechanisms underlying training effects and offline consolidation processes remain unclear. This study examined the correspondence between behavioral and neural changes during WM training and how individual diffe...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42950947/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261029040736&v=2.20.1

Dyadic fNIRS Hyperscanning: Experimental Paradigms and What They Reveal About Interbrain Synchrony

👤 Miguel Castelo-Branco 📅 2026-10-28 📖 1 min 📝 65 words FNIRS

Summary: Functional near-infrared spectroscopy (fNIRS) hyperscanning enables the simultaneous measurement of the brain activities of various individuals performing shared or interactive tasks and estimating their interbrain synchrony (IBS). IBS is believed to reflect different mechanisms, including social bo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42950963/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261029040736&v=2.20.1

Neural, vascular, motor and autonomic habituation to high-intensity auditory stimulation: A multimodal characterization of the human defensive responses

👤 Carlos M Gómez
📅 2026-10-28 📖 1 min 📝 55 words FNIRS

Summary: The defense reflex has been extensively characterized across sensory modalities; however, the integration of electrical, neurovascular, motor, and peripheral autonomic responses involving the central and autonomic nervous systems remains poorly understood. This study investigated defensive responses...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42951257/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261029040736&v=2.20.1

Applications of functional ultrasound brain-computer interfaces in neurological diseases: research progress based on neurovascular coupling

👤 Chengshan Zhang
📅 2026-10-28 📖 1 min 📝 55 words

BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) bypass peripheral nerves and muscles to establish direct information pathways between the brain and external devices, providing novel approaches for motor function reconstruction, communication assistance, and neurorehabilitation in patients with neurological disorde...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947961/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261029040733&v=2.20.1

Terahertz Photons Ameliorate Traumatic Brain Injury and Reduce Neuroinflammation via the Cyclic Adenosine Monophosphate Signaling Pathway

👤 Chao Chang
📅 2026-10-28 📖 1 min 📝 73 words

BRAIN COMPUTER INTERFACE

Summary: Traumatic brain injury (TBI) has the highest incidence among common neurological disorders and remains the leading cause of mortality and disability among all kinds of injuries, yet specific therapies for TBI are still unavailable. In the present study, we found that short-term administration of ter...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948201/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261029040733&v=2.20.1

Expert consensus on neuro-musculoskeletal functional reconstruction for post-stroke spastic hemiplegia

👤 “Zunyi Brain-Computer Interface Medical Center” Working Group

📅 2026-10-28 📖 1 min

📝 59 words

BRAIN COMPUTER INTERFACE

Summary: Persistent motor dysfunction in post-stroke spastic hemiplegia may result from impaired voluntary motor control, abnormal muscle overactivity, fixed structural restriction, muscle force imbalance, or their combination. Advances in brain-computer interfaces (BCIs), peripheral nerve surgery, and tendo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948598/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261029040733&v=2.20.1

Conducting polymers for implantable neural electrodes in brain-computer interfaces

Yang Wang 2026-10-28 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: As core elements of brain-computer interfaces, neural electrodes play a vital role in neural signal recording and stimulation. Conventional metallic electrodes, however, suffer from several inherent drawbacks, including mechanical mismatch with soft neural tissues, high interfacial impedance, and ch...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948971/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261029040733&v=2.20.1

Recognizing Brain Pioneering as Hard Work: Strategies for Making Research Participation More Meaningful

Eran Klein 2026-10-28 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) research offers novel opportunities for new tools for mobility and communication for those limited by injury or illness. Despite this promise, current BCI research remains largely early stage and without direct therapeutic benefits for the trial participants with motor...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949596/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261029040733&v=2.20.1

Two Kinds of Mental Privacy: What Speech Decoding Does and Does Not Threaten

👤 Marcello Ienca 📅 2026-10-28 📖 1 min 📝 66 words BRAIN COMPUTER INTERFACE

Summary: In the article "Speech Restoration and Mental Surveillance as Conflicting Ethical Concerns in Neural Speech-Decoding Research," Narayan Sankaran and Winston Chiong distinguish speech-decoding strategies by whether they read brain signals tied to communicative intent, softening the common equation of...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949611/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261029040733&v=2.20.1

Speech Restoration & Mental Surveillance as Conflicting Ethical Concerns in Neural Speech-Decoding Research

👤 Winston Chiong 📅 2026-10-28 📖 1 min 📝 68 words BRAIN COMPUTER INTERFACE

Summary: The research program of speech decoding-"reading" words directly from brain activity-elicits an unusually wide range of public responses, reflecting a tension between the ethical goal of restoring speech for people with serious communication disabilities and the ethical goal of protecting mental pri...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949614/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261029040733&v=2.20.1

Promises but Also Perils: Why We Should Not Dismiss Mental Surveillance Risks in Neurotechnology Research

Andrea Lavazza 2026-10-28 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: Speech brain-computer interfaces may restore communication to people who have lost the ability to speak. In the article "Speech Restoration and Mental Surveillance as Conflicting Ethical Concerns in Neural Speech-Decoding Research," Narayan Sankaran and Winston Chiong distinguish systems that decode...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949615/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261029040733&v=2.20.1

Susceptibility to task-irrelevant auditory distractors in relation to visual working memory in children with and without attention-deficit/hyperactivity disorder

Yan Song 2026-10-28 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Attention-deficit/hyperactivity disorder (ADHD) is a prevalent neurodevelopmental disorder associated with increased distractibility. Previous research has reported that susceptibility to task-irrelevant visual distraction is linked with visual working memory (WM) capacity in adults. Here, we examin...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949653/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261029040733&v=2.20.1

A synchronized dual-scale platform combining flexible micro-ECoG and skull-screw EEG reveals frequency-dependent cortex-to-skull spectral gain in awake mice

Zhaohui Song 2026-10-28 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) is the most scalable window onto brain dynamics, yet how cortical activity is transformed before reaching extracranial electrodes has largely been inferred from forward models rather than directly measured. We built a dual-scale sensing platform in awake mice that record...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42950624/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261029040733&v=2.20.1

Implicit learning of melodic structure: A role for pitch?

2024-01-22 1 min 180 words PSYCHOMUSICOLOGY

Summary: Growing evidence suggests that pitch influences musical processing, with melodic processing being enhanced in higher pitch ranges (e.g., Fujioka et al., 2005) and rhythmic processing being enhanced in lower pitches, and these effects may have a basis in elementary properties of the auditory system (...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000303

To those who feel unloved, misunderstood, or afraid

📅 2026-09-22 📖 3 min 📝 687 words FMHY

> **Summary:** I understand the world can feel devoid of love, but I swear, there is a spark within humanity that has the capability to care deeply. To love unconditionally. It doesn't always shine through us all, but it exists, it's real, it's tangible, and no one is exempt from it. We often see it shin...

🔗 **Read full article:**
https://fmhy.net/posts/love

Term frequency across 1,360 Software Engineer job postings

👤 /u/Enough_Charge2845 📅 2026-10-29 📖 1 min 📝 202 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Pulled open requisitions from public job board endpoints across 72 tech companies to see what people are actually listing. The interesting part is that practice words and concepts totally dwarf individual tech stacks. Scalability shows up in about 53 percent ...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xi5wdz/term_frequency_across_1360_software_engineer_job/

Claude Code regularly sends each background agent’s transcript to its model

👤 NikZaal 📅 2026-10-29 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://gist.github.com/flamerix-ai/c3905705cc4205e5de3011fa2e74efce Comments URL: https://news.ycombinator.com/item?id=50284255 <...

🔗 **Read full article:**
https://gist.github.com/flamerix-ai/c3905705cc4205e5de3011fa2e74efce

Zillow to pay $212.5M to settle class action for real estate venture

👤 WaitWaitWha 📅 2026-10-29 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.reuters.com/legal/government/zillow-pay-2125-million-settle-class-action-over-failed-real-estate-venture-2026-10-28/ Comme...

🔗 **Read full article:**
https://www.reuters.com/legal/government/zillow-pay-2125-million-settle-class-action-over-failed-real-estate-venture-2026-10-28/

We Won't "Just Hang Out"

barry-cotter | 2026-10-29 | 1 min | 13 words | HACKER NEWS

> **Summary:** Article URL: https://borretti.me/article/we-wont-just-hang-out Comments URL: https://news.ycombinator.com/item?id=50285154 Points: 16 # Comments: 10

🔗 **Read full article:**
https://borretti.me/article/we-wont-just-hang-out

Dogmatic Bayesianism Disorder

noreply@blogger.com (Daniel Lakens) | 2025-12-06 | 4 min | 910 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Note: This humorously intended sarcastic blog post directly mimics classifications of psychological disorders in the Diagnostic and Statistical Manual of the American Psychological Association, but Dogmatic Bayesianism is not in the DSM-5 as an actual psychologic...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/12/dogmatic-bayesianism-disorder.html

On the reliability and reproducibility of qualitative research

noreply@blogger.com (Daniel Lakens)
2026-02-08 14 min 2928 words

TWENTY PERCENT STATISTICIAN

Summary: <p class="MsoNormal">With my collaborators, I am increasingly performing qualitative research. I find qualitative research projects a useful way to improve my understanding of behaviors that I want to explore with other methods in the future. For example, some years ago I performe...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/02/on-reliability-and-reproducibility-of.html

New Papers: Optimal Filter Settings for ERP Research

Steve Luck
2024-02-04 2 min 569 words

ERP BOOT CAMP

Summary: <p class="">Zhang, G., Garrett, D. R., & Luck, S. J. (in press). Optimal filters for ERP research I: A general approach for selecting filter settings. Psychophysiology. https://doi.org/10.1111/psyp.14531 [<a href="https://www...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/2/4/optimal-filters

Q&A with Dr. Richard Carson, Professor of Biomedical Engineering and Radiology & Biomedical Imaging, Yale University and Yale School of Medicine

Adriel Carridice | 2025-11-04 | 1 min | 0 words | BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-richard-carson-professor-of-biomedical-engineering-and-radiology-biomedical-imaging-yale-university-and-yale-school-of-medicine/

Single-cell RNA sequencing reveals vascular heterogeneity and cell–cell communication remodeling in the glioma blood–tumor barrier

1 min | 31 words | BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Zhipeng Zhao, Jiachen Lang, Haiyan Gu, Luodan Wang, Jianchang Luo, Jiawang Lang, Qiongying Yao, Xinyu Qiao, Zida Wang, Ling Yin, Caidi He

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004579?dgcid=rss_sd_all

In search of Perruchet’s dissociation between expectancy and control in the flanker task

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-78183-0

Evaluation of the feasibility of classifying various upper limb movements using multichannel upper-arm sEMG signals

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-78316-5

Synaptic transmission and plasticity: molecular mechanisms, biological functions and therapeutic opportunities in neurodegenerative diseases

📅 2026-10-29 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41392-026-02776-4

Circuit plasticity coordinates action selection and temporal compression during operant conditioning in Aplysia

📅 2026-10-29 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11274-w

From pose estimation accuracy to motor biomarker reliability: a reliability-centred evaluation framework for markerless movement analysis

👤 Christine Fernandez-Maloigne 📅 2026-10-01 📖 1 min 📝 246 words

FRONTIERS NEUROINFORMATICS

Summary: Markerless pose estimation has become an essential tool for quantitative movement analysis in neuroscience. However, current evaluation frameworks primarily rely on frame-wise spatial accuracy metrics, which provide only limited information about the reliability of the downstream kinematic biomarker...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1790241

Altered interictal alpha-band organization in childhood absence epilepsy

👤 Justyna Paprocka 📅 2026-10-23 📖 1 min 📝 318 words

FRONTIERS NEUROINFORMATICS

Summary: ObjectiveInterictal background EEG in childhood absence epilepsy (CAE) may appear normal. We asked whether drug-naive children with CAE differ from controls in alpha frequency and the regional distribution of phase-lagged synchronization, and how these differences depend on participant-specific surr...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1875377

Short-term synaptic plasticity modulates the outcome of neurodegenerative diseases

👤 Chi Chung Alan Fung 📅 2026-10-23 📖 1 min 📝 264 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAlzheimer's disease (AD) is characterized by progressive neuronal loss and synaptic dysfunction that disrupt memory-encoding neural circuits. Short-term synaptic plasticity (STP), which governs transient activity-dependent changes in synaptic efficacy on millisecond-to-second timescales,...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1923602

Orientation detection for self-calibrating neuromorphic systems

Elisabetta Chicca
2026-10-23 1 min 183 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Bioinspired sensory architectures help build adaptive touch systems with efficient hardware implementations. Orientation detection is a fundamental capability for manipulating objects in real-world scenarios, with diverse applications in robotic manipulation and manufacturing assessment processes. I...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1878557

A thalamus-centered network model of restless legs syndrome: an integrative hypothesis illustrated by three clinical phenotypes

Szabolcs Várbíró
2026-10-23 1 min 231 words

FRONTIERS HUMAN NEUROSCIENCE

Summary: Restless legs syndrome (RLS) is a clinically heterogeneous sensorimotor disorder whose pathophysiology reflects interacting biological mechanisms rather than a single causal pathway. Established influences—including brain iron biology, dopaminergic regulation, glutamatergic and GABAergic balance, ad...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1977941

Cortical responses to lipread and text-based audiovisual stimuli in adults with and without dyslexia

👤 Milene Bonte 📅 2026-10-23 📖 1 min 📝 235 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Phonetic recalibration enables listeners to adjust their perception of ambiguous speech sounds based on visual contextual information. In line with a specific difficulty in forming letter-speech sound associations, adults with dyslexia have been found to show no text-based, but intact lipread-based,...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1806964

A systems-level energetic framework for conscious processing: energetic organization as a constraint on large-scale neural integration

👤 Holly Pollard-Wright 📅 2026-10-23 📖 1 min 📝 259 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Conscious processing is associated with coordinated interactions among distributed neural systems, yet the biological constraints that enable and sustain large-scale neural integration remain incompletely understood. Contemporary theories emphasize computational, informational, and network mechanism...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1966072

Editorial: New horizons in stroke management, volume II

Nasser Kashou 2026-10-23 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.2023091

Immediate effect of gait training using the hybrid assistive limb in patients in the acute phase of stroke: a prospective pilot study

Yasushi Takagi 2026-10-23 1 min 360 words FRONTIERS NEUROROBOTICS

Summary: IntroductionThe effectiveness of Hybrid Assistive Limb (HAL) gait training in patients in the acute phase of stroke remains unclear. We prospectively assessed the immediate effects of HAL gait training in this patient population.MethodsIn this prospective pilot study, four patients with acute-phase ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1985500

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028213613&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39870674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028213613&v=2.20.1

Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028213613&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028213613&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028213613&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41358599/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028213613&v=2.20.1

Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41676644/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028213613&v=2.20.1

Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41678848/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028213613&v=2.20.1

The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41703344/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028213613&v=2.20.1

EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42538927/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028213613&v=2.20.1

Communication patterns in home medication review for individuals with vision impairment: A qualitative study

👤 Woojin Park
📅 2026-10-27 📖 1 min 📝 37 words LOW VISION

Summary: CONCLUSIONS: In these consultations, professional dialogue supported verification and reassurance, while direct regimen review revealed unverified beliefs and gaps in medication knowledge. The findings suggest a role for accessible counseling that combines information provision with professional ver...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945707/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028213610&v=2.20.1

Global, regional, and national burden of unintentional poisoning, 1990-2023: results from the Global Burden of Disease Study 2023

👤 GBD 2023 Unintentional Poisoning Collaborators

📅 2026-10-27 📖 1 min 📝 65 words

LOW VISION

Summary: BACKGROUND: Unintentional poisoning is a preventable cause of injury-related morbidity and mortality, yet global patterns, trends, and inequalities remain incompletely characterised. Within the Global Burden of Disease (GBD) framework, unintentional poisoning includes poisoning by carbon monoxide an...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945982/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028213610&v=2.20.1

Directing Ru-based acidic oxygen evolution catalysts toward a universal dual-site pathway

👤 Chuangang Hu 📅 2026-10-28 📖 1 min 📝 62 words LOW VISION

Summary: The oxide path mechanism offers vision possibility from kinetics for the anodic oxygen evolution reaction, yet its universal activation remains elusive due to the difficulty of constructing globally appropriate interatomic distances. Here, we address this challenge in the integrated-screened Y(0.1)I...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42946705/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028213610&v=2.20.1

Biomimetic computational modelling of retinal ganglion cell responses to contact lens designs with refractive and opaque non-refractive features

👤 Praveen K Bandela 📅 2026-10-28 📖 1 min 📝 67 words LOW VISION

Summary: CONCLUSIONS: In this computational model, both BF and AR lens designs produced measurable changes in simulated RGC temporal activity relative to SV, with AR inducing greater and more configuration-independent modulation. These findings must be considered hypothesis-generating only; they are consiste...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947083/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028213610&v=2.20.1

Growth, nutritional status, and physical fitness trajectories among primary and secondary school students and their associations with poor vision: a school-based study in Chongqing, China

Yong Zhao 2026-10-28 1 min 42 words LOW VISION

Summary: CONCLUSION: From 2022 to 2026, growth and development indicators among primary and secondary school students in Yuzhong District improved, while poor vision remained prevalent. BMI and VCI trajectories were associated with poor vision, highlighting the importance of integrated monitoring and targete...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947915/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028213610&v=2.20.1

From Awareness to Action : Fall Prevention in Practice

Jola Jimoh 2026-10-28 1 min 73 words LOW VISION

Summary: Falls represent a public health and economic risk for older adults, serving as the leading cause of fatal and nonfatal injuries nationally and within Delaware. This article describes the efforts of a multidisciplinary team of subject matter experts from area health care systems and academic institut...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948503/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028213610&v=2.20.1

Leveraging Artificial Intelligence in Healthcare in Cambodia: Enhancing Access and Patient Outcomes

👤 Virak Sorn 📅 2026-10-28 📖 1 min 📝 58 words LOW VISION

Summary: Artificial intelligence (AI) is increasingly reshaping healthcare globally and may offer opportunities to strengthen health systems in low- and middle-income countries (LMICs), including Cambodia. AI-enabled technologies, including machine learning, deep learning, predictive analytics, computer visi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948795/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028213610&v=2.20.1

Marine Carbonate System Observing and mCDR: New Processes, Needs, and Systems

👤 Zhaohui Aleck Wang 📅 2026-10-28 📖 1 min 📝 70 words LOW VISION

Summary: Marine carbon dioxide removal (mCDR) techniques have the potential to remove legacy emitted carbon dioxide from the atmosphere and store it durably in ocean reservoirs. While the ocean's vast surface area and buffering capacity offer benefits for scaling removal, monitoring remains challenging. This...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949060/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028213610&v=2.20.1

Thirty-year follow-up of retinal vasculitis at a tertiary eye care center in India

Parthopratim Dutta Majumder 2026-10-28 1 min 47 words LOW VISION

Summary: CONCLUSION: Retinal vasculitis has a chronic and unpredictable course, warranting long-term follow-up to detect late bilateral involvement and recurrence. While recurrence becomes less frequent over time, it may still occur after many years. Most patients continue to have idiopathic disease without ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949077/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028213610&v=2.20.1

Visual and Patient-Reported Outcomes of a Bilaterally Implanted Enhanced Monofocal IOL: A Prospective, Multicenter Study

Eric Donnenfeld 2026-10-28 1 min 40 words LOW VISION

Summary: CONCLUSION: Bilateral implantation of enVista Aspire partial-depth-of-field enhanced monofocal IOL demonstrated very good uncorrected visual acuity at far and intermediate and can be a good option for patients who prioritize good distance and intermediate vision and a favorable dysphotopsia profile.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949788/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028213610&v=2.20.1

Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42470115/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028213608&v=2.20.1

A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42532383/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028213608&v=2.20.1

Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42590984/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028213608&v=2.20.1

Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42596634/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028213608&v=2.20.1

Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42614895/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028213608&v=2.20.1

A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42787568/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028213608&v=2.20.1

Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42788316/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028213608&v=2.20.1

Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42817759/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028213608&v=2.20.1

Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42923236/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028213608&v=2.20.1

Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42926384/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028213608&v=2.20.1

Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42711906/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028213606&v=2.20.1

Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42768901/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028213606&v=2.20.1

Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42775591/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028213606&v=2.20.1

Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42789611/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028213606&v=2.20.1

Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42845092/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028213606&v=2.20.1

Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42848198/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028213606&v=2.20.1

Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42881664/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028213606&v=2.20.1

Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885551/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028213606&v=2.20.1

An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885828/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028213606&v=2.20.1

Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42941752/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028213606&v=2.20.1

Non-Invasive Brain Stimulation Techniques and Effects on Neuromodulation and Athletic Performance in Combat Sports: A Systematic Review

👤 Bianca Miarka 📅 2026-10-27 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSIONS: tDCS may influence selected cognitive, neuromuscular, and fatigue-related outcomes in combat sport athletes; however, the direction and magnitude of these effects are inconsistent across studies. Given the substantial methodological heterogeneity and the low-to-very-low certainty of evi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939305/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028213604&v=2.20.1

Non-Invasive Brain Stimulation for Central Neuropathic Pain: An Exploratory Randomized Sham-Controlled Crossover Trial

👤 Eunhee Sohn 📅 2026-10-27 📖 1 min 📝 64 words TDCS TACS TRNS

Summary: Background/Objectives: Central neuropathic pain (CNP) is difficult to manage, and comparative evidence for non-invasive brain stimulation is limited. We explored the short-term efficacy and safety of repetitive transcranial magnetic stimulation (rTMS) and transcranial direct current stimulation (tDC...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939331/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028213604&v=2.20.1

Noninvasive Neuromodulation for Cognitive Dysfunction in Parkinson's Disease: A Scoping Review

Reginald B O'Hara 2026-10-27 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: Evidence specific to PD-MCI is limited and mixed. Several studies pairing stimulation with cognitive training reported favorable outcomes, but this pattern was inconsistent and does not establish superiority over stimulation alone. No VNS trial enrolled participants with confirmed PD-MC...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939339/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028213604&v=2.20.1

Comparative Plastid Phylogenomics of 29 Newly Sequenced Bougainvillea Cultivars Resolves the Phylogenetic Placement of Double-Bracted Cultivars

Huiqi Zhao 2026-10-27 1 min 59 words TDCS TACS TRNS

Summary: Bougainvillea (Nyctaginaceae) cultivars are horticulturally prized worldwide for their vivid bracts; however, repeated interspecific hybridization and somatic mutations have obscured their maternal origins, particularly for double-bracted forms. Here, we sequenced and assembled the complete plastome...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939474/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028213604&v=2.20.1

Anodal Transcranial Direct Current Stimulation Reduces Neuroinflammation and Cholinergic Dysregulation After Young Adult Traumatic Brain Injury

👤 Guven Akcay
📅 2026-10-27 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Traumatic brain injury (TBI) during childhood and adolescence can result in persistent motor and cognitive dysfunction due to secondary injury cascades involving neuroinflammation and disruption of neurotransmitter homeostasis. Although transcranial Direct Current Stimulation (tDCS) is a promising n...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939682/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028213604&v=2.20.1

Assessing brain-behaviour coupling in non-invasive brain stimulation: Insights from reliable change indices and Monte Carlo simulations

👤 Alexander D Tang
📅 2026-10-27 📖 1 min 📝 64 words

TDCS TACS TRNS

Summary: Non-invasive brain stimulation (NiBS) studies frequently report exploratory brain-behaviour correlations following the application of NiBS. However, these analyses are typically underpowered and rely on ratio-based change scores with known statistical limitations. We addressed these limitations by p...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945749/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028213604&v=2.20.1

The application of brain-computer interface combined with neuromodulation for stroke upper-limb rehabilitation: a systematic review and network meta-analysis

Zhongjian Luo
2026-10-28 1 min 42 words TDCS TACS TRNS

Summary: CONCLUSION: Some non-invasive MI-BCI combined interventions like neuromodulation, ROB, or FES might improve upper-limb function and daily activities in stroke patients. However, rankings don't imply clinical importance. Due to small samples, limited evidence, and differences, these findings need con...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947035/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028213604&v=2.20.1

Comparison of the efficacy of neuromodulation techniques based on corticospinal excitability in improving exercise-induced fatigue in athletes: a systematic review and network meta-analysis

👤 Li Qing 📅 2026-10-28 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSION: TEAS combined with acupuncture may be associated with the most favorable SUCRA ranking, but this should be interpreted as exploratory for improving EIF in athletes. However, the overall quality of current evidence is low, and SUCRA rankings should be interpreted as exploratory. Future hi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947696/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028213604&v=2.20.1

Across established non-invasive neuromodulation and emerging deep-targeting strategies: a comparative bibliometric analysis of TI, TMS, tDCS, and tFUS/TUS/LIFU

👤 Shuang Wu
📅 2026-10-28 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Non-invasive brain stimulation (NIBS) is expanding from conventional cortical modulation toward approaches capable of engaging deeper brain targets, but the literature structures of major modalities have rarely been compared within a common framework. We conducted a comparative bibliometric mapping ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947882/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028213604&v=2.20.1

mPFC-Dependent Valuation Shapes Information Sharing: Causal Evidence from tDCS and a Dyadic Social Media Paradigm

👤 Jinjin Wang
📅 2026-10-28 📖 1 min 📝 72 words

TDCS TACS TRNS

Summary: Information sharing drives the spread of both accurate and misleading content, yet its psychological and neural mechanisms remain unclear. We investigated whether perceived self- and social relevance influence sharing through downstream content evaluations, and whether this process depends on medial...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949098/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028213604&v=2.20.1

Neural-Discursive Dynamics in Collaborative Spatial Problem Solving: Evidence from fNIRS Hyperscanning and Accountable Talk

👤 Xingguang Guo
📅 2026-10-27 📖 1 min 📝 65 words FNIRS

Summary: High-quality collaboration cannot be inferred simply from how much students talk or how strongly their brain activity is synchronized during joint work. This study examined relations among accountable talk, inter-brain synchrony (IBS), and group task performance in fifth-grade students' collaborativ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42938505/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028213601&v=2.20.1

Probing Executive Function in the Toddler Data Desert: A Touchscreen fNIRS Study of Working Memory and ADHD-Related Symptoms

Caitie Busch 2026-10-27 1 min 60 words FNIRS

Summary: Attention-deficit/hyperactivity disorder (ADHD) is associated with early differences in attentional control and working memory (WM), yet relatively few paradigms are developmentally feasible across the toddler-to-preschool period while also distinguishing spatial from featural information use. Addre...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42938612/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028213601&v=2.20.1

Mapping the Global Landscape and Thematic Evolution of fNIRS-Gait Research: A Systematic Bibliometric and Network Visualization Analysis

Naime Uluğ 2026-10-27 1 min 20 words FNIRS

Summary: CONCLUSIONS: Establishing standardized preprocessing pipelines and short-separation regressions may prove pivotal for transitioning mobile neuroimaging into objective, closed-loop clinical neurorehabilitation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939053/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028213601&v=2.20.1

Self-Selected Versus Assigned Emotion Regulation in Pain Expectancy: Behavioral and fNIRS Evidence

👤 Ping Shi 📅 2026-10-27 📖 1 min 📝 63 words FNIRS

Summary: Background: Expectation is a powerful cognitive modulator of pain, but how different types of emotion regulation approaches interact with expectancy to influence pain processing remains unclear. The present study compared self-selected habitual strategies (cognitive reappraisal or deep breathing) wi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939283/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028213601&v=2.20.1

Comparative Effects of Acute HIIE and MICE on Executive Function, Prefrontal Oxyhaemoglobin Responses, and Related Biomarkers in Sedentary Young Adults: A Randomized Crossover Trial

👤 Xinyue Yang
📅 2026-10-27 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSIONS: Both exercise sessions were followed by faster reaction times on selected tasks. The small sample, multiple analyses, and potential systemic contributions to fNIRS signals preclude firm conclusions about exercise-mode superiority or biological mediation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939307/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028213601&v=2.20.1

Cerebral Hemodynamic Patterns in People with Subjective Cognitive Decline: A Systematic Review Based on Functional Near-Infrared Spectroscopy

👤 Israel Contador
📅 2026-10-27 📖 1 min 📝 62 words FNIRS

Summary: Background/Objectives: Subjective cognitive decline (SCD) has been proposed as an early symptomatic manifestation of dementia/Alzheimer's disease. However, its associated brain patterns are not yet fully characterized. In recent years, functional near-infrared spectroscopy (fNIRS) has emerged as a c...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939380/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028213601&v=2.20.1

Resting-State Functional Brain Connectivity and Neurodevelopmental Profiles in 3-Year-Olds with Congenital Heart Disease

👤 Anne Gallagher
📅 2026-10-27 📖 1 min 📝 67 words FNIRS

Summary: Background/Objectives: Congenital heart disease (CHD) is associated with altered brain development and an increased risk of neurodevelopmental difficulties, particularly in more severe forms of the condition. Although alterations in functional brain connectivity have been reported before and after t...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42940032/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028213601&v=2.20.1

Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan
📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944288/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028213601&v=2.20.1

Multimodal Sensing and Digital Biomarkers in Virtual Reality-Based Cognitive Rehabilitation: Clinical Validation, Adaptive Systems, and Real-World Translation

👤 Raffaele Ferri
📅 2026-10-27 📖 1 min 📝 58 words FNIRS

Summary: Virtual reality (VR) is evolving from a platform for cognitive exercises into an instrumented rehabilitation environment capable of simultaneously delivering treatment and quantifying behavior. This critical narrative review examines how behavioral, ocular, kinematic, physiological, and neurophysiol...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944342/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028213601&v=2.20.1

Unpredictable human screams elicit anticipatory and phasic cortical responses that affect subsequent interpersonal space regulation

👤 Alessandro Grecucci
📅 2026-10-27 📖 1 min 📝 59 words

FNIRS

Summary: Unpredictable threats can sustain vigilance states that bias perception, cognition, and behaviour beyond immediate stimulus processing. Human screams, as evolutionarily salient alarm signals, are especially effective in eliciting defensive responses, yet their cortical dynamics and consequences for ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42946138/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028213601&v=2.20.1

The application of brain-computer interface combined with neuromodulation for stroke upper-limb rehabilitation: a systematic review and network meta-analysis

Zhongjian Luo
2026-10-28 1 min 42 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Some non-invasive MI-BCI combined interventions like neuromodulation, ROB, or FES might improve upper-limb function and daily activities in stroke patients. However, rankings don't imply clinical importance. Due to small samples, limited evidence, and differences, these findings need con...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947035/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028213559&v=2.20.1

Applications of functional ultrasound brain-computer interfaces in neurological diseases: research progress based on neurovascular coupling

👤 Chengshan Zhang
📅 2026-10-28 📖 1 min 📝 55 words

BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) bypass peripheral nerves and muscles to establish direct information pathways between the brain and external devices, providing novel approaches for motor function reconstruction, communication assistance, and neurorehabilitation in patients with neurological disorde...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947961/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028213559&v=2.20.1

Terahertz Photons Ameliorate Traumatic Brain Injury and Reduce Neuroinflammation via the Cyclic Adenosine Monophosphate Signaling Pathway

👤 Chao Chang
📅 2026-10-28 📖 1 min 📝 73 words

BRAIN COMPUTER INTERFACE

Summary: Traumatic brain injury (TBI) has the highest incidence among common neurological disorders and remains the leading cause of mortality and disability among all kinds of injuries, yet specific therapies for TBI are still unavailable. In the present study, we found that short-term administration of ter...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948201/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028213559&v=2.20.1

Expert consensus on neuro-musculoskeletal functional reconstruction for post-stroke spastic hemiplegia

👤 “Zunyi Brain-Computer Interface Medical Center” Working Group

📅 2026-10-28 📖 1 min

📝 59 words

BRAIN COMPUTER INTERFACE

Summary: Persistent motor dysfunction in post-stroke spastic hemiplegia may result from impaired voluntary motor control, abnormal muscle overactivity, fixed structural restriction, muscle force imbalance, or their combination. Advances in brain-computer interfaces (BCIs), peripheral nerve surgery, and tendo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948598/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028213559&v=2.20.1

Conducting polymers for implantable neural electrodes in brain-computer interfaces

Yang Wang 2026-10-28 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: As core elements of brain-computer interfaces, neural electrodes play a vital role in neural signal recording and stimulation. Conventional metallic electrodes, however, suffer from several inherent drawbacks, including mechanical mismatch with soft neural tissues, high interfacial impedance, and ch...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948971/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028213559&v=2.20.1

Recognizing Brain Pioneering as Hard Work: Strategies for Making Research Participation More Meaningful

Eran Klein 2026-10-28 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) research offers novel opportunities for new tools for mobility and communication for those limited by injury or illness. Despite this promise, current BCI research remains largely early stage and without direct therapeutic benefits for the trial participants with motor...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949596/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028213559&v=2.20.1

Two Kinds of Mental Privacy: What Speech Decoding Does and Does Not Threaten

Marcello Ienca 2026-10-28 1 min 66 words BRAIN COMPUTER INTERFACE

Summary: In the article "Speech Restoration and Mental Surveillance as Conflicting Ethical Concerns in Neural Speech-Decoding Research," Narayan Sankaran and Winston Chiong distinguish speech-decoding strategies by whether they read brain signals tied to communicative intent, softening the common equation of...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949611/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028213559&v=2.20.1

Speech Restoration & Mental Surveillance as Conflicting Ethical Concerns in Neural Speech-Decoding Research

Winston Chiong 2026-10-28 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: The research program of speech decoding-"reading" words directly from brain activity-elicits an unusually wide range of public responses, reflecting a tension between the ethical goal of restoring speech for people with serious communication disabilities and the ethical goal of protecting mental pri...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949614/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028213559&v=2.20.1

Promises but Also Perils: Why We Should Not Dismiss Mental Surveillance Risks in Neurotechnology Research

Andrea Lavazza 2026-10-28 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: Speech brain-computer interfaces may restore communication to people who have lost the ability to speak. In the article "Speech Restoration and Mental Surveillance as Conflicting Ethical Concerns in Neural Speech-Decoding Research," Narayan Sankaran and Winston Chiong distinguish systems that decode...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949615/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028213559&v=2.20.1

Susceptibility to task-irrelevant auditory distractors in relation to visual working memory in children with and without attention-deficit/hyperactivity disorder

Yan Song 2026-10-28 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Attention-deficit/hyperactivity disorder (ADHD) is a prevalent neurodevelopmental disorder associated with increased distractibility. Previous research has reported that susceptibility to task-irrelevant visual distraction is linked with visual working memory (WM) capacity in adults. Here, we examin...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949653/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028213559&v=2.20.1

Thursday Daily Thread: Python Careers, Courses, and Furthering Education!

👤 /u/AutoModerator 📅 2026-10-29 📖 1 min 📝 219 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <h1>Weekly Thread: Professional Use, Jobs, and Education 🏢</h1> Welcome to this week's discussion on Python in the professional world! This is your spot to talk about job hunting, career growth, and educational resources in Python. Please note, this thread is <stron...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xi4t9j/thursday_daily_thread_python_careers_courses_and/

A Health Update

👤 frizlab 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://furbo.org/2026/10/28/a-health-update/ Comments URL: https://news.ycombinator.com/item?id=50280974 Points: 92 # Comments: 8

🔗 **Read full article:**
https://furbo.org/2026/10/28/a-health-update/

StackOverflow now gets fewer than 1000 questions a month

👤 babelfish 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://data.stackexchange.com/stackoverflow/query/1926661#graph?t=1 Comments URL: https://news.ycombinator.com/item?id=50282454 <...

🔗 **Read full article:**
https://data.stackexchange.com/stackoverflow/query/1926661#graph?t=1

The first wired XR glasses powered by Android XR

👤 yincrash 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.google/products-and-platforms/platforms/android/xreal-aura/ Comments URL: <a href="https://news.ycombinator.com/item?id=50282538">https://news.ycombinator.com/item?id=50282538</...

🔗 **Read full article:**
https://blog.google/products-and-platforms/platforms/android/xreal-aura/

GitLab Security Incident: Attempted Exploitation of CVE-2026-97229

👤 sashk 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://alpinelinux.org/posts/GitLab-security-incident-CVE-2026-97229.html Comments URL: <a href="https://news.ycombinator.com/item?id=50282544">https://news.ycombinator.com/item?id=502825...

🔗 **Read full article:**
https://alpinelinux.org/posts/GitLab-security-incident-CVE-2026-97229.html

New York Times adds more AI features to its Cooking app. Users aren't thrilled

👤 ilamont 📅 2026-10-29 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.niemanlab.org/2026/10/the-new-york-times-is-adding-more-ai-features-to-its-cooking-app-and-users-arent-thrilled/ Comments URL:...

🔗 **Read full article:**
https://www.niemanlab.org/2026/10/the-new-york-times-is-adding-more-ai-features-to-its-cooking-app-and-users-arent-thrilled/

Qwenputer: A functioning, deterministic RISC-V CPU inside Qwen3.5

👤 csmantle 📅 2026-10-29 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://divagr.com/blog/qwenputer Comments URL: https://news.ycombinator.com/item?id=50282838 Points: 29 # Comments: 2

🔗 **Read full article:**
https://divagr.com/blog/qwenputer

MicroZig to Adopt Strict No LLM / No AI Policy

👤 paaloeye 📅 2026-10-29 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/ZigEmbeddedGroup/microzig/blob/main/CODE_OF_CONDUCT.md Comments URL: <a href="https://news.ycombinator.com/item?id=50282865">https://news.ycombinator.com/item?id=50282865...

🔗 **Read full article:**
https://github.com/ZigEmbeddedGroup/microzig/blob/main/CODE_OF_CONDUCT.md

Evaluating Dr. Cuddy’s Claim that the Debunking of Power Posing is a Myth

noreply@blogger.com (Daniel Lakens)
2026-05-25 30 min 6008 words

TWENTY PERCENT STATISTICIAN

Summary: In this blog post I will analyse the arguments that Dr. Amy Cuddy provided in a LinkedIn post “The "Power Posing Was Debunked" Myth: What the Research Actually Shows — and Why Scientific Discourse Matters” on February 26. You can find the LinkedIn post here: <span lang="...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/evaluating-dr-cuddys-claim-that.html

Episode 29: Q&A with Dr. Anahita Kyani, Director of Neuroscience, Data Science and Digital Health at Johnson & Johnson Innovative Medicine

Adriel Carridice
2026-06-22 1 min 0 words

BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/episode-29-qa-with-dr-anahita-kyani-director-of-neuroscience-data-science-and-digital-health-at-johnson-johnson-innovative-medicine/

Q&A with Felix Schwock, PhD student at University of Washington and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-felix-schwock-phd-student-at-university-of-washington-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor in Chief Search

👤 Deidre Artis 📅 2026-07-17 📖 1 min 📝 34 words EMBS

> **Summary:** Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor-in-Chief Search Submit your Application The post <a href="https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/">Transactions on Neural Syst...

🔗 **Read full article:**
https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/

State-specific dynamic functional connectivity disturbances in spinal cord injury-related neuropathic pain

1 min | 29 words | BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Chuchu Sun, Haotian Xin, Ling Wang, Qian Chen, Yu Wang, Beining Yang, Xin Chen, Fang Li, Jie Lu, Nan Chen

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004658?dgcid=rss_sd_all

Towards a scalable effective connectivity estimation in fMRI

Mariani, B., Taffarello, L., Baron, G., Barzon, G., Calesella, F., Gilson, M., de Marchi, S., Chiuso, A., Vallesi, A., Suweis, S., Zorzi, M., Bertoldo, A., Allegra, M.

2026-10-28 | 1 min | 229 words | BIORXIV NEUROSCIENCE

Summary: Characterizing directed connections between brain regions is fundamental for understanding the brains network organization and how it supports cognitive processes. However, a reliable methodology allowing assessment of effective connectivity (EC) at the whole-brain level and across high-resolution p...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.761780v1?rss=1

An information-theoretic analysis of the tsetse fly (Glossina morsitans) antenna reveals trichoid sensilla as the main carriers of odor identity

👤 Soni, N. 📅 2026-10-28 📖 1 min 📝 293 words BIORXIV NEUROSCIENCE

Summary: Tsetse flies locate vertebrate hosts largely by smell, and odor-baited devices are central to tsetse and trypanosomiasis control. A published electrophysiological survey of the Glossina morsitans antenna defined seven classes of olfactory sensilla, but descriptive clustering cannot reveal how much i...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.761766v1?rss=1

How large language models handle ethical dilemmas when providing mental health and emotional support

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s44184-026-00255-w

Serotonin signaling enhances phagocytosis by brain macrophages through Htr2b and alleviates brain injury after stroke

Peiying Li 2026-10-28 1 min 45 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 28 October 2026; doi:10.1038/s41593-026-02482-y The serotonin receptor Htr2b activation enhances lysosome biogenesis and phagocytic clearance of debris to mitigate neuroinflammation and brai...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02482-y

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028173807&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39870674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028173807&v=2.20.1

Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028173807&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028173807&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028173807&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41358599/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028173807&v=2.20.1

Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41676644/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028173807&v=2.20.1

Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41678848/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028173807&v=2.20.1

The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41703344/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028173807&v=2.20.1

EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42538927/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028173807&v=2.20.1

Communication patterns in home medication review for individuals with vision impairment: A qualitative study

👤 Woojin Park 📅 2026-10-27 📖 1 min 📝 37 words LOW VISION

Summary: CONCLUSIONS: In these consultations, professional dialogue supported verification and reassurance, while direct regimen review revealed unverified beliefs and gaps in medication knowledge. The findings suggest a role for accessible counseling that combines information provision with professional ver...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945707/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028173805&v=2.20.1

Global, regional, and national burden of unintentional poisoning, 1990-2023: results from the Global Burden of Disease Study 2023

👤 GBD 2023 Unintentional Poisoning Collaborators

📅 2026-10-27 📖 1 min 📝 65 words

LOW VISION

Summary: BACKGROUND: Unintentional poisoning is a preventable cause of injury-related morbidity and mortality, yet global patterns, trends, and inequalities remain incompletely characterised. Within the Global Burden of Disease (GBD) framework, unintentional poisoning includes poisoning by carbon monoxide an...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945982/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028173805&v=2.20.1

Directing Ru-based acidic oxygen evolution catalysts toward a universal dual-site pathway

Chuangang Hu 2026-10-28 1 min 62 words LOW VISION

Summary: The oxide path mechanism offers vision possibility from kinetics for the anodic oxygen evolution reaction, yet its universal activation remains elusive due to the difficulty of constructing globally appropriate interatomic distances. Here, we address this challenge in the integrated-screened Y(0.1)I...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42946705/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028173805&v=2.20.1

Biomimetic computational modelling of retinal ganglion cell responses to contact lens designs with refractive and opaque non-refractive features

Praveen K Bandela 2026-10-28 1 min 67 words LOW VISION

Summary: CONCLUSIONS: In this computational model, both BF and AR lens designs produced measurable changes in simulated RGC temporal activity relative to SV, with AR inducing greater and more configuration-independent modulation. These findings must be considered hypothesis-generating only; they are consiste...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947083/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028173805&v=2.20.1

Growth, nutritional status, and physical fitness trajectories among primary and secondary school students and their associations with poor vision: a school-based study in Chongqing, China

Yong Zhao 2026-10-28 1 min 42 words LOW VISION

Summary: CONCLUSION: From 2022 to 2026, growth and development indicators among primary and secondary school students in Yuzhong District improved, while poor vision remained prevalent. BMI and VCI trajectories were associated with poor vision, highlighting the importance of integrated monitoring and targete...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947915/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028173805&v=2.20.1

From Awareness to Action : Fall Prevention in Practice

Jola Jimoh 2026-10-28 1 min 73 words LOW VISION

Summary: Falls represent a public health and economic risk for older adults, serving as the leading cause of fatal and nonfatal injuries nationally and within Delaware. This article describes the efforts of a multidisciplinary team of subject matter experts from area health care systems and academic institut...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948503/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028173805&v=2.20.1

Leveraging Artificial Intelligence in Healthcare in Cambodia: Enhancing Access and Patient Outcomes

👤 Virak Sorn 📅 2026-10-28 📖 1 min 📝 58 words LOW VISION

Summary: Artificial intelligence (AI) is increasingly reshaping healthcare globally and may offer opportunities to strengthen health systems in low- and middle-income countries (LMICs), including Cambodia. AI-enabled technologies, including machine learning, deep learning, predictive analytics, computer visi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948795/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028173805&v=2.20.1

Marine Carbonate System Observing and mCDR: New Processes, Needs, and Systems

👤 Zhaohui Aleck Wang 📅 2026-10-28 📖 1 min 📝 70 words LOW VISION

Summary: Marine carbon dioxide removal (mCDR) techniques have the potential to remove legacy emitted carbon dioxide from the atmosphere and store it durably in ocean reservoirs. While the ocean's vast surface area and buffering capacity offer benefits for scaling removal, monitoring remains challenging. This...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949060/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028173805&v=2.20.1

Thirty-year follow-up of retinal vasculitis at a tertiary eye care center in India

👤 Parthopratim Dutta Majumder 📅 2026-10-28 📖 1 min 📝 47 words LOW VISION

Summary: CONCLUSION: Retinal vasculitis has a chronic and unpredictable course, warranting long-term follow-up to detect late bilateral involvement and recurrence. While recurrence becomes less frequent over time, it may still occur after many years. Most patients continue to have idiopathic disease without ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949077/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028173805&v=2.20.1

Visual and Patient-Reported Outcomes of a Bilaterally Implanted Enhanced Monofocal IOL: A Prospective, Multicenter Study

👤 Eric Donnenfeld 📅 2026-10-28 📖 1 min 📝 40 words LOW VISION

Summary: CONCLUSION: Bilateral implantation of enVista Aspire partial-depth-of-field enhanced monofocal IOL demonstrated very good uncorrected visual acuity at far and intermediate and can be a good option for patients who prioritize good distance and intermediate vision and a favorable dysphotopsia profile.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949788/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028173805&v=2.20.1

Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42470115/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028173802&v=2.20.1

A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42532383/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028173802&v=2.20.1

Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42590984/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028173802&v=2.20.1

Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42596634/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028173802&v=2.20.1

Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42614895/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028173802&v=2.20.1

A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42787568/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028173802&v=2.20.1

Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42788316/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028173802&v=2.20.1

Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42817759/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028173802&v=2.20.1

Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42923236/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028173802&v=2.20.1

Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42926384/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028173802&v=2.20.1

Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42711906/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028173800&v=2.20.1

Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42768901/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028173800&v=2.20.1

Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42775591/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028173800&v=2.20.1

Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42789611/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028173800&v=2.20.1

Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42845092/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028173800&v=2.20.1

Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42848198/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028173800&v=2.20.1

Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42881664/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028173800&v=2.20.1

Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885551/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028173800&v=2.20.1

An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885828/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028173800&v=2.20.1

Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42941752/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028173800&v=2.20.1

Non-Invasive Brain Stimulation Techniques and Effects on Neuromodulation and Athletic Performance in Combat Sports: A Systematic Review

👤 Bianca Miarka 📅 2026-10-27 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSIONS: tDCS may influence selected cognitive, neuromuscular, and fatigue-related outcomes in combat sport athletes; however, the direction and magnitude of these effects are inconsistent across studies. Given the substantial methodological heterogeneity and the low-to-very-low certainty of evi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939305/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028173758&v=2.20.1

Non-Invasive Brain Stimulation for Central Neuropathic Pain: An Exploratory Randomized Sham-Controlled Crossover Trial

👤 Eunhee Sohn 📅 2026-10-27 📖 1 min 📝 64 words TDCS TACS TRNS

Summary: Background/Objectives: Central neuropathic pain (CNP) is difficult to manage, and comparative evidence for non-invasive brain stimulation is limited. We explored the short-term efficacy and safety of repetitive transcranial magnetic stimulation (rTMS) and transcranial direct current stimulation (tDC...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939331/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028173758&v=2.20.1

Noninvasive Neuromodulation for Cognitive Dysfunction in Parkinson's Disease: A Scoping Review

Reginald B O'Hara 2026-10-27 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: Evidence specific to PD-MCI is limited and mixed. Several studies pairing stimulation with cognitive training reported favorable outcomes, but this pattern was inconsistent and does not establish superiority over stimulation alone. No VNS trial enrolled participants with confirmed PD-MC...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939339/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028173758&v=2.20.1

Comparative Plastid Phylogenomics of 29 Newly Sequenced Bougainvillea Cultivars Resolves the Phylogenetic Placement of Double-Bracted Cultivars

Huiqi Zhao 2026-10-27 1 min 59 words TDCS TACS TRNS

Summary: Bougainvillea (Nyctaginaceae) cultivars are horticulturally prized worldwide for their vivid bracts; however, repeated interspecific hybridization and somatic mutations have obscured their maternal origins, particularly for double-bracted forms. Here, we sequenced and assembled the complete plastome...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939474/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028173758&v=2.20.1

Anodal Transcranial Direct Current Stimulation Reduces Neuroinflammation and Cholinergic Dysregulation After Young Adult Traumatic Brain Injury

Guven Akcay
2026-10-27 1 min 61 words TDCS TACS TRNS

Summary: Traumatic brain injury (TBI) during childhood and adolescence can result in persistent motor and cognitive dysfunction due to secondary injury cascades involving neuroinflammation and disruption of neurotransmitter homeostasis. Although transcranial Direct Current Stimulation (tDCS) is a promising n...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939682/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028173758&v=2.20.1

Assessing brain-behaviour coupling in non-invasive brain stimulation: Insights from reliable change indices and Monte Carlo simulations

👤 Alexander D Tang
📅 2026-10-27 📖 1 min 📝 64 words

TDCS TACS TRNS

Summary: Non-invasive brain stimulation (NiBS) studies frequently report exploratory brain-behaviour correlations following the application of NiBS. However, these analyses are typically underpowered and rely on ratio-based change scores with known statistical limitations. We addressed these limitations by p...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945749/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028173758&v=2.20.1

The application of brain-computer interface combined with neuromodulation for stroke upper-limb rehabilitation: a systematic review and network meta-analysis

👤 Zhongjian Luo
📅 2026-10-28 📖 1 min 📝 42 words

TDCS TACS TRNS

Summary: CONCLUSION: Some non-invasive MI-BCI combined interventions like neuromodulation, ROB, or FES might improve upper-limb function and daily activities in stroke patients. However, rankings don't imply clinical importance. Due to small samples, limited evidence, and differences, these findings need con...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947035/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028173758&v=2.20.1

Comparison of the efficacy of neuromodulation techniques based on corticospinal excitability in improving exercise-induced fatigue in athletes: a systematic review and network meta-analysis

👤 Li Qing 📅 2026-10-28 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSION: TEAS combined with acupuncture may be associated with the most favorable SUCRA ranking, but this should be interpreted as exploratory for improving EIF in athletes. However, the overall quality of current evidence is low, and SUCRA rankings should be interpreted as exploratory. Future hi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947696/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028173758&v=2.20.1

Across established non-invasive neuromodulation and emerging deep-targeting strategies: a comparative bibliometric analysis of TI, TMS, tDCS, and tFUS/TUS/LIFU

👤 Shuang Wu
📅 2026-10-28 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Non-invasive brain stimulation (NIBS) is expanding from conventional cortical modulation toward approaches capable of engaging deeper brain targets, but the literature structures of major modalities have rarely been compared within a common framework. We conducted a comparative bibliometric mapping ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947882/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028173758&v=2.20.1

mPFC-Dependent Valuation Shapes Information Sharing: Causal Evidence from tDCS and a Dyadic Social Media Paradigm

👤 Jinjin Wang
📅 2026-10-28 📖 1 min 📝 72 words

TDCS TACS TRNS

Summary: Information sharing drives the spread of both accurate and misleading content, yet its psychological and neural mechanisms remain unclear. We investigated whether perceived self- and social relevance influence sharing through downstream content evaluations, and whether this process depends on medial...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949098/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028173758&v=2.20.1

Neural-Discursive Dynamics in Collaborative Spatial Problem Solving: Evidence from fNIRS Hyperscanning and Accountable Talk

👤 Xingguang Guo
📅 2026-10-27 📖 1 min 📝 65 words FNIRS

Summary: High-quality collaboration cannot be inferred simply from how much students talk or how strongly their brain activity is synchronized during joint work. This study examined relations among accountable talk, inter-brain synchrony (IBS), and group task performance in fifth-grade students' collaborativ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42938505/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028173755&v=2.20.1

Probing Executive Function in the Toddler Data Desert: A Touchscreen fNIRS Study of Working Memory and ADHD-Related Symptoms

👤 Caitie Busch 📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Attention-deficit/hyperactivity disorder (ADHD) is associated with early differences in attentional control and working memory (WM), yet relatively few paradigms are developmentally feasible across the toddler-to-preschool period while also distinguishing spatial from featural information use. Addre...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42938612/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028173755&v=2.20.1

Mapping the Global Landscape and Thematic Evolution of fNIRS-Gait Research: A Systematic Bibliometric and Network Visualization Analysis

👤 Naime Uluğ 📅 2026-10-27 📖 1 min 📝 20 words FNIRS

Summary: CONCLUSIONS: Establishing standardized preprocessing pipelines and short-separation regressions may prove pivotal for transitioning mobile neuroimaging into objective, closed-loop clinical neurorehabilitation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939053/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028173755&v=2.20.1

Self-Selected Versus Assigned Emotion Regulation in Pain Expectancy: Behavioral and fNIRS Evidence

👤 Ping Shi 📅 2026-10-27 📖 1 min 📝 63 words FNIRS

Summary: Background: Expectation is a powerful cognitive modulator of pain, but how different types of emotion regulation approaches interact with expectancy to influence pain processing remains unclear. The present study compared self-selected habitual strategies (cognitive reappraisal or deep breathing) wi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939283/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028173755&v=2.20.1

Comparative Effects of Acute HIIE and MICE on Executive Function, Prefrontal Oxyhaemoglobin Responses, and Related Biomarkers in Sedentary Young Adults: A Randomized Crossover Trial

👤 Xinyue Yang
📅 2026-10-27 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSIONS: Both exercise sessions were followed by faster reaction times on selected tasks. The small sample, multiple analyses, and potential systemic contributions to fNIRS signals preclude firm conclusions about exercise-mode superiority or biological mediation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939307/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028173755&v=2.20.1

Cerebral Hemodynamic Patterns in People with Subjective Cognitive Decline: A Systematic Review Based on Functional Near-Infrared Spectroscopy

👤 Israel Contador
📅 2026-10-27 📖 1 min 📝 62 words FNIRS

Summary: Background/Objectives: Subjective cognitive decline (SCD) has been proposed as an early symptomatic manifestation of dementia/Alzheimer's disease. However, its associated brain patterns are not yet fully characterized. In recent years, functional near-infrared spectroscopy (fNIRS) has emerged as a c...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939380/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028173755&v=2.20.1

Resting-State Functional Brain Connectivity and Neurodevelopmental Profiles in 3-Year-Olds with Congenital Heart Disease

👤 Anne Gallagher
📅 2026-10-27 📖 1 min 📝 67 words FNIRS

Summary: Background/Objectives: Congenital heart disease (CHD) is associated with altered brain development and an increased risk of neurodevelopmental difficulties, particularly in more severe forms of the condition. Although alterations in functional brain connectivity have been reported before and after t...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42940032/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028173755&v=2.20.1

Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan
📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944288/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028173755&v=2.20.1

Multimodal Sensing and Digital Biomarkers in Virtual Reality-Based Cognitive Rehabilitation: Clinical Validation, Adaptive Systems, and Real-World Translation

Raffaele Ferri
2026-10-27 1 min 58 words FNIRS

Summary: Virtual reality (VR) is evolving from a platform for cognitive exercises into an instrumented rehabilitation environment capable of simultaneously delivering treatment and quantifying behavior. This critical narrative review examines how behavioral, ocular, kinematic, physiological, and neurophysiol...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944342/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028173755&v=2.20.1

Unpredictable human screams elicit anticipatory and phasic cortical responses that affect subsequent interpersonal space regulation

👤 Alessandro Grecucci
📅 2026-10-27 📖 1 min 📝 59 words

FNIRS

Summary: Unpredictable threats can sustain vigilance states that bias perception, cognition, and behaviour beyond immediate stimulus processing. Human screams, as evolutionarily salient alarm signals, are especially effective in eliciting defensive responses, yet their cortical dynamics and consequences for ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42946138/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028173755&v=2.20.1

The application of brain-computer interface combined with neuromodulation for stroke upper-limb rehabilitation: a systematic review and network meta-analysis

Zhongjian Luo
2026-10-28 1 min 42 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Some non-invasive MI-BCI combined interventions like neuromodulation, ROB, or FES might improve upper-limb function and daily activities in stroke patients. However, rankings don't imply clinical importance. Due to small samples, limited evidence, and differences, these findings need con...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947035/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028173752&v=2.20.1

Applications of functional ultrasound brain-computer interfaces in neurological diseases: research progress based on neurovascular coupling

Chengshan Zhang
2026-10-28 1 min 55 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) bypass peripheral nerves and muscles to establish direct information pathways between the brain and external devices, providing novel approaches for motor function reconstruction, communication assistance, and neurorehabilitation in patients with neurological disorde...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947961/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028173752&v=2.20.1

Terahertz Photons Ameliorate Traumatic Brain Injury and Reduce Neuroinflammation via the Cyclic Adenosine Monophosphate Signaling Pathway

👤 Chao Chang
📅 2026-10-28 📖 1 min 📝 73 words

BRAIN COMPUTER INTERFACE

Summary: Traumatic brain injury (TBI) has the highest incidence among common neurological disorders and remains the leading cause of mortality and disability among all kinds of injuries, yet specific therapies for TBI are still unavailable. In the present study, we found that short-term administration of ter...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948201/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028173752&v=2.20.1

Expert consensus on neuro-musculoskeletal functional reconstruction for post-stroke spastic hemiplegia

👤 “Zunyi Brain-Computer Interface Medical Center” Working Group

📅 2026-10-28 📖 1 min

📝 59 words

BRAIN COMPUTER INTERFACE

Summary: Persistent motor dysfunction in post-stroke spastic hemiplegia may result from impaired voluntary motor control, abnormal muscle overactivity, fixed structural restriction, muscle force imbalance, or their combination. Advances in brain-computer interfaces (BCIs), peripheral nerve surgery, and tendo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948598/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028173752&v=2.20.1

Conducting polymers for implantable neural electrodes in brain-computer interfaces

Yang Wang 2026-10-28 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: As core elements of brain-computer interfaces, neural electrodes play a vital role in neural signal recording and stimulation. Conventional metallic electrodes, however, suffer from several inherent drawbacks, including mechanical mismatch with soft neural tissues, high interfacial impedance, and ch...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948971/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028173752&v=2.20.1

Recognizing Brain Pioneering as Hard Work: Strategies for Making Research Participation More Meaningful

Eran Klein 2026-10-28 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) research offers novel opportunities for new tools for mobility and communication for those limited by injury or illness. Despite this promise, current BCI research remains largely early stage and without direct therapeutic benefits for the trial participants with motor...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949596/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028173752&v=2.20.1

Two Kinds of Mental Privacy: What Speech Decoding Does and Does Not Threaten

Marcello Ienca 2026-10-28 1 min 66 words BRAIN COMPUTER INTERFACE

Summary: In the article "Speech Restoration and Mental Surveillance as Conflicting Ethical Concerns in Neural Speech-Decoding Research," Narayan Sankaran and Winston Chiong distinguish speech-decoding strategies by whether they read brain signals tied to communicative intent, softening the common equation of...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949611/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028173752&v=2.20.1

Speech Restoration & Mental Surveillance as Conflicting Ethical Concerns in Neural Speech-Decoding Research

Winston Chiong 2026-10-28 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: The research program of speech decoding-"reading" words directly from brain activity-elicits an unusually wide range of public responses, reflecting a tension between the ethical goal of restoring speech for people with serious communication disabilities and the ethical goal of protecting mental pri...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949614/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028173752&v=2.20.1

Promises but Also Perils: Why We Should Not Dismiss Mental Surveillance Risks in Neurotechnology Research

Andrea Lavazza 2026-10-28 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: Speech brain-computer interfaces may restore communication to people who have lost the ability to speak. In the article "Speech Restoration and Mental Surveillance as Conflicting Ethical Concerns in Neural Speech-Decoding Research," Narayan Sankaran and Winston Chiong distinguish systems that decode...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949615/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028173752&v=2.20.1

Susceptibility to task-irrelevant auditory distractors in relation to visual working memory in children with and without attention-deficit/hyperactivity disorder

Yan Song 2026-10-28 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Attention-deficit/hyperactivity disorder (ADHD) is a prevalent neurodevelopmental disorder associated with increased distractibility. Previous research has reported that susceptibility to task-irrelevant visual distraction is linked with visual working memory (WM) capacity in adults. Here, we examin...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949653/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028173752&v=2.20.1

The sound of manufactured music: Reviewing the role of artificial stimuli in music cognition research.

📅 2024-01-22 📖 1 min 📝 259 words PSYCHOMUSICOLOGY

Summary: Having participants listen and react to musical stimuli is one of music cognition's foundational methods. Whereas most researchers have used stimuli adapted from existing musical traditions in such work, others have incorporated artificial stimuli (i.e., stimuli generated specifically for research t...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000304

Music-evoked nostalgia and charitable giving: A cross-cultural study in the United States and Mexico.

📅 2024-01-22 📖 1 min 📝 192 words PSYCHOMUSICOLOGY

Summary: Nostalgia, a past-oriented emotion characterized by complex affective responses, is a pervasive and fundamental human experience. Prior research has demonstrated that nostalgia serves various socioemotional functions, such as promoting a sense of belonging, enhancing one's perception of meaning in l...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000302

Preferred music listening does not affect cognitive inhibition in young and older adults.

📅 2023-10-12 📖 1 min 📝 227 words PSYCHOMUSICOLOGY

Summary: Previous literature has found links between music listening and cognitive performance. Specifically, background music may play a role in modulating cognitive inhibition. However, determining what type of background music affects cognitive inhibition throughout the lifespan has not been studied. The ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000300

Absolute pitch: A literature review of underlying factors, with special regard to music pedagogy.

📅 2023-07-10 📖 1 min 📝 202 words PSYCHOMUSICOLOGY

Summary: Absolute pitch (AP) is a fairly rare and special phenomenon that has relevance for musicology, psychology, genetics, and neuroscience. AP possessors are able to identify the pitch of an isolated sound or to produce that sound without a reference point. The authors' aim is to review the literature on...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000298

Capturing coordination and intentionality in joint musical improvisation.

📅 2023-08-03 📖 1 min 📝 217 words PSYCHOMUSICOLOGY

Summary: Humans collaborate with each other on a wide variety of tasks that are often largely improvised and unscripted. In this study, we investigated the dynamics of coordination in a joint musical improvisation task, what the effect of intentions is on coordination, and how musicians propagate these inten...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000299

Continuous affect responses to a large diverse set of unfamiliar music: Bayesian time-series and cluster analyses.

📅 2023-04-20 📖 1 min 📝 252 words PSYCHOMUSICOLOGY

Summary: Sixty-nine participants made continuous response judgments of perceived arousal and valence while listening to 30-s extracts of 100 unfamiliar pieces within a novel recommender system. Our purpose was to take advantage of the relatively large number of participants and pieces studied (compared with ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000295

Monthly Updates [July]

📅 2026-07-01 📖 3 min 📝 640 words FMHY

> **Summary:** <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   Added FMHY Plugin to <a href="https://www.flowlau...

🔗 **Read full article:**
https://fmhy.net/posts/july-2026

Humble x Pragmatic Programmers = elite Python ebooks!

👤 /u/qokyoshi 📅 2026-10-28 📖 1 min 📝 23 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> https://www.humblebundle.com/books/python-mastery-pragmatic-programmers-books <!-- SC_ON -->   submitted by   <a href="https://www.reddit.com/user/qokyoshi"...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xi0rmk/humble_x_pragmatic_programmers_elite_python_ebooks/

Show HN: SlideBySide – Present slides without sharing your screen

👤 denismenace 📅 2026-10-28 📖 1 min 📝 69 words HACKER NEWS

> **Summary:** Why is it useful? - Present slides without sharing the screen and have every one be on the same page. - Present together with other cohosts and all control the current page. - Other features such as presentation collab features. Why should I not just vibe code my own solution? - Its a hassle to...

🔗 **Read full article:**
https://slidebyside.com/

MacBook Pro with PROLED XDR Display and Touch

👤 etothet 📅 2026-10-28 📖 1 min 📝 14 words HACKER NEWS

> **Summary:** https://www.apple.com/newsroom/2026/10/apple-unveils-groundb... <hr /> Comments URL: <a href="https://news.ycombinator.com/item?id=50276940">https://news.yc...

🔗 **Read full article:**
https://www.apple.com/macbook-pro/proled-xdr-display/

Why is this fingerprinting script creating bioweapons?

👤 hazebooth 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://nullpt.rs/fingerprinting-bioweapons Comments URL: https://news.ycombinator.com/item?id=50276998 Points: 453 # Comments: 98

🔗 **Read full article:**
https://nullpt.rs/fingerprinting-bioweapons

Spiders Spin 'Silk Dope' into Nature's Toughest Material

👤 ibobev 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.quantamagazine.org/how-spiders-spin-silk-dope-into-natures-toughest-material-20261028/ Comments URL: <a href="https://news.ycombinator.com/item?id=50277...

🔗 **Read full article:**
https://www.quantamagazine.org/how-spiders-spin-silk-dope-into-natures-toughest-material-20261028/

Google AI gets cagey when asked who won the 2020 elections

👤 jacquesm 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://arstechnica.com/ai/2026/10/google-ai-gets-cagey-when-asked-who-won-the-2020-elections/ Comments URL: <a href="https://news.ycombinator.com/item?id=50280178">htt...

🔗 **Read full article:**
https://arstechnica.com/ai/2026/10/google-ai-gets-cagey-when-asked-who-won-the-2020-elections/

Leak website releases data on DHS employees

👤 wondersky 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.reuters.com/legal/government/leak-website-releases-data-tens-thousands-dhs-employees-2026-10-28/ Comments URL: <a href="https://news.ycombinat...

🔗 **Read full article:**
https://www.reuters.com/legal/government/leak-website-releases-data-tens-thousands-dhs-employees-2026-10-28/

The case of the 9 bit AES key

👤 ledoge 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://v12.sh/blog/the-case-of-the-9bit-aes-key Comments URL: https://news.ycombinator.com/item?id=50280722 Points: 4 # Comments: 1

🔗 **Read full article:**
https://v12.sh/blog/the-case-of-the-9bit-aes-key

Nango (YC W23) Is Hiring Across Eng, FDE, Marketing, DevRel

👤 bastienbeurier 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://nango.dev/careers Comments URL: https://news.ycombinator.com/item?id=50280824 Points: 0 # Comments: 0

🔗 **Read full article:**
https://nango.dev/careers

Leak Exposes 41,000 Homeland Security Workers–Including Undercover ICE Agents

👤 Aboutplants 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.wired.com/story/massive-leak-exposes-41000-homeland-security-workers-including-undercover-ice-agents/ Comments URL: <a href="https://news...

🔗 **Read full article:**
https://www.wired.com/story/massive-leak-exposes-41000-homeland-security-workers-including-undercover-ice-agents/

Formlabs Studio 1 $699 Resin Printer

👤 mhb 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://formlabs.com/products/form-studio-1-basic-founders-edition/ Comments URL: https://news.ycombinator.com/item?id=50281044 ...

🔗 **Read full article:**
https://formlabs.com/products/form-studio-1-basic-founders-edition/

There are 13 ways to analyse a replication study, but only one of them is coherent.

noreply@blogger.com (Daniel Lakens)
2026-05-03 6 min 1361 words

TWENTY PERCENT STATISTICIAN

Summary: Paul Simon’s says there are 50 ways to leave your lover, and a similar abundance characterizes the contemporary literature on how to analyse a replication study. The recent SCORE replication project in Nature makes this explicit by reporting “13 replication success metrics along with the number o...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/there-are-13-ways-to-analyse.html

Which Data Repository Should You Use?

noreply@blogger.com (Daniel Lakens)
2026-08-15 17 min 3434 words

TWENTY PERCENT STATISTICIAN

Summary: Update 28-10-2026: ResearchBox now has an API. This new information has been incorporated in the text below. The Center for Open Science has announced that from November 16, 2026, no new projects can be created on the Open Sci...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/08/which-data-repository-should-you-use.html

Neuroethical Perspectives on Neurotechnology in Latin America

👤 Adriel Carridice 📅 2026-09-23 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/publications/research-and-white-papers/neuroethical-perspectives-on-neurotechnology-in-latin-america/

T-MRB Call for Nominations for Associate Editors

👤 Deidre Artis 📅 2026-04-22 📖 1 min 📝 16 words EMBS

Summary: The post T-MRB Call for Nominations for Associate Editors appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/t-mrb-call-for-nominations-ae/#new_tab

2026 Adcom Petition is Open

Deidre Artis | 2026-09-11 | 1 min | 14 words | EMBS

Summary: The post 2026 Adcom Petition is Open appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/about/embs-officers/2026-embs-administrative-committee-adcom-election-2/#new_tab

Analysis of the dynamic evolution and immunosuppressive functions of myeloid cells in gliomas based on single-cell transcriptomics

1 min | 23 words | BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Hao Yuan, Jingyi Yang, Zili Bai, Hongyan Wang, Jiaxuan Tong, Peng Chen, Shouping Gong

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S000689932600466X?dgcid=rss_sd_all

Altered language network microstructure links with impaired language function and development in very preterm/very low birth weight born adults

📖 1 min 📝 31 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Lena Gambarte, Jil Wendt, Rebecca Hippen, Marcel Daamen, Henning Boecker, Peter Bartmann, Dieter Wolke, Benita Schmitz-Koep, Dennis M. Hedderich, Christian Sorg, Aurore Menegaux

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S105381192600621X?dgcid=rss_sd_all

Common phenomenal and neural substrate geometry in visual motion perception

📖 1 min 📝 18 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Kallum Robinson, Ariel Zeleznikow-Johnston, Jiahao Wu, Yumiko Yoshimura, Naotsugu Tsuchiya

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006282?dgcid=rss_sd_all

Flexible compression of sensory evidence makes human decisions more resilient to noisy neural computations

Drevet, J., Benjamin, L., Rondot, L., Drugowitsch, J., Wyart, V.

2026-10-28 1 min

150 words

BIORXIV NEUROSCIENCE

Summary: Categorizing ambiguous stimuli requires accumulating sensory evidence over time through noisy neural computations. Theories disagree on whether this evidence is accumulated in its native sensory space or in an abstract and compressed category space that retains only decision-relevant information. He...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.762195v1?rss=1

Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Rajeswaran, P., Scholl, L. R., Smith, L. I., Orsborn, A. L.

2026-10-28 1 min 241 words

BIORXIV NEUROSCIENCE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.762093v1?rss=1

Multi-scale spatial transcriptomic atlas of the whole mouse spinal cord

Zhang, G.-W., Li, J., Huang, J. J., Tao, H. W., Zhang, K. K., Zhang, L. I.

2026-10-28 1 min

181 words

BIORXIV NEUROSCIENCE

Summary: The spinal cord coordinates sensory, motor and autonomic functions through diverse neuronal and non-neuronal populations, yet how molecularly defined cell types are organized across its full extent remains incompletely understood. Here we use Stereo-seq to generate a genome-wide, single-cell-resolut...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.24.761397v1?rss=1

The Aging Brain Matrix: Cross-Species Evidence Linking FREM3 to Brain and Cognitive Aging

Clifford, K. P., Prevot, T. D., Sumitomo, A., Bansal, Y., Das, S., Marcotte, M., Zhou, R., Yeung, J., Ding, M., Felsky, D., Nieman, B., Banasr, M., Sibille, E., Tomoda, T., Nikolova, Y.

2026-10-28 1 min 151 words

BIORXIV NEUROSCIENCE

Summary: The human extracellular-matrix-related FREM3 gene is strongly age-dependent, enriched in age-sensitive excitatory long-range-projecting neurons, and implicated in neurodegenerative diseases. To establish the first translationally relevant causal evidence for the role of FREM3 in brain structure and ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.762265v1?rss=1

Charting vulnerability of brainstem functional pathways in aging and prodromal synucleinopathy

Hua, L., Singh, K., Garcia-Gomar, M. G., Hannanu, F. F., Cauzzo, S., Koley, S., Stefani, A., Grimaldi, S., Videnovic, A., Bianciardi, M.

2026-10-28 1 min 150 words BIORXIV NEUROSCIENCE

Summary: The brainstem is vulnerable to aging and prodromal synucleinopathy, yet whether these processes disrupt shared or distinct functional pathways linking the brainstem to the cortex remains unknown. Using high-resolution 7-Tesla fMRI, we mapped connectivity between 58 brainstem nuclei and cortical regi...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.27.760186v1?rss=1

Neuronal Hyperactivity Induces Tau Phosphorylation Selectively in Excitatory Pyramidal Neurons Which Contributes to GluA2 Regulation

👤 Sun, M. N., Vaughey, K., Liu, B., Guo, H., Huganir, R. L., Morris, M.

📅 2026-10-28 📖 1 min

📝 234 words

BIORXIV NEUROSCIENCE

Summary: While tau plays a role in regulating neuronal network excitability, a process thought to be dysregulated in Alzheimers disease (AD), the molecular mechanisms linking tau and hyperexcitability remain unclear. To investigate interactions between endogenous tau and neuronal activity, we increased neuro...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.762028v1?rss=1

Multi-region two-photon calcium imaging in freely moving animals

Zong, W., Ma, S., Arbaciauskaite, J., Saxena, R., Camara, A. L., Jenssen, K. J., Manfred, G., Kjellevold-Skallerud, E.

2026-10-28 1 min 146 words BIORXIV NEUROSCIENCE

Summary: Complex behaviors emerge from the coordinated activity of neural circuits distributed across widely separated brain regions. Existing optical approaches for monitoring neural activity lack the combined capability to record from anatomically distributed regions with high spatiotemporal resolution in ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.760324v1?rss=1

Causal reasoning is domain-specific in the human brain

Pramod, R., Chomik-Morales, J., Schulz, L., Kanwisher, N. G.

2026-10-28 1 min 205 words BIORXIV NEUROSCIENCE

Summary: Causal reasoning enables humans to explain the past, predict the future, and intervene on the present. Despite extensive interest in causal inference by philosophers, cognitive scientists, and developmental psychologists, few studies have investigated causal reasoning in the human brain. Here we use...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.762213v1?rss=1

Human-calibrated automated intelligibility evaluation of speech neuroprostheses

👤 Srinivasan, A., Wairagkar, M., Wakeb, R., Farias, D., Miller, L. M., Brandman, D. M., Stavisky, S. D.

📅 2026-10-28 📖 1 min 📝 219 words BIORXIV NEUROSCIENCE

Summary: Objective: Brain-to-voice neuroprostheses can synthesize speech from the neural activity of people who have lost the ability to speak, enabling more natural and expressive communication. Assessing the intelligibility of synthesized speech, however, remains challenging because there is no ground-trut...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.762196v1?rss=1

Loss of SUR1 and Kir6.2 ATP-sensitive potassium channel subunits produces comparable neuronal excitability and morphine antinociception

Song, S. C., Beck, B., Johnson, K., Mujteba, S., Coetzee, W. A., Klein, A. H.

2026-10-28 1 min

246 words

BIORXIV NEUROSCIENCE

Summary: ATP-sensitive potassium (KATP) channels are integral membrane proteins that play a crucial role in linking cellular metabolism to neuronal excitability. KATP channels in peripheral nociception and pain emphasize metabolic gating as a determinant of dorsal root ganglia and spinal cord excitability, p...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.27.761978v1?rss=1

Nigrosome-1 neuromelanin and iron coupling in idiopathic and GBA-associated Parkinson disease

2026-10-28 1 min 0 words

NATURE NEUROSCIENCE SUBJECTS

Read full article:
https://www.nature.com/articles/s41531-026-01599-y

The brain liquidates billions of mutant neurons around the time we’re born

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/d41586-026-03444-x

Alzheimer’s disease genetics converges on microglial and neuronal dysregulation

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41586-026-11206-y

A multivariate approach to understanding cognitive functioning in high-performance athletes

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-75737-0

Presynaptic homeostasis mediated by exocytosis of a SNARE protein

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41586-026-11201-3

Emotion regulation capacity lacks a shared neural signature

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41586-026-11203-1

Connectome simulations identify a central pattern generator for fly walking

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41586-026-11195-y

Microglial NUS1 deficiency drives dopaminergic neurodegeneration through intercellular iron dysregulation in Parkinson’s disease models

Haigang Ren 2026-10-26 1 min 39 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 26 October 2026; doi:10.1038/s41593-026-02499-3 Using cellular and mouse models, this study reports that deficiency of the Parkinson’s disease risk gene NUS1 in microglia disrupts intercellu...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02499-3

Chronic progressive demyelination induces a microglial state with lysosomal dysfunction

Mikael Simons 2026-10-26 1 min 39 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 26 October 2026; doi:10.1038/s41593-026-02489-5 Acute myelin injury responses are well defined, but microglia adaptations to chronic damage remain unclear. Mouse models reveal a chronic demy...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02489-5

Calcium permeability–enhancing mutations in GluRIIC suppress sustained presynaptic homeostatic plasticity triggered by loss of GluRIIA

Olfat A. MalakLindsay GrayAmmar AlyRobin W. BallJessica Gustin-MurrayMyriam MoujahidineGuadalupe CarrascoAndrew RoskoEhud Y. IsacoffA. Pejmun Haghighiahttps://ror.org/050sv4x28Buck Institute for Research on Aging, Novato, CA 94945bhttps://ror.org/04b6nzv94Brigham and Women's Hospital, Harvard Medical School, Boston, MA 02115cInscopix Inc., Mountain View, CA 94043dhttps://ror.org/03taz7m60Leonard Davis School of Gerontology, University of Southern California, Los Angeles, CA 90089ehttps://ror.org/01pxwe438Department of Physiology, McGill University, Montreal, QC H3G1Y6, Canadafhttps://ror.org/01an7q238Helen Wills Neuroscience Institute, University of California, Berkeley, CA 94720ghttps://ror.org/01an7q238Department of Neuroscience, University of California, Berkeley, CA 94720hhttps://ror.org/01an7q238Department of Molecular and Cell Biology, University of California, Berkeley, CA 94720

📅 2026-10-27 📖 1 min 📝 43 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceAccumulating experimental evidence indicates that neurons adjust their synaptic strength, in response to genetic or environmental/ pharmacological perturbations, to ensure stability in neural circu...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2625954123?af=R

VAP organizes presynaptic Ca2+ signaling and synaptic vesicle mobilization through Kv2 membrane contact sites

👤 Cameron D. PatonIn Ha ChoMichael B. Hoppaahttps://ror.org/049s0rh22Division of Arts and Sciences, Department of Biological Sciences, Dartmouth College, Hanover, NH 03755

📅 2026-10-27 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceThe endoplasmic reticulum (ER) supports neuronal function by regulating Ca2+, lipid, and protein homeostasis, yet how ER–plasma membrane contact sites contribute to synaptic transmission remains p...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2602106123?af=R

Intraventricular clindamycin enhances interstitial brain fluid drainage and the blood–brain barrier after a traumatic brain injury

👤 Landon S. AshbyMegan G. CroweDaveraj ChakravartySanjeev MedamGabriel FramptonAyesha B. DasDavid A. HormuthThomas E. YankeelovVincent Y. C. WangLinda J. Noble-HaeussleinBenjamin J. Umlaufahttps://ror.org/00hj54h04Department of Neurosurgery, The University of Texas at Austin, Austin, TX 78712bhttps://ror.org/00hj54h04Department of Biomedical Engineering, The University of Texas at Austin, Austin, TX 78712chttps://ror.org/00hj54h04Oden Institute for Computational Engineering and Sciences, The University of Texas at Austin, Austin, TX 78712dhttps://ror.org/00hj54h04Cancer Research Center, The University of Texas at Austin, Austin, TX 78712ehttps://ror.org/00hj54h04Department of Diagnostic Medicine, The University of Texas at Austin, Austin, TX 78712fDepartment of Imaging Physics, The University of Texas M.D. Anderson Cancer Center, Houston, TX 77030ghttps://ror.org/00hj54h04Department of Neurology, The University of Texas at Austin, Austin, TX 78712hhttps://ror.org/00hj54h04Department of Psychology, The University of Texas at Austin, Austin, TX 78712ihttps://ror.org/00hj54h04Mulva Clinic for the Neurosciences, The University of Texas at Austin, Austin, TX 78712

📅 2026-10-27 📖 1 min 📝 44 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDespite advances in the acute clinical management of severe traumatic brain injuries, treating injury-derived secondary pathogenesis remains a major challenge. We describe an alternative strategy ...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2536755123?af=R

CLSP-REQA: a real-time quality-aware closed-loop seizure prediction framework with Mamba-BiLSTM and confidence-gated intervention

Quansheng Ren 2026-09-30 1 min 304 words FRONTIERS NEUROINFORMATICS

Summary: IntroductionReliable seizure prediction is a prerequisite for closed-loop neurostimulation therapy, yet existing methods rarely account for the variability in EEG signal quality encountered in real-world deployment, and the overwhelming majority adopt non-strict evaluation protocols that overestimat...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1888743

Digital health meets Parkinson's disease: a bibliometric exploration of wearable sensors research and emerging frontiers (2005–2026)

Lan Xiong 2026-10-08 1 min 228 words FRONTIERS NEUROINFORMATICS

Summary: BackgroundWearable and sensor-based monitoring technologies are increasingly used to support objective, longitudinal assessment in Parkinson's disease (PD). However, the global development, knowledge structure, and emerging research frontiers of this interdisciplinary field remain incompletely mappe...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1937585

HQML-ICU: hybrid quantum-classical machine learning for ICU mortality prediction using MIMIC-IV electronic health records

👤 Mohammed Farsi 📅 2026-10-09 📖 1 min 📝 277 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Accurate prediction of mortality in the intensive care unit (ICU) is crucial for informing clinical decision-making and optimizing resource allocation. Although machine learning methodologies have enhanced traditional severity-of-illness scoring systems, they encounter intrinsic limitations in model...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1900553

Adaptive neuro-fuzzy risk index for safe and interpretable lane-change decisions in Level-3 autonomous driving

👤 Jianwei Li 📅 2026-10-09 📖 1 min 📝 250 words

FRONTIERS NEUROROBOTICS

Summary: IntroductionHighway lane-change maneuvers pose a safety challenge for SAE Level 3 autonomous driving because conventional threshold-based decisions and isolated surrogate safety measures may inadequately capture the coupling between longitudinal braking demand and evolving inter-vehicle gaps. This s...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1918469

Clinical safety assessment of implanted brain-computer interfaces: current evidence and a pathway to standardization

Wen-Jun Tu 2026-10-28 1 min 212 words FRONTIERS NEUROSCIENCE

Summary: Implanted brain-computer interfaces are beginning to move beyond short laboratory sessions into long-term and home use. The evidence used to judge clinical safety remains fragmented. This review examines penetrating intracortical, cortical-surface (including subpial) and cerebral-venous systems, wit...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1994636

MTDANet: lightweight multi-scale temporal hierarchy and distance-regularized linear attention network for motor imagery brain–computer interfaces

Binbin Chen, Chen Deng, Huanmin Ge, Xinhua Su and Yanfei Shen 2026-10-26 1 min 239 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Motor-imagery (MI) electroencephalography (EEG) offers a non-invasive route to brain–computer interaction, yet reliable single-trial decoding remains difficult because task-related patterns are weak, nonstationary, and distributed over multiple temporal scales. This study investigates whe...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb330

AOR: adaptive orthogonal regression for depression feature selection with incomplete EEG channels

👤 Wenjia Dong, Zhijian Gong, Xueyuan Xu, Ke Liu, Fulin Wei and Li Zhuo

📅 2026-10-26 📖 1 min

📝 272 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Depression induces severe mental and physical impairments, highlighting the clinical need for rapid and reliable detection. Electroencephalography (EEG) directly reflects neural electrical activity and serves as a promising tool for depression investigation. However, the inherent non-stat...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb32f

Multimodal regulation of neurite extension by extracellular matrix composition and electrical stimulation on aligned collagen fiber scaffolds

👤 Madalynn J Thompson, Joanna Borowska-Fielding, Ivy A Murphy, Ying Zhang, Victor Rafuse and John P Frampton

📅 2026-10-26 📖 1 min 📝 279 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Peripheral nerve regeneration is regulated by structural, biochemical, and electrical cues, yet their combined effects on neurite outgrowth remain unclear. In this study, an aligned fibrous scaffold was developed to evaluate how extracellular matrix (ECM) composition and electrical stimul...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb32c

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt

📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028130507&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39870674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028130507&v=2.20.1

Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028130507&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028130507&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

Robert Oostenveld 2025-08-13 1 min 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028130507&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41358599/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028130507&v=2.20.1

Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41676644/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028130507&v=2.20.1

Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41678848/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028130507&v=2.20.1

The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41703344/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028130507&v=2.20.1

EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42538927/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028130507&v=2.20.1

An open-source platform for measuring optomotor responses in mice

👤 Shogo Soma 📅 2026-10-27 📖 1 min 📝 64 words LOW VISION

Summary: The optomotor response (OMR) is a valuable behavioral index for assessing visual function in basic research; however, commercially available OMR evaluation equipment is expensive and lacks flexibility, as proprietary software and fixed hardware geometry restrict custom stimulus protocols and system ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945590/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028130504&v=2.20.1

Communication patterns in home medication review for individuals with vision impairment: A qualitative study

👤 Woojin Park
📅 2026-10-27 📖 1 min 📝 37 words

LOW VISION

Summary: CONCLUSIONS: In these consultations, professional dialogue supported verification and reassurance, while direct regimen review revealed unverified beliefs and gaps in medication knowledge. The findings suggest a role for accessible counseling that combines information provision with professional ver...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945707/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028130504&v=2.20.1

Global, regional, and national burden of unintentional poisoning, 1990-2023: results from the Global Burden of Disease Study 2023

👤 GBD 2023 Unintentional Poisoning Collaborators

📅 2026-10-27 📖 1 min 📝 65 words

LOW VISION

Summary: BACKGROUND: Unintentional poisoning is a preventable cause of injury-related morbidity and mortality, yet global patterns, trends, and inequalities remain incompletely characterised. Within the Global Burden of Disease (GBD) framework, unintentional poisoning includes poisoning by carbon monoxide an...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945982/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028130504&v=2.20.1

Directing Ru-based acidic oxygen evolution catalysts toward a universal dual-site pathway

Chuangang Hu 2026-10-28 1 min 62 words LOW VISION

Summary: The oxide path mechanism offers vision possibility from kinetics for the anodic oxygen evolution reaction, yet its universal activation remains elusive due to the difficulty of constructing globally appropriate interatomic distances. Here, we address this challenge in the integrated-screened Y(0.1)I...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42946705/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028130504&v=2.20.1

Biomimetic computational modelling of retinal ganglion cell responses to contact lens designs with refractive and opaque non-refractive features

Praveen K Bandela 2026-10-28 1 min 67 words LOW VISION

Summary: CONCLUSIONS: In this computational model, both BF and AR lens designs produced measurable changes in simulated RGC temporal activity relative to SV, with AR inducing greater and more configuration-independent modulation. These findings must be considered hypothesis-generating only; they are consiste...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947083/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028130504&v=2.20.1

Growth, nutritional status, and physical fitness trajectories among primary and secondary school students and their associations with poor vision: a school-based study in Chongqing, China

Yong Zhao 2026-10-28 1 min 42 words LOW VISION

Summary: CONCLUSION: From 2022 to 2026, growth and development indicators among primary and secondary school students in Yuzhong District improved, while poor vision remained prevalent. BMI and VCI trajectories were associated with poor vision, highlighting the importance of integrated monitoring and targete...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947915/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028130504&v=2.20.1

From Awareness to Action : Fall Prevention in Practice

Jola Jimoh 2026-10-28 1 min 73 words LOW VISION

Summary: Falls represent a public health and economic risk for older adults, serving as the leading cause of fatal and nonfatal injuries nationally and within Delaware. This article describes the efforts of a multidisciplinary team of subject matter experts from area health care systems and academic institut...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948503/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028130504&v=2.20.1

Leveraging Artificial Intelligence in Healthcare in Cambodia: Enhancing Access and Patient Outcomes

👤 Virak Sorn 📅 2026-10-28 📖 1 min 📝 58 words LOW VISION

Summary: Artificial intelligence (AI) is increasingly reshaping healthcare globally and may offer opportunities to strengthen health systems in low- and middle-income countries (LMICs), including Cambodia. AI-enabled technologies, including machine learning, deep learning, predictive analytics, computer visi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948795/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028130504&v=2.20.1

Marine Carbonate System Observing and mCDR: New Processes, Needs, and Systems

👤 Zhaohui Aleck Wang 📅 2026-10-28 📖 1 min 📝 70 words LOW VISION

Summary: Marine carbon dioxide removal (mCDR) techniques have the potential to remove legacy emitted carbon dioxide from the atmosphere and store it durably in ocean reservoirs. While the ocean's vast surface area and buffering capacity offer benefits for scaling removal, monitoring remains challenging. This...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949060/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028130504&v=2.20.1

Thirty-year follow-up of retinal vasculitis at a tertiary eye care center in India

👤 Parthopratim Dutta Majumder 📅 2026-10-28 📖 1 min 📝 47 words LOW VISION

Summary: CONCLUSION: Retinal vasculitis has a chronic and unpredictable course, warranting long-term follow-up to detect late bilateral involvement and recurrence. While recurrence becomes less frequent over time, it may still occur after many years. Most patients continue to have idiopathic disease without ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949077/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028130504&v=2.20.1

Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux 📅 2026-07-18 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42470115/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028130502&v=2.20.1

A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42532383/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028130502&v=2.20.1

Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42590984/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028130502&v=2.20.1

Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42596634/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028130502&v=2.20.1

Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42614895/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028130502&v=2.20.1

A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42787568/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028130502&v=2.20.1

Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42788316/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028130502&v=2.20.1

Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42817759/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028130502&v=2.20.1

Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

Zhifei Xie
2026-10-24 1 min 66 words TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42923236/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028130502&v=2.20.1

Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42926384/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028130502&v=2.20.1

Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42711906/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028130459&v=2.20.1

Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42768901/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028130459&v=2.20.1

Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42775591/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028130459&v=2.20.1

Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42789611/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028130459&v=2.20.1

Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42845092/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028130459&v=2.20.1

Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42848198/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028130459&v=2.20.1

Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42881664/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028130459&v=2.20.1

Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885551/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028130459&v=2.20.1

An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885828/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028130459&v=2.20.1

Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42941752/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028130459&v=2.20.1

Non-Invasive Brain Stimulation Techniques and Effects on Neuromodulation and Athletic Performance in Combat Sports: A Systematic Review

👤 Bianca Miarka 📅 2026-10-27 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSIONS: tDCS may influence selected cognitive, neuromuscular, and fatigue-related outcomes in combat sport athletes; however, the direction and magnitude of these effects are inconsistent across studies. Given the substantial methodological heterogeneity and the low-to-very-low certainty of evi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939305/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028130456&v=2.20.1

Non-Invasive Brain Stimulation for Central Neuropathic Pain: An Exploratory Randomized Sham-Controlled Crossover Trial

👤 Eunhee Sohn 📅 2026-10-27 📖 1 min 📝 64 words TDCS TACS TRNS

Summary: Background/Objectives: Central neuropathic pain (CNP) is difficult to manage, and comparative evidence for non-invasive brain stimulation is limited. We explored the short-term efficacy and safety of repetitive transcranial magnetic stimulation (rTMS) and transcranial direct current stimulation (tDC...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939331/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028130456&v=2.20.1

Noninvasive Neuromodulation for Cognitive Dysfunction in Parkinson's Disease: A Scoping Review

Reginald B O'Hara 2026-10-27 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: Evidence specific to PD-MCI is limited and mixed. Several studies pairing stimulation with cognitive training reported favorable outcomes, but this pattern was inconsistent and does not establish superiority over stimulation alone. No VNS trial enrolled participants with confirmed PD-MC...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939339/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028130456&v=2.20.1

Comparative Plastid Phylogenomics of 29 Newly Sequenced Bougainvillea Cultivars Resolves the Phylogenetic Placement of Double-Bracted Cultivars

Huiqi Zhao 2026-10-27 1 min 59 words TDCS TACS TRNS

Summary: Bougainvillea (Nyctaginaceae) cultivars are horticulturally prized worldwide for their vivid bracts; however, repeated interspecific hybridization and somatic mutations have obscured their maternal origins, particularly for double-bracted forms. Here, we sequenced and assembled the complete plastome...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939474/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028130456&v=2.20.1

Anodal Transcranial Direct Current Stimulation Reduces Neuroinflammation and Cholinergic Dysregulation After Young Adult Traumatic Brain Injury

👤 Guven Akcay
📅 2026-10-27 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Traumatic brain injury (TBI) during childhood and adolescence can result in persistent motor and cognitive dysfunction due to secondary injury cascades involving neuroinflammation and disruption of neurotransmitter homeostasis. Although transcranial Direct Current Stimulation (tDCS) is a promising n...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939682/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028130456&v=2.20.1

Assessing brain-behaviour coupling in non-invasive brain stimulation: Insights from reliable change indices and Monte Carlo simulations

👤 Alexander D Tang
📅 2026-10-27 📖 1 min 📝 64 words

TDCS TACS TRNS

Summary: Non-invasive brain stimulation (NiBS) studies frequently report exploratory brain-behaviour correlations following the application of NiBS. However, these analyses are typically underpowered and rely on ratio-based change scores with known statistical limitations. We addressed these limitations by p...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945749/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028130456&v=2.20.1

The application of brain-computer interface combined with neuromodulation for stroke upper-limb rehabilitation: a systematic review and network meta-analysis

Zhongjian Luo
2026-10-28 1 min 42 words TDCS TACS TRNS

Summary: CONCLUSION: Some non-invasive MI-BCI combined interventions like neuromodulation, ROB, or FES might improve upper-limb function and daily activities in stroke patients. However, rankings don't imply clinical importance. Due to small samples, limited evidence, and differences, these findings need con...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947035/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028130456&v=2.20.1

Comparison of the efficacy of neuromodulation techniques based on corticospinal excitability in improving exercise-induced fatigue in athletes: a systematic review and network meta-analysis

👤 Li Qing 📅 2026-10-28 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSION: TEAS combined with acupuncture may be associated with the most favorable SUCRA ranking, but this should be interpreted as exploratory for improving EIF in athletes. However, the overall quality of current evidence is low, and SUCRA rankings should be interpreted as exploratory. Future hi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947696/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028130456&v=2.20.1

Across established non-invasive neuromodulation and emerging deep-targeting strategies: a comparative bibliometric analysis of TI, TMS, tDCS, and tFUS/TUS/LIFU

👤 Shuang Wu
📅 2026-10-28 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Non-invasive brain stimulation (NIBS) is expanding from conventional cortical modulation toward approaches capable of engaging deeper brain targets, but the literature structures of major modalities have rarely been compared within a common framework. We conducted a comparative bibliometric mapping ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947882/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028130456&v=2.20.1

mPFC-Dependent Valuation Shapes Information Sharing: Causal Evidence from tDCS and a Dyadic Social Media Paradigm

👤 Jinjin Wang
📅 2026-10-28 📖 1 min 📝 72 words

TDCS TACS TRNS

Summary: Information sharing drives the spread of both accurate and misleading content, yet its psychological and neural mechanisms remain unclear. We investigated whether perceived self- and social relevance influence sharing through downstream content evaluations, and whether this process depends on medial...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949098/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028130456&v=2.20.1

Neural-Discursive Dynamics in Collaborative Spatial Problem Solving: Evidence from fNIRS Hyperscanning and Accountable Talk

👤 Xingguang Guo
📅 2026-10-27 📖 1 min 📝 65 words FNIRS

Summary: High-quality collaboration cannot be inferred simply from how much students talk or how strongly their brain activity is synchronized during joint work. This study examined relations among accountable talk, inter-brain synchrony (IBS), and group task performance in fifth-grade students' collaborativ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42938505/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028130454&v=2.20.1

Probing Executive Function in the Toddler Data Desert: A Touchscreen fNIRS Study of Working Memory and ADHD-Related Symptoms

👤 Caitie Busch 📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Attention-deficit/hyperactivity disorder (ADHD) is associated with early differences in attentional control and working memory (WM), yet relatively few paradigms are developmentally feasible across the toddler-to-preschool period while also distinguishing spatial from featural information use. Addre...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42938612/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028130454&v=2.20.1

Mapping the Global Landscape and Thematic Evolution of fNIRS-Gait Research: A Systematic Bibliometric and Network Visualization Analysis

👤 Naime Uluğ 📅 2026-10-27 📖 1 min 📝 20 words FNIRS

Summary: CONCLUSIONS: Establishing standardized preprocessing pipelines and short-separation regressions may prove pivotal for transitioning mobile neuroimaging into objective, closed-loop clinical neurorehabilitation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939053/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028130454&v=2.20.1

Self-Selected Versus Assigned Emotion Regulation in Pain Expectancy: Behavioral and fNIRS Evidence

👤 Ping Shi 📅 2026-10-27 📖 1 min 📝 63 words FNIRS

Summary: Background: Expectation is a powerful cognitive modulator of pain, but how different types of emotion regulation approaches interact with expectancy to influence pain processing remains unclear. The present study compared self-selected habitual strategies (cognitive reappraisal or deep breathing) wi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939283/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028130454&v=2.20.1

Comparative Effects of Acute HIIE and MICE on Executive Function, Prefrontal Oxyhaemoglobin Responses, and Related Biomarkers in Sedentary Young Adults: A Randomized Crossover Trial

👤 Xinyue Yang
📅 2026-10-27 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSIONS: Both exercise sessions were followed by faster reaction times on selected tasks. The small sample, multiple analyses, and potential systemic contributions to fNIRS signals preclude firm conclusions about exercise-mode superiority or biological mediation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939307/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028130454&v=2.20.1

Cerebral Hemodynamic Patterns in People with Subjective Cognitive Decline: A Systematic Review Based on Functional Near-Infrared Spectroscopy

👤 Israel Contador
📅 2026-10-27 📖 1 min 📝 62 words FNIRS

Summary: Background/Objectives: Subjective cognitive decline (SCD) has been proposed as an early symptomatic manifestation of dementia/Alzheimer's disease. However, its associated brain patterns are not yet fully characterized. In recent years, functional near-infrared spectroscopy (fNIRS) has emerged as a c...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939380/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028130454&v=2.20.1

Resting-State Functional Brain Connectivity and Neurodevelopmental Profiles in 3-Year-Olds with Congenital Heart Disease

👤 Anne Gallagher
📅 2026-10-27 📖 1 min 📝 67 words FNIRS

Summary: Background/Objectives: Congenital heart disease (CHD) is associated with altered brain development and an increased risk of neurodevelopmental difficulties, particularly in more severe forms of the condition. Although alterations in functional brain connectivity have been reported before and after t...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42940032/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028130454&v=2.20.1

Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan
📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944288/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028130454&v=2.20.1

Multimodal Sensing and Digital Biomarkers in Virtual Reality-Based Cognitive Rehabilitation: Clinical Validation, Adaptive Systems, and Real-World Translation

👤 Raffaele Ferri
📅 2026-10-27 📖 1 min 📝 58 words FNIRS

Summary: Virtual reality (VR) is evolving from a platform for cognitive exercises into an instrumented rehabilitation environment capable of simultaneously delivering treatment and quantifying behavior. This critical narrative review examines how behavioral, ocular, kinematic, physiological, and neurophysiol...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944342/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028130454&v=2.20.1

Unpredictable human screams elicit anticipatory and phasic cortical responses that affect subsequent interpersonal space regulation

👤 Alessandro Grecucci
📅 2026-10-27 📖 1 min 📝 59 words

FNIRS

Summary: Unpredictable threats can sustain vigilance states that bias perception, cognition, and behaviour beyond immediate stimulus processing. Human screams, as evolutionarily salient alarm signals, are especially effective in eliciting defensive responses, yet their cortical dynamics and consequences for ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42946138/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028130454&v=2.20.1

DSGTANet: A semi-supervised dual-stream graph-temporal adaptation network for cross-subject continuous EEG emotion recognition

👤 Wang-Yang Xu
📅 2026-10-27 📖 1 min 📝 61 words

BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: Continuous emotion recognition using electroencephalogram (EEG) is a critical topic in brain-computer inter-faces (BCIs), with applications in mental disorder regulation and human-computer interaction. However, its practical appli-cation is significantly constrained by several key issues,...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945587/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028130451&v=2.20.1

The application of brain-computer interface combined with neuromodulation for stroke upper-limb rehabilitation: a systematic review and network meta-analysis

Zhongjian Luo
2026-10-28 1 min 42 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Some non-invasive MI-BCI combined interventions like neuromodulation, ROB, or FES might improve upper-limb function and daily activities in stroke patients. However, rankings don't imply clinical importance. Due to small samples, limited evidence, and differences, these findings need con...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947035/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028130451&v=2.20.1

Applications of functional ultrasound brain-computer interfaces in neurological diseases: research progress based on neurovascular coupling

Chengshan Zhang
2026-10-28 1 min 55 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) bypass peripheral nerves and muscles to establish direct information pathways between the brain and external devices, providing novel approaches for motor function reconstruction, communication assistance, and neurorehabilitation in patients with neurological disorde...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947961/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028130451&v=2.20.1

Terahertz Photons Ameliorate Traumatic Brain Injury and Reduce Neuroinflammation via the Cyclic Adenosine Monophosphate Signaling Pathway

👤 Chao Chang
📅 2026-10-28 📖 1 min 📝 73 words

BRAIN COMPUTER INTERFACE

Summary: Traumatic brain injury (TBI) has the highest incidence among common neurological disorders and remains the leading cause of mortality and disability among all kinds of injuries, yet specific therapies for TBI are still unavailable. In the present study, we found that short-term administration of ter...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948201/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028130451&v=2.20.1

Expert consensus on neuro-musculoskeletal functional reconstruction for post-stroke spastic hemiplegia

👤 “Zunyi Brain-Computer Interface Medical Center” Working Group

📅 2026-10-28 📖 1 min

📝 59 words

BRAIN COMPUTER INTERFACE

Summary: Persistent motor dysfunction in post-stroke spastic hemiplegia may result from impaired voluntary motor control, abnormal muscle overactivity, fixed structural restriction, muscle force imbalance, or their combination. Advances in brain-computer interfaces (BCIs), peripheral nerve surgery, and tendo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948598/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028130451&v=2.20.1

Conducting polymers for implantable neural electrodes in brain-computer interfaces

Yang Wang 2026-10-28 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: As core elements of brain-computer interfaces, neural electrodes play a vital role in neural signal recording and stimulation. Conventional metallic electrodes, however, suffer from several inherent drawbacks, including mechanical mismatch with soft neural tissues, high interfacial impedance, and ch...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42948971/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028130451&v=2.20.1

Recognizing Brain Pioneering as Hard Work: Strategies for Making Research Participation More Meaningful

Eran Klein 2026-10-28 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) research offers novel opportunities for new tools for mobility and communication for those limited by injury or illness. Despite this promise, current BCI research remains largely early stage and without direct therapeutic benefits for the trial participants with motor...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949596/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028130451&v=2.20.1

Two Kinds of Mental Privacy: What Speech Decoding Does and Does Not Threaten

Marcello Ienca 2026-10-28 1 min 66 words BRAIN COMPUTER INTERFACE

Summary: In the article "Speech Restoration and Mental Surveillance as Conflicting Ethical Concerns in Neural Speech-Decoding Research," Narayan Sankaran and Winston Chiong distinguish speech-decoding strategies by whether they read brain signals tied to communicative intent, softening the common equation of...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949611/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028130451&v=2.20.1

Speech Restoration & Mental Surveillance as Conflicting Ethical Concerns in Neural Speech-Decoding Research

Winston Chiong 2026-10-28 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: The research program of speech decoding-"reading" words directly from brain activity-elicits an unusually wide range of public responses, reflecting a tension between the ethical goal of restoring speech for people with serious communication disabilities and the ethical goal of protecting mental pri...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949614/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028130451&v=2.20.1

Promises but Also Perils: Why We Should Not Dismiss Mental Surveillance Risks in Neurotechnology Research

Andrea Lavazza 2026-10-28 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: Speech brain-computer interfaces may restore communication to people who have lost the ability to speak. In the article "Speech Restoration and Mental Surveillance as Conflicting Ethical Concerns in Neural Speech-Decoding Research," Narayan Sankaran and Winston Chiong distinguish systems that decode...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42949615/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028130451&v=2.20.1

Advancing neuropsychology training: Insights and advice from a qualitative analysis in 102 brain-behavior experts.

2026-07-30 1 min 266 words NEUROPSYCHOLOGY

Summary: Objective: Neuropsychologists possess knowledge that could benefit trainees as they navigate important career decisions within their unique training pathways. This wisdom is typically passed down in an individual mentorship model wherein supervisors and advisors relay advice directly. While this mod...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001116

Sex-specific differences in amygdala resting-state functional connectivity with trait anxiety.

📅 2026-06-04 📖 1 min 📝 267 words NEUROPSYCHOLOGY

Summary: Objective: Anxiety disorders are the most common mental health conditions in the world, disproportionately affecting females and associated with significant morbidity. Neuroimaging can help us to understand the underpinnings of anxiety behaviors. The amygdala and its subnuclei have been strongly imp...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001100

Effects of simulated hearing loss on cognitive task performance: A scoping review.

📅 2026-07-30 📖 1 min 📝 235 words NEUROPSYCHOLOGY

Summary: Objective: Hearing loss (HL) and cognitive decline often co-occur in older adults. Many cognitive tests are administered using spoken instructions or stimuli, potentially biasing test results. The current scoping review aimed to provide an overview of studies using simulated HL in cognitively health...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001111

Normative cognitive test performance and typical error patterns among American Indian older adults: The Strong Heart Study.

📅 2026-06-15 📖 1 min 📝 245 words NEUROPSYCHOLOGY

Summary: Objective: The Strong Heart Study, a large cohort of American Indians from three U.S. regions, collected standardized neuropsychological assessments from participants aged 65–95 years at two visits (2010–2019), with prior documentation of lower overall score distributions and higher prevalence of co...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001093

The unity and diversity of executive functions across adulthood in a diverse Brazilian sample with varying educational attainment.

📅 2026-04-16 📖 1 min 📝 261 words NEUROPSYCHOLOGY

Summary: Objective: Executive functions (EFs) are a set of cognitive abilities associated with many social and clinical outcomes that are believed to be sensitive to culture, age, and educational attainment. Yet, this was determined using EF theoretical models and tests developed for highly educated young ad...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001085

Everyday executive functions in children with neurofibromatosis type 1: Data from nine institutions.

📅 2026-04-16 📖 1 min 📝 278 words NEUROPSYCHOLOGY

Summary: Objective: Executive function (EF) difficulties are prevalent in neurogenetic syndromes such as Neurofibromatosis Type 1 (NF1), the most common autosomal dominant condition associated with nervous system tumors. However, age-related differences in EF problems in children with NF1 remain poorly under...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001087

Integrity in six key brain regions predicts numeric working memory performance in 30,000 middle-aged and older adults: A UK Biobank magnetic resonance imaging study.

📅 2026-06-18 📖 1 min 📝 262 words NEUROPSYCHOLOGY

Summary: Objective: This study investigated the neuroanatomical correlates of working memory using the UK Biobank Numeric Memory Test (NMT; a visual backward digit-span task) in a large sample of middle-aged and older adults. Volumes of brain regions previously implicated in digit-span tasks were hypothesize...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001096

Pushing the boundary on dataclasses

/u/ GianniMariani | 2026-10-28 | 1 min | 75 words | REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Has anyone else wanted more out of dataclass? Like transactions, undo/redo, lambda initializers that take parameters (as other field values) and call them in the correct order and no post_init (does not survive a diamond inheritance)? Transactional field values star...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xhll8t/pushing_the_boundary_on_dataclasses/

Python Typing Survey 2026 Results

/u/ BeamMeUpBiscotti | 2026-10-28 | 1 min | 135 words | REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> The results are out for the 2026 Python Typing Survey (1,794 responses). The full writeup can be read at: <a href="https://engineering.fb.com/2026/10/27...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xhnvs0/python_typing_survey_2026_results/

If you run Python 3.10 on AWS Lambda: it's deprecated this Friday (Oct 31). Gotchas moving to 3.12+

👤 /u/timmywick 📅 2026-10-28 📖 1 min 📝 206 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Heads-up for anyone with Lambda functions still on python3.10. AWS deprecates the runtime on Oct 31, 2026. Functions keep running, but they stop getting patches, the console won't let you edit them, and in 2027 AWS blocks creating new ones (Jul 29) and then updating...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xhoetl/if_you_run_python_310_on_aws_lambda_its/

ICE Is Running One of the Largest Anti-Left Surveillance Ops in US History

👤 zzzeek 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.wired.com/story/ice-is-running-one-of-the-largest-anti-left-surveillance-operations-in-us-history/ Comments URL: <a href="https://news.ycomb...

🔗 **Read full article:**
https://www.wired.com/story/ice-is-running-one-of-the-largest-anti-left-surveillance-operations-in-us-history/

Emacs Turns 50

👤 internet_points 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.technologyreview.com/2026/10/20/1145399/emacs-turns-50/ Comments URL: https://news.ycombinator.com/item?id=50273171 ...

🔗 **Read full article:**
https://www.technologyreview.com/2026/10/20/1145399/emacs-turns-50/

Kentucky man becomes the first humanist granted religious exemption from AI

👤 danso 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://religionnews.com/2026/10/26/in-a-first-a-humanist-granted-a-religious-exemption-from-using-ai/ Comments URL: <a href="https://news.ycombinator.com/item?...

🔗 **Read full article:**
https://religionnews.com/2026/10/26/in-a-first-a-humanist-granted-a-religious-exemption-from-using-ai/

An architectural comparison of Obsidian Canvas and Heptabase

👤 ezrasu 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.fluxlinks.com/blog/obsidian-canvas-vs-heptabase Comments URL: https://news.ycombinator.com/item?id=50274139 Points: 26 <...

🔗 **Read full article:**
https://www.fluxlinks.com/blog/obsidian-canvas-vs-heptabase

CNC Stone Carving

👤 bschne 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://carmelitegothic.com/cnc-stone-carving/ Comments URL: https://news.ycombinator.com/item?id=50274409 Points: 161 # Comments: 44

🔗 **Read full article:**
https://carmelitegothic.com/cnc-stone-carving/

Tesla Semi catches fire on I-880

👤 metadat 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://electrek.co/2026/10/28/tesla-semi-fire-i-880-hayward-no-crash/ Comments URL: https://news.ycombinator.com/item?id=50274706</...

🔗 **Read full article:**
https://electrek.co/2026/10/28/tesla-semi-fire-i-880-hayward-no-crash/

Nvidia Nemotron 3.5 Transcribe: open ASR for 40 languages

👤 wondersky 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://huggingface.co/blog/nvidia/nemotron-transcribe-granary Comments URL: https://news.ycombinator.com/item?id=50275128 Points: 15...

🔗 **Read full article:**
https://huggingface.co/blog/nvidia/nemotron-transcribe-granary

r/Seattle mods discover malware injection by archive.today, ban site

👤 consumer451 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.reddit.com/r/Seattle/comments/1xhf6qp/archivetoday_links_going_forward/ Comments URL: <a href="https://news.ycombinator.com/item?id=50275373">https://news.ycombinator....

🔗 **Read full article:**
https://www.reddit.com/r/Seattle/comments/1xhf6qp/archivetoday_links_going_forward/

26,400 live credentials found scanning 7.25M repos on China's Gitee

👤 kernelrocks 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.mongodb.com/company/blog/technical/scanning-7-point-25-million-public-repositories-chinas-largest-git-platform-for-secrets<...

🔗 **Read full article:**
https://www.mongodb.com/company/blog/technical/scanning-7-point-25-million-public-repositories-chinas-largest-git-platform-for-secrets

Dayjob: Tax – evaluating AI agents on 50 professional tax assignments

👤 landonxi 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://surgehq.ai/blog/dayjob-tax Comments URL: https://news.ycombinator.com/item?id=50275626 Points: 37 # Comments: 22

🔗 **Read full article:**
https://surgehq.ai/blog/dayjob-tax

Are meta-scientists ignoring philosophy of science?

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-07-04 📖 8 min 📝 1681 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Are meta-scientists ignoring philosophy of science (PoS)? Are they re-inventing the wheel? A recent panel at the Metascience conference engaged with this question, and the first sentence of the abstract states “Critics argue t...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/07/are-meta-scientists-ignoring-philosophy.html

Type S and M errors as a “rhetorical tool”

noreply@blogger.com (Daniel Lakens)
2025-09-28 17 min 3572 words

TWENTY PERCENT STATISTICIAN

Summary: Update 30/09/2025: I have added a reply by Andrew Gelman below my original blog post. We recently posted a preprint criticizing the idea of Type S and M errors (https://osf.io/2phzb_v1). From our abstract: “While these concepts have been pr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/09/type-s-and-m-errors-as-rhetorical-tool.html

Why we should stop using statistical techniques that have not been adequately vetted by experts in psychology

noreply@blogger.com (Daniel Lakens)
2025-10-29 27 min 5516 words

TWENTY PERCENT STATISTICIAN

Summary: In a recent post on Bluesky, where Richard Morey reflects on a paper he published with Clintin Davis-Stober that points out concerns with the p-curve method (Morey & Davis-Stober, 2025), he writes: <p cla...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/10/why-we-should-stop-using-statistical.html

ERP Decoding for Everyone: Software and Webinar

👤 Steve Luck 📅 2023-06-23 📖 2 min 📝 420 words ERP BOOT CAMP

Summary: You can access the recording h.
You can access the final PDF of the slides <a href="https://ucdavis.box.com/s/f...

🔗 **Read full article:**
https://erpinfo.org/blog/2023/6/23/decoding-webinar

Important Changes to the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-05 📖 2 min 📝 444 words ERP BOOT CAMP

Summary: We are disappointed to announce that we will not be holding a regular 10-day ERP Boot Camp this summer. We have held Boot Camps nearly every summer since 2007, supported by a series of generous grants from NIMH that allowed us to provide scholarships for all attendees. Unf...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/5/changes-to-the-2024-erp-boot-camp

Standardized Measurement Error now available in BrainVision Analyzer and MNE-Python

👤 Steve Luck 📅 2025-12-19 📖 1 min 📝 253 words ERP BOOT CAMP

Summary: <p class="">A few years ago, we developed a universal metric of data quality for averaged ERPs called the standardized measurement error (click here for more information or watch our <a href="https://www.youtube.com/watch?v=tEKsx0p53rs" t...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/12/19/sme-now-available-in-bva-and-mne-python

Transcranial Focused Ultrasound: A Transformative Tool for Intracranial Ablation, Drug Delivery, and Neuromodulation

📅 2025-11-11 📖 1 min 📝 136 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Transcranial focused ultrasound (tFUS) is an emerging neuromodulation and therapeutic technology offering noninvasive, submillimeter precision for targeting deep brain structures. Unlike transcranial magnetic stimulation (TMS) and transcranial electric stimulation (tES), which are limited by depth-f...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11241049

Network Models of Neurodegeneration: Bridging Neuronal Dynamics and Disease Progression

📅 2026-01-16 📖 1 min 📝 196 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Neurodegenerative diseases are characterized by the accumulation of misfolded proteins and widespread disruptions in brain function. Computational modeling has advanced our understanding of these processes, but efforts have traditionally focused on either neuronal dynamics or the biological processe...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11357276

Optogenetics: Pinpoint Light on Precise Neuromodulation

📅 2025-11-06 📖 1 min 📝 137 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Optogenetics has emerged as a pivotal tool in neuroscience, enabling intricate modulation of targeted neurons within the nervous system. Despite its transformative potential, achieving high spatiotemporal resolution in neuromodulation remains a significant challenge, particularly in free-behaving an...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11230544

Emerging Noninvasive Approaches for the Suppression of Pathological Tremor

📅 2025-12-16 📖 1 min 📝 215 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Pathological tremor affects over 40 million people worldwide, significantly impairing daily activities and quality of life. Pharmacological treatments show limited efficacy, with up to 30% discontinuation rates, while surgical interventions like deep brain stimulation achieve significant tremor redu...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11301615

Assistive Trajectory Planning for Lower Limb Exoskeletons: Strategies From Laboratory-Optimized Gait to Environmentally-Adaptive Locomotion Through Multimodal Parameter Awareness

📅 2025-12-30 📖 1 min 📝 182 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Lower limb assistive exoskeletons (LLEs) show great potential to reduce metabolic cost, improve walking performance, and correct abnormal gait patterns. Among their core control architectures, assistive trajectory planning is key to determining system responsiveness and effectiveness under varying l...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11319164

Join the Mailing List

👤 dziura 📅 2025-12-10 📖 1 min 📝 13 words EMBS

> **Summary:** The post Join the Mailing List appeared first on IEEE EMBS.

🔗 **Read full article:**
https://embc.embs.org/2026/#new_tab

EMBS MentorLink: Fostering Careers Together

👤 Deidre Artis 📅 2026-08-18 📖 1 min 📝 14 words EMBS

> **Summary:** The post EMBS MentorLink: Fostering Careers Together appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/embs-mentorlink-fostering-careers-together/

Alterations in the populations of peripheral immune cells in the blood of a mouse model of progressive myoclonus epilepsy of Lafora type

📖 1 min 📝 17 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): L. Baños-Carrión, A. Campos-Rodriguez, M.A. Garcia-Gimeno, P. Sanz

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004580?dgcid=rss_sd_all

Addressing the between-group comparison problem: Detecting differences between correlation matrix populations due to single-variable perturbations for resting state fMRI

📖 1 min 📝 16 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Itamar Faran, Michael Peer, Shahar Arzy, Yuval Benjamini

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926005549?dgcid=rss_sd_all

A comparison of continuous-wave fNIRS with quantitative fMRI-derived indices of brain function in the visual cortex

📖 1 min 📝 42 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Giulia Rocco, Lucie Chalet, Elizabeth J. Fear, Sara Pomante, Francesca Graziano, Davide Di Censo, Manuela Carriero, Édouard Delaire, Fabrizio Esposito, Mauro Gianni Perrucci, Cosimo Del Gratta, David P...

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006051?dgcid=rss_sd_all

Circadian and sleep-wake modulation of EEG functional connectivity across brain states and Its associations with cognition in humans

📖 1 min 📝 24 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Alpar S Lazar, Zsolt I Lazar, Nayantara Santhi, June C Lo, John A Groeger, Derk-Jan Dijk

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006166?dgcid=rss_sd_all

Cortical pyramidal tract neurons constrain pathological dopaminergic motor responses

Atudorei, M., Olsson, G. E. D., del Agua Villa, C., Corcea, C., Gether, U., Cenci, M. A., Siebner, H., Rickhag, M.

2026-10-28 1 min 148 words BIORXIV NEUROSCIENCE

Summary: Pyramidal tract (PT) neurons constitute a major cortical output pathway to subcortical structures, yet their contribution to parkinsonian and dyskinetic motor states remains unclear. Here, we show that chemogenetic activation of motor cortical PT neurons selectively confines pathological dopaminergi...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.758468v1?rss=1

Activity dependent regulation of Angiomotin modulates dendritic complexity in hippocampal neurons

Krueger, S. R., Fawcett, J. P., Wigerius, M.

2026-10-28 1 min 151 words BIORXIV NEUROSCIENCE

Summary: Neuronal activity alters synaptic protein composition, which is important for maintaining dendritic complexity. Multiple signaling networks have been implicated in these processes, but how increased neuronal activity regulates their expression remains incompletely understood. The long isoform of Ang...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.761988v1?rss=1

Post-intervention high-beta synchronization tracks stable-manifold alignment during stick-balancing learning

👤 Shimokado, H., Hayashi, K., Nomura, T.
📅 2026-10-28 📖 1 min 📝 226 words

BIORXIV NEUROSCIENCE

Summary: Learning to stabilize an unstable object may depend not only on the accuracy of active corrective interventions but also on when to suspend intervention and allow the object's intrinsic dynamics to evolve. We investigated learning-related changes in control dynamics and associated beta-band electroe...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.762123v1?rss=1

Claustrum Morphometry Reveals Spatially Heterogeneous Alterations in 22q11DS

👤 Althauser, A. K., Delavari, F., Latreche, C., Van De Ville, D., Eliez, S.
📅 2026-10-28 📖 1 min 📝 228 words

BIORXIV NEUROSCIENCE

Summary: The human claustrum is a small, irregular, and complexly positioned structure, yet it carries long range projections to most cortical regions responsible for executive functioning. Prior structural MRI research has largely relied on global volume as the metric of its morphology. Such measures may ob...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.762130v1?rss=1

Variability and Landmark Quality in Anatomical Co-Registration in Neuronavigation for Brain Stimulation

Ebrahimi, Z., Farahmandrad, M., Seyedi, B., Xie, Z., Di Martino, M., Peterchev, A. V., Kammer, T., Goetz, S. M.

2026-10-28 1 min 212 words BIORXIV NEUROSCIENCE

Summary: Accurate neuronavigation depends not only on the technical performance of the tracking system but also on the reproducibility with which anatomical landmarks are physically localized. We quantified operator- and landmark-specific variability in repeated facial and cranial landmark localization in 19...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.24.761558v1?rss=1

Birth weight, cognitive reserve and outcomes after first-episode psychosis

2026-10-28 1 min 0 words NATURE NEUROSCIENCE SUBJECTS

Read full article:
https://www.nature.com/articles/s44220-026-00751-0

Cellular-resolution three-photon ultra-deep imaging (TrUDI) of neuronal population activity at 2-mm depth in the mammalian brain

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41592-026-03251-2

ADP-ribose induces neuronal ferroptosis through ENT2-mediated nucleotide rewiring

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41589-026-02355-2

The grand plan to transform psychiatry — by personalizing mental-health care

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/d41586-026-03391-7

Resting-state electroencephalographic characteristics in patients with atopic dermatitis accompanied by chronic pruritus

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-76915-w

Computational modeling and preclinical evaluation support targeting somatic instability for Huntington’s disease treatment

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-78769-2

Editorial: Women in interacting minds and brains II

👤 Anna Ilona Roberts 📅 2026-10-26 📖 1 min 📝 0 words FRONTIERS HUMAN NEUROSCIENCE

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.2001078

Imaging findings and clinical outcomes in neonates with isolated subependymal echogenicity on cranial ultrasound: a retrospective cohort study

Xuehua Zhang 2026-10-28 1 min 255 words FRONTIERS HUMAN NEUROSCIENCE

Summary: ObjectiveTo describe clinical and imaging findings in neonates with isolated subependymal echogenicity on cranial ultrasound and examine associations with recorded mortality and cerebral palsy.MethodsThis retrospective single-center cohort included 317 neonates admitted between January 2015 and Dece...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1945002

Effects of non-invasive brain stimulation combined with exercise training on pain and physical function in knee osteoarthritis: a systematic review and meta-analysis

Zhaoxiang Meng 2026-10-28 1 min 345 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundNon-invasive brain stimulation (NIBS) combined with exercise training (ET) is a promising treatment for knee osteoarthritis (KOA), but its effects on pain and physical function lack systematic evaluation.PurposeTo evaluate the effects of NIBS combined with ET on pain and physical function ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1902298

Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-16 1 min 222 words FRONTIERS NEUROROBOTICS

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1925089

Is the oxytocin system a keystone in human wellbeing?

Hans P. Nazarloo 2026-10-28 1 min 149 words FRONTIERS NEUROSCIENCE

Summary: It has also taken over a century for science to fully realize that oxytocin is far more than a “female reproductive hormone.” Both the measurement and interpretation of the biochemical and functional properties of oxytocin have raised controversy and remain works in progress. However, oxytocin is no...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1989082

Asymmetric encoder-decoder U-net with hybrid attention and residual dilated convolution for brain tumor segmentation

Ganqin Du 2026-10-28 1 min 187 words FRONTIERS NEUROSCIENCE

Summary: InroductionThe accurate and automatic segmentation of brain tumors from magnetic resonance imaging is regarded as essential for clinical diagnosis and surgical planning. Nevertheless, the concurrent extraction of both local fine details and global contextual cues remains challenging, primarily due t...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1999080

Effects of auditory sensory stimulation on consciousness and physiological parameters among individuals with disorders of consciousness: a systematic review and meta-analysis

Takamichi Taniguchi 2026-10-28 1 min 339 words FRONTIERS NEUROSCIENCE

Summary: ObjectiveTo systematically evaluate the effects of auditory sensory stimulation (SS) on consciousness levels and physiological parameters in patients with disorders of consciousness (DOC). Furthermore, this study aimed to explore intervention characteristics based on the frequency, intensity, time, ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1879145

Source-resolved electroencephalogram microstates reveal opposing prefrontal-posterior dynamics and a candidate temporal-to-precuneus pathway in disorders of consciousness

Hua Yuan 2026-10-28 1 min 149 words FRONTIERS NEUROSCIENCE

Summary: IntroductionObjective and circuit-level biomarkers of consciousness are urgently needed for disorders of consciousness.MethodsWe recorded high-density electroencephalography from 33 patients with disorders of consciousness (17 vegetative state/unresponsive wakefulness syndrome, 16 minimally consciou...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1995232

De novo POLR2A p.(L1127F) variant presenting with neurodevelopmental regression and dystonia, with favorable response to low-dose dopamine therapy: a case report and literature review

Xulai Shi 2026-10-28 1 min 247 words FRONTIERS NEUROSCIENCE

Summary: BackgroundNeurodevelopmental disorder with hypotonia and variable intellectual and behavioral abnormalities (NEDHIB) is a rare autosomal-dominant condition caused by pathogenic POLR2A variants. It typically presents in infancy with hypotonia, developmental delay, and multisystem manifestations. Fewe...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1997676

Development and feasibility testing of the taVNS adverse events questionnaire: a multistage methodological study

Karsten Witt 2026-10-28 1 min 301 words FRONTIERS NEUROSCIENCE

Summary: Transcutaneous auricular vagus nerve stimulation (taVNS) is a non-invasive neuromodulatory therapy gaining interest for its neural mechanisms and therapeutic effects. Although taVNS is considered safe, adverse event (AE) reporting remains inconsistent, lacking standardized procedures for assessment ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1974496

Psychosocial distress is a stronger correlate of psychiatric symptom burden than cannabis dependence in young adults: a functional connectome and structural equation modeling study

Jacob Levman 2026-10-28 1 min 343 words FRONTIERS NEUROSCIENCE

Summary: IntroductionCannabis is widely used among young adults and often viewed as a primary driver of adverse psychiatric and neural outcomes, yet the psychosocial and socioeconomic context in which use occurs is rarely modeled jointly with neural and behavioral measures.MethodsUsing the Human Connectome P...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1922823

Assessing consciousness by near-nerve electrostimulation in intensive rehabilitation

Piergiuseppe Liuzzi, Sara Ballanti, Andrea Mannini, Bahia Hakiki, Rachele Burali, Tanita Toci, Maria Chiara Carrozza, Calogero Maria Oddo and Antonello Grippo

2026-10-15 1 min 172 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Consciousness diagnosis in prolonged disorders of consciousness (pDoC) relies on the Coma Recovery Scale–Revised, yet consciousness or its potential for recovery may escape behavioural detection. We tested whether near-nerve stimulation (NNS) of the median nerve, compared to standard surf...

Read full article:
https://iopscience.iop.org/article/10.1088/1741-2552/aea645

Prioritized nonlinear modeling of shared dynamics across neural populations

Trisha Jha, Dong Song and Maryam M Shanechi

2026-10-15 1 min 284 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Advances in neural recording technology enable simultaneous measurements across multiple brain regions, providing new opportunities to study inter-regional interactions. However, several challenges arise when developing nonlinear dynamical models of cross-region interactions. In particula...

Read full article:
https://iopscience.iop.org/article/10.1088/1741-2552/ae9bee

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39870674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 2025-06-26 1 min 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41358599/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41676644/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41678848/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41703344/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42538927/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

Fidelity-oriented pelvic CBCT-to-CT synthesis using an anatomical information-attention diffusion model

Kunyuan Xu 2026-10-27 1 min 63 words LOW VISION

Summary: Cone-beam computed tomography (CBCT) is widely used in image-guided radiotherapy and adaptive radiotherapy, but its direct quantitative use is limited by scatter artifacts, low soft-tissue contrast, and inaccurate Hounsfield unit (HU) values. Diffusion models provide stable image generation capabili...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944823/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028064524&v=2.20.1

Poros: Perception-Enabled Interactive Assistance for Appliance Use by Blind and Low Vision Users

Wei Liang 2026-10-27 1 min 62 words LOW VISION

Summary: Modern home appliances pose significant accessibility barriers for individuals with visual impairments, driven by increasing functional complexity and minimalist design trends. Drawing on formative interviews with blind and low vision (BLV) participants, we propose Poros, a novel system providing pe...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945092/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028064524&v=2.20.1

DELUGE: Decomposed Entropy-coded Live Unstructured Geometry Exchange for Real-time Particle Streaming

Takefumi Hiraki 2026-10-27 1 min 74 words LOW VISION

Summary: Particle-based physics simulations, including fluids, smoke, and granular media, are fundamental to visual realism in immersive VR and AR. With the growing adoption of social VR and digital twins, demand is increasing for shared experiences in which multiple users interact with the same simulation i...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945105/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028064524&v=2.20.1

A multi-attention mechanism-improved ResNet18 approach for lithology identification and classification

Ruifeng Li 2026-10-27 1 min 61 words LOW VISION

Summary: Lithology identification is vital for improving geological exploration accuracy, optimizing petroleum exploration schemes, and ensuring construction engineering safety. Traditional identification methods suffer from low efficiency, high cost, and heavy dependence on domain expertise. This study prop...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945382/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028064524&v=2.20.1

An open-source platform for measuring optomotor responses in mice

👤 Shogo Soma 📅 2026-10-27 📖 1 min 📝 64 words LOW VISION

Summary: The optomotor response (OMR) is a valuable behavioral index for assessing visual function in basic research; however, commercially available OMR evaluation equipment is expensive and lacks flexibility, as proprietary software and fixed hardware geometry restrict custom stimulus protocols and system ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945590/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028064524&v=2.20.1

Communication patterns in home medication review for individuals with vision impairment: A qualitative study

👤 Woojin Park 📅 2026-10-27 📖 1 min 📝 37 words LOW VISION

Summary: CONCLUSIONS: In these consultations, professional dialogue supported verification and reassurance, while direct regimen review revealed unverified beliefs and gaps in medication knowledge. The findings suggest a role for accessible counseling that combines information provision with professional ver...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945707/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028064524&v=2.20.1

Global, regional, and national burden of unintentional poisoning, 1990-2023: results from the Global Burden of Disease Study 2023

👤 GBD 2023 Unintentional Poisoning Collaborators

📅 2026-10-27 📖 1 min 📝 65 words

LOW VISION

Summary: BACKGROUND: Unintentional poisoning is a preventable cause of injury-related morbidity and mortality, yet global patterns, trends, and inequalities remain incompletely characterised. Within the Global Burden of Disease (GBD) framework, unintentional poisoning includes poisoning by carbon monoxide an...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945982/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028064524&v=2.20.1

Directing Ru-based acidic oxygen evolution catalysts toward a universal dual-site pathway

Chuangang Hu 2026-10-28 1 min 62 words LOW VISION

Summary: The oxide path mechanism offers vision possibility from kinetics for the anodic oxygen evolution reaction, yet its universal activation remains elusive due to the difficulty of constructing globally appropriate interatomic distances. Here, we address this challenge in the integrated-screened Y(0.1)I...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42946705/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028064524&v=2.20.1

Biomimetic computational modelling of retinal ganglion cell responses to contact lens designs with refractive and opaque non-refractive features

Praveen K Bandela 2026-10-28 1 min 67 words LOW VISION

Summary: CONCLUSIONS: In this computational model, both BF and AR lens designs produced measurable changes in simulated RGC temporal activity relative to SV, with AR inducing greater and more configuration-independent modulation. These findings must be considered hypothesis-generating only; they are consiste...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947083/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028064524&v=2.20.1

Growth, nutritional status, and physical fitness trajectories among primary and secondary school students and their associations with poor vision: a school-based study in Chongqing, China

👤 Yong Zhao
📅 2026-10-28 📖 1 min 📝 42 words

LOW VISION

Summary: CONCLUSION: From 2022 to 2026, growth and development indicators among primary and secondary school students in Yuzhong District improved, while poor vision remained prevalent. BMI and VCI trajectories were associated with poor vision, highlighting the importance of integrated monitoring and targete...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947915/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028064524&v=2.20.1

Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42470115/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028064522&v=2.20.1

A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42532383/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028064522&v=2.20.1

Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42590984/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028064522&v=2.20.1

Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42596634/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028064522&v=2.20.1

Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42614895/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028064522&v=2.20.1

A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42787568/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028064522&v=2.20.1

Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42788316/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028064522&v=2.20.1

Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42817759/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028064522&v=2.20.1

Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

Zhifei Xie
2026-10-24 1 min 66 words TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42923236/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028064522&v=2.20.1

Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42926384/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028064522&v=2.20.1

Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42711906/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028064519&v=2.20.1

Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42768901/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028064519&v=2.20.1

Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42775591/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028064519&v=2.20.1

Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42789611/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028064519&v=2.20.1

Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42845092/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028064519&v=2.20.1

Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42848198/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028064519&v=2.20.1

Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42881664/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028064519&v=2.20.1

Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885551/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028064519&v=2.20.1

An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885828/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028064519&v=2.20.1

Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42941752/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028064519&v=2.20.1

Combined Transcranial Direct Current Stimulation and Transcutaneous Cervical Vagus Nerve Stimulation for Anxiety and Depression: An Exploratory Pilot Study

👤 Maria-Manuela Apostol
📅 2026-10-27 📖 1 min 📝 60 words

TDCS TACS TRNS

Summary: Anxiety and depressive disorders are among the leading causes of disability worldwide, yet current treatments remain limited by modest efficacy and restricted accessibility; non-invasive neuromodulation techniques such as transcranial direct current stimulation (tDCS) and transcutaneous cervical vag...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939258/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028064517&v=2.20.1

Non-Invasive Brain Stimulation Techniques and Effects on Neuromodulation and Athletic Performance in Combat Sports: A Systematic Review

Bianca Miarka 2026-10-27 1 min 59 words TDCS TACS TRNS

Summary: CONCLUSIONS: tDCS may influence selected cognitive, neuromuscular, and fatigue-related outcomes in combat sport athletes; however, the direction and magnitude of these effects are inconsistent across studies. Given the substantial methodological heterogeneity and the low-to-very-low certainty of evi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939305/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028064517&v=2.20.1

Non-Invasive Brain Stimulation for Central Neuropathic Pain: An Exploratory Randomized Sham-Controlled Crossover Trial

Eunhee Sohn 2026-10-27 1 min 64 words TDCS TACS TRNS

Summary: Background/Objectives: Central neuropathic pain (CNP) is difficult to manage, and comparative evidence for non-invasive brain stimulation is limited. We explored the short-term efficacy and safety of repetitive transcranial magnetic stimulation (rTMS) and transcranial direct current stimulation (tDC...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939331/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028064517&v=2.20.1

Noninvasive Neuromodulation for Cognitive Dysfunction in Parkinson's Disease: A Scoping Review

Reginald B O'Hara 2026-10-27 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: Evidence specific to PD-MCI is limited and mixed. Several studies pairing stimulation with cognitive training reported favorable outcomes, but this pattern was inconsistent and does not establish superiority over stimulation alone. No VNS trial enrolled participants with confirmed PD-MC...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939339/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028064517&v=2.20.1

Comparative Plastid Phylogenomics of 29 Newly Sequenced Bougainvillea Cultivars Resolves the Phylogenetic Placement of Double-Bracted Cultivars

Huiqi Zhao 2026-10-27 1 min 59 words TDCS TACS TRNS

Summary: Bougainvillea (Nyctaginaceae) cultivars are horticulturally prized worldwide for their vivid bracts; however, repeated interspecific hybridization and somatic mutations have obscured their maternal origins, particularly for double-bracted forms. Here, we sequenced and assembled the complete plastome...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939474/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028064517&v=2.20.1

Anodal Transcranial Direct Current Stimulation Reduces Neuroinflammation and Cholinergic Dysregulation After Young Adult Traumatic Brain Injury

👤 Guven Akcay
📅 2026-10-27 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Traumatic brain injury (TBI) during childhood and adolescence can result in persistent motor and cognitive dysfunction due to secondary injury cascades involving neuroinflammation and disruption of neurotransmitter homeostasis. Although transcranial Direct Current Stimulation (tDCS) is a promising n...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939682/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028064517&v=2.20.1

Assessing brain-behaviour coupling in non-invasive brain stimulation: Insights from reliable change indices and Monte Carlo simulations

👤 Alexander D Tang
📅 2026-10-27 📖 1 min 📝 64 words

TDCS TACS TRNS

Summary: Non-invasive brain stimulation (NiBS) studies frequently report exploratory brain-behaviour correlations following the application of NiBS. However, these analyses are typically underpowered and rely on ratio-based change scores with known statistical limitations. We addressed these limitations by p...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945749/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028064517&v=2.20.1

The application of brain-computer interface combined with neuromodulation for stroke upper-limb rehabilitation: a systematic review and network meta-analysis

Zhongjian Luo
2026-10-28 1 min 42 words TDCS TACS TRNS

Summary: CONCLUSION: Some non-invasive MI-BCI combined interventions like neuromodulation, ROB, or FES might improve upper-limb function and daily activities in stroke patients. However, rankings don't imply clinical importance. Due to small samples, limited evidence, and differences, these findings need con...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947035/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028064517&v=2.20.1

Comparison of the efficacy of neuromodulation techniques based on corticospinal excitability in improving exercise-induced fatigue in athletes: a systematic review and network meta-analysis

👤 Li Qing 📅 2026-10-28 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSION: TEAS combined with acupuncture may be associated with the most favorable SUCRA ranking, but this should be interpreted as exploratory for improving EIF in athletes. However, the overall quality of current evidence is low, and SUCRA rankings should be interpreted as exploratory. Future hi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947696/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028064517&v=2.20.1

Across established non-invasive neuromodulation and emerging deep-targeting strategies: a comparative bibliometric analysis of TI, TMS, tDCS, and tFUS/TUS/LIFU

👤 Shuang Wu
📅 2026-10-28 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Non-invasive brain stimulation (NIBS) is expanding from conventional cortical modulation toward approaches capable of engaging deeper brain targets, but the literature structures of major modalities have rarely been compared within a common framework. We conducted a comparative bibliometric mapping ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947882/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028064517&v=2.20.1

Neural-Discursive Dynamics in Collaborative Spatial Problem Solving: Evidence from fNIRS Hyperscanning and Accountable Talk

👤 Xingguang Guo
📅 2026-10-27 📖 1 min 📝 65 words FNIRS

Summary: High-quality collaboration cannot be inferred simply from how much students talk or how strongly their brain activity is synchronized during joint work. This study examined relations among accountable talk, inter-brain synchrony (IBS), and group task performance in fifth-grade students' collaborativ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42938505/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028064515&v=2.20.1

Probing Executive Function in the Toddler Data Desert: A Touchscreen fNIRS Study of Working Memory and ADHD-Related Symptoms

👤 Caitie Busch 📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Attention-deficit/hyperactivity disorder (ADHD) is associated with early differences in attentional control and working memory (WM), yet relatively few paradigms are developmentally feasible across the toddler-to-preschool period while also distinguishing spatial from featural information use. Addre...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42938612/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028064515&v=2.20.1

Mapping the Global Landscape and Thematic Evolution of fNIRS-Gait Research: A Systematic Bibliometric and Network Visualization Analysis

👤 Naime Uluğ 📅 2026-10-27 📖 1 min 📝 20 words FNIRS

Summary: CONCLUSIONS: Establishing standardized preprocessing pipelines and short-separation regressions may prove pivotal for transitioning mobile neuroimaging into objective, closed-loop clinical neurorehabilitation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939053/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028064515&v=2.20.1

Self-Selected Versus Assigned Emotion Regulation in Pain Expectancy: Behavioral and fNIRS Evidence

👤 Ping Shi 📅 2026-10-27 📖 1 min 📝 63 words FNIRS

Summary: Background: Expectation is a powerful cognitive modulator of pain, but how different types of emotion regulation approaches interact with expectancy to influence pain processing remains unclear. The present study compared self-selected habitual strategies (cognitive reappraisal or deep breathing) wi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939283/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028064515&v=2.20.1

Comparative Effects of Acute HIIE and MICE on Executive Function, Prefrontal Oxyhaemoglobin Responses, and Related Biomarkers in Sedentary Young Adults: A Randomized Crossover Trial

👤 Xinyue Yang
📅 2026-10-27 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSIONS: Both exercise sessions were followed by faster reaction times on selected tasks. The small sample, multiple analyses, and potential systemic contributions to fNIRS signals preclude firm conclusions about exercise-mode superiority or biological mediation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939307/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028064515&v=2.20.1

Cerebral Hemodynamic Patterns in People with Subjective Cognitive Decline: A Systematic Review Based on Functional Near-Infrared Spectroscopy

👤 Israel Contador
📅 2026-10-27 📖 1 min 📝 62 words FNIRS

Summary: Background/Objectives: Subjective cognitive decline (SCD) has been proposed as an early symptomatic manifestation of dementia/Alzheimer's disease. However, its associated brain patterns are not yet fully characterized. In recent years, functional near-infrared spectroscopy (fNIRS) has emerged as a c...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939380/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028064515&v=2.20.1

Resting-State Functional Brain Connectivity and Neurodevelopmental Profiles in 3-Year-Olds with Congenital Heart Disease

👤 Anne Gallagher
📅 2026-10-27 📖 1 min 📝 67 words FNIRS

Summary: Background/Objectives: Congenital heart disease (CHD) is associated with altered brain development and an increased risk of neurodevelopmental difficulties, particularly in more severe forms of the condition. Although alterations in functional brain connectivity have been reported before and after t...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42940032/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028064515&v=2.20.1

Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan
📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944288/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028064515&v=2.20.1

Multimodal Sensing and Digital Biomarkers in Virtual Reality-Based Cognitive Rehabilitation: Clinical Validation, Adaptive Systems, and Real-World Translation

Raffaele Ferri
2026-10-27 1 min 58 words FNIRS

Summary: Virtual reality (VR) is evolving from a platform for cognitive exercises into an instrumented rehabilitation environment capable of simultaneously delivering treatment and quantifying behavior. This critical narrative review examines how behavioral, ocular, kinematic, physiological, and neurophysiol...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944342/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028064515&v=2.20.1

Unpredictable human screams elicit anticipatory and phasic cortical responses that affect subsequent interpersonal space regulation

👤 Alessandro Grecucci 📅 2026-10-27 📖 1 min 📝 59 words FNIRS

Summary: Unpredictable threats can sustain vigilance states that bias perception, cognition, and behaviour beyond immediate stimulus processing. Human screams, as evolutionarily salient alarm signals, are especially effective in eliciting defensive responses, yet their cortical dynamics and consequences for ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42946138/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028064515&v=2.20.1

EOG Artifact Removal from EEG Using a Gated WGAN with Multi-Scale and Multi-Domain Feature Extraction

👤 Hao Tang 📅 2026-10-27 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: Background: Electroencephalogram (EEG) signals provide important information for brain-computer interfaces (BCIs) and the diagnosis of neurological disorders but are susceptible to electrooculographic (EOG) artifacts. Traditional methods often rely on multichannel data or relatively long windows, li...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939376/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028064512&v=2.20.1

EEG Features of Cortical Intent for Urination and Defecation in Healthy Aging and Spinal Cord Injury: A Descriptive Feasibility Study for BCI Neurorehabilitation

👤 Jie Jia 📅 2026-10-27 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: Background/Objectives: Systematically elucidating the neural mechanisms of bowel and bladder control is crucial for developing closed-loop brain-computer interface (BCI) neurorehabilitation for patients with spinal cord injury (SCI). This study aims to characterize the spatiotemporal Electroencephal...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939378/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028064512&v=2.20.1

The Development and Evolution of Substrate Materials for Brain-Computer Interfaces

👤 Yen Wei 📅 2026-10-27 📖 1 min 📝 58 words BRAIN COMPUTER INTERFACE

Summary: Brain-Computer Interfaces (BCIs) hold tremendous potential for neurological therapeutics and neuroprosthetics; however, their long-term clinical viability is fundamentally limited by chronic interfacial failure. This failure originates from the profound biomechanical mismatch between synthetic micro...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42942690/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028064512&v=2.20.1

Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan 📅 2026-10-27 📖 1 min 📝 60 words BRAIN COMPUTER INTERFACE

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944288/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028064512&v=2.20.1

Agentic EEG Pipeline Synthesis Using Large Language Models and the Model Context Protocol

👤 Katy Tarrit 📅 2026-10-27 📖 1 min 📝 56 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography analysis typically requires multistage processing pipelines for which configuration depends on recording characteristics and the intended analytical objective. Although automated pipelines improve reproducibility, they generally implement predefined processing sequences. We pr...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944335/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028064512&v=2.20.1

Overt and Covert Speech-State Classification from Tripolar Concentric Ring Electrode EEG: A 4 mm and 10 mm Configuration Study

Shoaib Memon 2026-10-27 1 min 76 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) speech-state classification is a noninvasive method to identify the brain's overt and covert behavior. The EEG achieved through the scalp is contaminated by artifacts and volume conduction and suffers from poor spatial resolution. We present the first application of smal...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944450/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028064512&v=2.20.1

The compositional encoding of hand-eye coordinated movements for single neurons in the posterior parietal cortex

Richard A Andersen 2026-10-27 1 min 77 words BRAIN COMPUTER INTERFACE

Summary: Human posterior parietal cortex (PPC) is known to play a role in hand-eye (H-E) coordination, yet the underlying encoding mechanisms remain uncertain. We recorded 412 single neurons across 11 sessions from motor cortex (MC; n = 251) and PPC (n = 161) in a single human participant performing an H-E c...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945213/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028064512&v=2.20.1

NCEL: Convolution-Enhanced Lightweight Neural Architecture for Robust EEG Signal Decoding in Noninvasive Brain-Computer Interfaces

👤 Liangliang Hu
📅 2026-10-27 📖 1 min 📝 57 words

BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) is a cornerstone of brain-computer interfaces (BCIs) and clinical neuro-diagnostics due to its non-invasiveness and high temporal resolution. However, robust EEG decoding remains challenging, particularly for resource-constrained deployments. While lightweight architectu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945585/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028064512&v=2.20.1

DSGTANet: A semi-supervised dual-stream graph-temporal adaptation network for cross-subject continuous EEG emotion recognition

👤 Wang-Yang Xu
📅 2026-10-27 📖 1 min 📝 61 words

BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: Continuous emotion recognition using electroencephalogram (EEG) is a critical topic in brain-computer inter-faces (BCIs), with applications in mental disorder regulation and human-computer interaction. However, its practical appli-cation is significantly constrained by several key issues,...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945587/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028064512&v=2.20.1

The application of brain-computer interface combined with neuromodulation for stroke upper-limb rehabilitation: a systematic review and network meta-analysis

Zhongjian Luo
2026-10-28 1 min 42 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Some non-invasive MI-BCI combined interventions like neuromodulation, ROB, or FES might improve upper-limb function and daily activities in stroke patients. However, rankings don't imply clinical importance. Due to small samples, limited evidence, and differences, these findings need con...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42947035/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028064512&v=2.20.1

Psychomusicology: A resounding closing cadence.

2024-01-22 1 min 256 words PSYCHOMUSICOLOGY

Summary: From 2012 to 2023, the American Psychological Association served as publisher of Psychomusicology: Music, Mind, and Brain. Annabel Cohen and Mark Schmuckler were the successive editors-in-chiefs during this time. As the journal is ceasing publication, the two editors reflect on the developm...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000305

Suggestibility and psychedelics: From therapeutics to social context.

📅 2024-12-12 📖 1 min 📝 175 words CLINICAL NEUROSCIENCE

Summary: Suggestibility is an important aspect of the psychedelic experience that merits greater attention in the expanding field of psychedelic research. This article provides a comprehensive road map for exploring the critical questions and issues at the intersection of suggestion, suggestibility, and psyc...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000412

Evaluating dispositional emotion-related factors in the effect of acceptance on affect reactivity: A secondary analysis of an experimental study.

📅 2024-12-30 📖 1 min 📝 257 words CLINICAL NEUROSCIENCE

Summary: Acceptance as a self-regulation strategy has beneficial effects on affect reactivity in response to stressors via reducing negative affect and state anxiety as well as improving positive affect. However, dispositional emotion-related factors, namely, experiential avoidance, emotional dysregulation, ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000416

Open monitoring meditation defined as the nonreactive meta-awareness of mind wandering.

📅 2024-12-12 📖 1 min 📝 128 words CLINICAL NEUROSCIENCE

Summary: Open monitoring meditation is generally defined as the nonreactive, meta-aware observation of moment-by-moment phenomenal experiences associated with mental contents without focusing on predetermined mental or physical objects. However, in the current literature, neither general definitions nor more...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000411

Multisensory interplay and theories of consciousness.

📅 2024-11-25 📖 1 min 📝 174 words CLINICAL NEUROSCIENCE

Summary: Sensory interplay typically refers to cases where the content of one sensory modality affects the content of another sensory modality. Although there is much evidence today of such interplay, it is unknown how to interpret them. Is it a one-way causal function between two separate states (e.g., a vi...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000407

How ghosting and stress impact vulnerable narcissism and maladaptive daydreaming: The role of mindfulness and rumination.

📅 2024-09-12 📖 1 min 📝 204 words CLINICAL NEUROSCIENCE

Summary: The present study aims to explore the influence of ghosting and stress on maladaptive daydreaming and vulnerable narcissism within a multivariate framework, using mindfulness and rumination as mediators. We employed a cross-sectional design and conducted it among 301 university students (M<...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000403

The conscious experience of thinking aloud: Are positive feelings associated with better think-aloud performance?

📅 2024-12-30 📖 1 min 📝 216 words CLINICAL NEUROSCIENCE

Summary: The think-aloud interview is a psychological procedure used to measure an individual’s cognition in solving a task. However, very little is known about how an individual’s conscious experience of the interview might be associated with their performance during the interview. This is consequential as ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000410

Mystical experiences in virtual reality.

📅 2024-04-04 📖 1 min 📝 176 words CLINICAL NEUROSCIENCE

Summary: Virtual reality (VR) has shown promise as a psychological means of inducing mystical experiences. Here, we compared the subjective responses of 48 undergraduate volunteers to script-guided multiuser perceptual experiences in VR (VR perception [VR-P]) with their responses to a control condit...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000396

Effect of a peer-led meditation club on adolescents' well-being, empathy, and compassion.

📅 2024-09-09 📖 1 min 📝 197 words CLINICAL NEUROSCIENCE

Summary: Many studies have examined the effect of meditation sessions led by experienced teachers on personal benefits for adults. The present study investigated the effect of broad-based mindfulness practices with an adolescent peer facilitator on prosocial outcomes in addition to individual gains. The 20 s...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000400

Prefrontal cortex activity of adolescents with problematic smartphone use in response to gains and losses: A functional near-infrared spectroscopy study.

📅 2026-09-10 📖 1 min 📝 237 words BEHAVIORAL NEUROSCIENCE

Summary: Problematic smartphone use (PSU) has become a widespread problem that is highly prevalent in adolescent population. Previous studies suggest that PSU is associated with impairment in reward processing system. Thus, increased sensitivity to rewards could make individuals more vulnerable to abusive be...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000668

Topiramate and anterior cingulate cortex high-frequency stimulation mitigate aggression and promote sociability in socially isolated male mice.

📅 2026-08-10 📖 1 min 📝 230 words BEHAVIORAL NEUROSCIENCE

Summary: Excessive aggression induced by social isolation (SI) has been closely linked to dysfunction in prefrontal circuits, particularly the anterior cingulate cortex (ACC). Topiramate, an antiepileptic drug known to enhance GABAergic and inhibit glutamatergic transmission, has demonstrated antiaggressive ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000663

Is memory spared in monkeys with selective lesions of the hippocampus?

📅 2026-08-03 📖 1 min 📝 239 words BEHAVIORAL NEUROSCIENCE

Summary: Normal explicit/declarative memory requires a system of anatomically related structures in the medial temporal lobe that includes the hippocampus and the entorhinal, perirhinal, and parahippocampal cortex. Although extensive medial temporal lobe damage in primates causes robust anterograde amnesia, ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000658

Distal pup cues attenuate defensive reactivity and fear-potentiated startle magnitude in postpartum rats without altering sensorimotor gating.

📅 2026-07-09 📖 1 min 📝 241 words BEHAVIORAL NEUROSCIENCE

Summary: Using a novel behavioral paradigm that integrates fear-potentiated startle (FPS) and prepulse inhibition testing, this study examined how motherhood and infant cues impact innate defensive reactivity (as measured by the magnitude of basal startle reflex), attentional processing (as measured by prepu...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000661

A novel rat spatial inference task reveals rapid deduction of transitive relationships using schemas and deliberation.

📅 2026-07-13 📖 1 min 📝 264 words BEHAVIORAL NEUROSCIENCE

Summary: Inferential reasoning is a vital cognitive ability that enables animals to navigate novel situations by leveraging existing relational knowledge of memory schema, with hypothesized roles of prefrontal cortical—hippocampal circuits. Transitive inference (TI) tasks test the ability of subjects to infe...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000660

Keep Android Open 🔓

📅 2026-02-26 📖 3 min 📝 625 words FMHY

Summary: <h3 id="android-will-become-a-locked-down-platform-in-less-than-200-days" tabindex="-1">Android will become a locked-down platform in less than 200 days. </h3> <p>In August 2025, Google announced th...

🔗 **Read full article:**
https://fmhy.net/posts/KeepAndroidOpen

Monthly Updates [April]

📅 2026-04-01 📖 4 min 📝 885 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/april-2026

Monthly Updates [May]

📅 2026-05-01 📖 4 min 📝 892 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/may-2026

Monthly Updates [October]

📅 2026-10-01 📖 5 min 📝 1097 words FMHY

> **Summary:** <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   Re-ordered Live Sports based on the <a href="https://challonge.com/ajqvt0wo/fullscreen" rel=...

🔗 **Read full article:**
https://fmhy.net/posts/oct-2026

Hunger beats tenure: Advice for junior engineers

👤 akavi 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://felixrieseberg.com/hunger-beats-tenure-advice-for-junior-engineers/ Comments URL: <a href="https://news.ycombinator.com/item?id=50266719">https://news.ycombinator.com/item?id=5026...

🔗 **Read full article:**
https://felixrieseberg.com/hunger-beats-tenure-advice-for-junior-engineers/

Wealth Inequality in America drawn to scale

👤 haraball 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://dreamagainusa.org/ Comments URL: https://news.ycombinator.com/item?id=50270764 Points: 4 # Comments: 0

🔗 **Read full article:**
https://dreamagainusa.org/

Spain train crash: 'At least 10 dead' in high-speed collision

👤 ultral 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.thetimes.com/world/europe/article/spain-train-crash-madrid-latest-deaths-victims-807kwv23l Comments URL: <a href="https://news.ycombinator.com/item?...

🔗 **Read full article:**
https://www.thetimes.com/world/europe/article/spain-train-crash-madrid-latest-deaths-victims-807kwv23l

Registration is now full for the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-16 📖 1 min 📝 106 words ERP BOOT CAMP

> **Summary:** The demand for the 2024 ERP Boot Camp was far beyond our expectations, and we reached our maximum registration of 30 people within one day. We already have a waiting list of over 30 people, so we have closed the registration site. <...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/15/registration-full

New Paper: Using Multivariate Pattern Analysis to Increase Effect Sizes for ERP Amplitude Comparisons

👤 Steve Luck 📅 2024-06-10 📖 2 min 📝 521 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Bahle, B., Simmons, A. M., & Luck, S. J. (2024). Using multivariate pattern analysis to increase effect sizes for event-related potential analyses. Psychophysiology, 61, e14570. https://doi.org/10.1111/psyp.14570 [<a h...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/10/erp-core-decoding-paper

Call for Student/YP Deputy Editor

Deidre Artis 2026-09-29 1 min 14 words EMBS

Summary: The post Call for Student/YP Deputy Editor appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/call-for-student-yp-deputy-editor/

Time-Dynamic Semantic-Activity Entanglement: Quantifying Communication-Based Synchrony in Collaborative VR

Pooja Pol, Jianing Zhang, Qing Huang 2026-10-28 1 min 187 words ARXIV CS HC

Summary: arXiv:2610.37427v1 Announce Type: new Abstract: Collaborative virtual reality (VR) produces rich multimodal traces of team behavior, yet practical methods for characterizing time-varying team communication remain limited. We present time-dynamic semantic-activity entanglement (TSAE), inspired by th...

🔗 **Read full article:**
https://arxiv.org/abs/2610.37427

Comparing Temporal Deep Learning Architectures for Discrete Cross-User Myoelectric Control

Ethan Eddy, Evan Campbell, Scott Bateman, Erik Scheme

2026-10-28 1 min 208 words

ARXIV CS HC

Summary: arXiv:2610.37374v1 Announce Type: new Abstract: The recognition of dynamic gestures via discrete myoelectric control is becoming an attractive input modality for event-based human-machine interaction. While recent work suggests that discrete cross-user models are possible (i.e., no calibration data...

🔗 **Read full article:**
https://arxiv.org/abs/2610.37374

Understanding Human-Robot Collaboration through Joint Action

Chengwen Zhang, Jianyang Wang, Chang Liu, Yukun Wang, Yuan Xu, Zhoutong Ye, Jie Cai, Chun Yu, Yuanchun Shi

2026-10-28 1 min 126 words

ARXIV CS HC

Summary: arXiv:2610.37298v1 Announce Type: new Abstract: Human-robot collaboration requires partners to coordinate interdependent contributions toward a shared goal. Joint Action theory provides concepts for understanding this coordination, but how these concepts are instantiated and combined across empiric...

🔗 **Read full article:**
https://arxiv.org/abs/2610.37298

An Interactive Mobile Augmented Reality System for Efficient Virtual Model-to-Patient Registration in Thoracic Surgery

👤 Emmanuel Ajaero, Pascal Spiegler, George Rakovich, Marta Kersten-Oertal, Yiming Xiao

📅 2026-10-28 📖 1 min 📝 192 words ARXIV CS HC

Summary: arXiv:2610.37271v1 Announce Type: new Abstract: Video-Assisted Thoracoscopic Surgery (VATS) is a minimally invasive procedure, where precise surgical port placement is vital, yet it relies heavily on surgeon experience and mental mapping of pre-operative imaging in the operating room. To address th...

🔗 **Read full article:**
https://arxiv.org/abs/2610.37271

Peers4U: Designing Learner-Centered Virtual Peers for Children's Multimodal Mathematics Learning

👤 Zhi Zheng, Binglin Liu, Weihao Chen, Lan Luo, Yu Mei, Jie Cai, Chun Yu, Yuanchun Shi

📅 2026-10-28 📖 1 min 📝 153 words ARXIV CS HC

Summary: arXiv:2610.37238v1 Announce Type: new Abstract: In early mathematics, children learn abstract concepts through concrete representations, such as manipulating objects, drawing, and speaking. Learning with peers can enrich this process by exposing children to diverse strategies and enabling them to e...

🔗 **Read full article:**
https://arxiv.org/abs/2610.37238

Science-Fiction Transformation as a Design Lens for Visualization Transformation

👤 Abdulhaq Adetunji Salako, Christian Tominski

📅 2026-10-28 📖 1 min 📝 222 words

ARXIV CS HC

Summary: arXiv:2610.37131v1 Announce Type: new Abstract: Science fiction often features characters or objects that undergo radical changes in form while still being recognizable as the same entity. For example, a vehicle might unfold into a robot, or one body might flow or shape-shift into another. We use s...

🔗 **Read full article:**
https://arxiv.org/abs/2610.37131

Externalizing Hand Grip Pressure Through Real-Time Visual Feedback

👤 Andrew L. Huard, Avinash Ajit Nargund, Misha Sra

📅 2026-10-28 📖 1 min 📝 202 words

ARXIV CS HC

Summary: arXiv:2610.36769v1 Announce Type: new Abstract: Grip pressure, including contact force and its distribution across the hand, is central to skilled grasping but is not directly visible. This makes it difficult to teach, learn, and self-correct in grip-dependent motor tasks, where verbal instruction ...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36769

Translating Poetic Emotion into Dynamic Mathematical Visuals: A Strange Attractor-Based Approach

Subhadra Nayak 2026-10-28 1 min 99 words ARXIV CS HC

Summary: arXiv:2610.36698v1 Announce Type: new Abstract: Poetry conveys emotional experiences through language. This project investigates how poetic emotional experiences can be represented through dynamic mathematical visualisations while further developing a pipeline for this purpose. By leveraging a pret...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36698

Beyond the 1D Reality-Virtuality Continuum: A 2D Framework for Seven Extended Reality Paradigms

Rajashekhar Vachiravelu Saminathan 2026-10-28 1 min 160 words ARXIV CS HC

Summary: arXiv:2610.36674v1 Announce Type: new Abstract: The Reality-Virtuality (RV) continuum helps in describing the relationship between physical and computer-generated environments. However, the increasing number of Extended Reality (XR) paradigms introduces new technologies. Their characteristics canno...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36674

Immersive Commons: Designing Ephemeral Spatial Dialogue for Co-Located Mixed Reality Environments

Wuwu Ran, Mohammed Faraj Al-Suwaidi
2026-10-28 1 min 150 words ARXIV CS HC

Summary: arXiv:2610.36014v1 Announce Type: new Abstract: Head-mounted mixed reality (MR) often prioritises individual immersion, potentially drawing attention away from others sharing the room. This paper asks how MR can support co-located spatial dialogue through embodied interaction and activity visible t...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36014

p-Adic statistical field theory of fluctuation effects in hierarchical neural networks

W. A. Z\'u\~niga-Galindo
2026-10-28 1 min 281 words ARXIV QBIO NC

Summary: arXiv:2610.36400v1 Announce Type: cross Abstract: The Buice-Cowan field theory [PRE 75, 051919 (2007)] describes fluctuations and correlations of neural activity through a path integral whose mean-field limit is the Wilson-Cowan equation and whose critical behavior belongs to the directed percolati...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36400

SOMA: A Benchmark for Fine-Grained Egocentric Behavior and Body-Motion Understanding

Matea Tashkovska, Na\"el Ouerghemi, Haozhe Qi, Alexander Mathis

2026-10-28 1 min

242 words

ARXIV QBIO NC

Summary: arXiv:2610.36384v1 Announce Type: cross Abstract: Perceiving human body motion is critical for multimodal systems that understand and assist people. Yet current evaluations of Multimodal Large Language Models (MLLMs) largely focus on objects, scenes, and coarse actions, leaving their ability to rea...

Read full article:
https://arxiv.org/abs/2610.36384

MtReNIRS: A Self-Supervised Foundation Model for Transferable Neurovascular Representations

Tony Chae, Kleanthis Avramidis, Dimitra Emmanouilidou, Ivan J. Tashev

2026-10-28 1 min

200 words

ARXIV QBIO NC

Summary: arXiv:2610.36159v1 Announce Type: cross Abstract: Functional near-infrared spectroscopy (fNIRS) offers a promising approach for monitoring the hemodynamic correlates of neural activity in naturalistic settings. However, existing modeling methods for fNIRS are largely confined to small-scale, single...

Read full article:
https://arxiv.org/abs/2610.36159

Every route counts, but some count more: a max-plus view of the C. elegans brain

Emma Reyner-Fuentes, Sebastian Rudolph, Carmen Pel\'aez-Moreno, Francisco J. Valverde-Albacete

2026-10-28 1 min 156 words ARXIV QBIO NC

Summary: arXiv:2610.37587v1 Announce Type: new Abstract: Brains do not simply add up every signal they receive: they select which routes dominate. Yet connectomes are usually analyzed with tools that sum over all paths and place neural states in Euclidean spaces, blurring the feedback circuits that may sust...

🔗 **Read full article:**
https://arxiv.org/abs/2610.37587

Sufficiency Theorem and Closed-Form Posterior for a Windowed Change Point Detection Task

Payton Hanks, Rishidev Chaudhuri, Timothy D. Hanks

2026-10-28 1 min 206 words ARXIV QBIO NC

Summary: arXiv:2610.37486v1 Announce Type: new Abstract: To function effectively in a changing world, agents must detect changes and act on them in a timely fashion. We study this in the context of a change point detection task in which clicks arrive from a Poisson process whose rate changes at an unknown t...

🔗 **Read full article:**
https://arxiv.org/abs/2610.37486

Local learning is all you need: brain-aligned predictive coding networks

👤 Dirk G\"utlin, Ryszard Auksztulewicz, Adrien Doerig

📅 2026-10-28 📖 1 min 📝 206 words

ARXIV QBIO NC

Summary: arXiv:2610.37327v1 Announce Type: new Abstract: Predictive Coding (PC) is an influential account of how brain representations emerge through hierarchical, local, self-supervised prediction-based learning, without requiring globally propagated error signals. Despite extensive evidence supporting PC,...

🔗 **Read full article:**
https://arxiv.org/abs/2610.37327

Exploring Neural Dynamics with Time Delays: A Numerical and Bifurcation Analysis Approach

👤 Martin R Nelson, Jonathan J Crofts

📅 2026-10-28 📖 1 min 📝 177 words

ARXIV QBIO NC

Summary: arXiv:2610.36920v1 Announce Type: new Abstract: The Wilson-Cowan model is a classic example of a ``neural mass model'' - a simplified yet powerful way to describe the activity of large networks of neurons. It is particularly interesting because, even in its relatively simple form, it can produce a ...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36920

Neuronal Identity as an Organizational Basis for Analyzing Neural Population Dynamics

Seungjae Han, Joshua Yedam You, Soi Kim, Minho Eom, Jinho Park, Seunghee Park, Byeongwook Lee, Young-Gyu Yoon

2026-10-28 1 min 237 words ARXIV QBIO NC

Summary: arXiv:2610.36745v1 Announce Type: new Abstract: Individual neurons exhibit diverse molecular, morphological, and electrophysiological properties that shape their distinct roles in neural circuits. While these multimodal features are largely shared across individuals, they remain an underutilized so...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36745

The geometry of cognition: from neural populations to neurotechnology

Lucas Bayones, Devika Narain, Roman Rossi-Pool

2026-10-28 1 min 225 words

ARXIV QBIO NC

Summary: arXiv:2610.36541v1 Announce Type: new Abstract: Cognition is implemented by distributed neural populations whose collective activity evolves through structured, low-dimensional state spaces. In this review, we survey the analytical and conceptual tools used to study latent population dynamics in bi...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36541

Temporal smoothness constraints on efficient neurobiological codes imply temporal specialization

John J. Vastola, Samuel J. Gershman
2026-10-28 1 min 170 words ARXIV QBIO NC

Summary: arXiv:2610.36272v1 Announce Type: new Abstract: Neurobiological circuits may be energy efficient in part because there is a division of labor: different subsystems compute or represent different things. In light of this, it is interesting that a major difference between subsystems is the time scale...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36272

Vitamin B6 augments antidepressant activity of venlafaxine by restoring cortical EEG oscillations, behavior and oxidative biomarkers in chronic unpredictable mild stress (CUMS)-induced model of depression-like behavior

1 min 30 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Fashwa Khan Tareen, Sana Javaid, Waseem Ashraf, Syed Muhammad Muneeb Anjum, Rana Muhammad Zahid Mushtaq, Asad Abrar, Faleh Alqahtani, Imran Imran

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004506?dgcid=rss_sd_all

Stage-specific exposure to an activity-permissive medium enhances neuronal activity in oligodendrocyte-enriched cortical organoids

📖 1 min 📝 34 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Clara Chung, Michelle J. Kim, Grace Field, Konstantina Pilarinos, Elizabeth K. Kharitonova, Natalie Baker Campbell, Marie Shi, Christopher V. Gabel, Joseph L. Orofino, Ella Zeldich

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004634?dgcid=rss_sd_all

Neurodegenerative diseases in Sub-Saharan Africa: diagnostic invisibility, biological heterogeneity and a precision-neuroscience agenda

📖 1 min 📝 11 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Maulilio John Kipanyula

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226006962?dgcid=rss_sd_all

Investigating fatigability and cortical activation during prolonged foot tapping in multiple sclerosis

📖 1 min 📝 28 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Zuhal Abasiyanik, Lousin Moumdjian, Natalia Skorupa, Daphne Kos, Bart Van Wijmeersch, Miguel D’Haeseleer, Inge Zijdewind, Peter Feys, Felipe Balistieri Santinelli

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226006949?dgcid=rss_sd_all

Neural dynamics of auditory attention: Interactions between top-down and bottom-up processes

📖 1 min 📝 12 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): S. Graceffo, M. Elhilali

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006002?dgcid=rss_sd_all

Neurovascular coupling in simultaneous EEG-fNIRS studies: A scoping review of analytical frameworks and clinical applications

📖 1 min 📝 18 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Yu Wu, Rajamanickam Yuvaraj, Stéphane Perrey, Hasan Ayaz, Wei-Peng Teo

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006130?dgcid=rss_sd_all

Ameliorating effects and mechanisms of therapeutic electrical stimulation on the recurrent laryngeal nerve injury model in rats

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-58788-1

Genetic mechanisms underlying the associations between brain structure and mental health

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-79115-2

Functional connectivity in cerebral visual impairment: implications for daily visual behaviors

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-76731-2

Compensation of the effects of temperature on a motor system in the crab, Cancer borealis

👤 Kathleen JacquerieJames M. DiMartinoAnanya DalalJi ZengEve Marderahttps://ror.org/05abbep66Volen Center and Biology Department, Brandeis University, Waltham, MA 02454

📅 2026-10-21 📖 1 min 📝 50 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceCold-blooded animals that live in climates with significant seasonal changes in ambient temperature must have myriad mechanisms to function over a wide range of environmental conditions. We explore...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2625185123?af=R

Action features dominate cortical representation during natural vision

Heejung JungHunar JainXiaochun HanMa FeilongJane HanDeepanshi ShokeenCara Van UdenIsabella HansonAndrew C. ConnollyTor D. WagerJames V. HaxbySamuel A. Nastaseahttps://ror.org/049s0rh22Department of Psychological and Brain Sciences, Dartmouth College, Hanover, NH 03755bIndependent Researcher, Sunnyvale, CA 94085chttps://ror.org/03taz7m60Department of Psychology, Center for Computational Language Sciences, University of Southern California, Los Angeles, CA 90089

2026-10-23 1 min 51 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceHuman cortex is functionally organized to prioritize the most important features of our external world. In perceiving the natural world, however, these features are often overlapping and intermixed...

Read full article:
https://www.pnas.org/doi/abs/10.1073/pnas.2625018123?af=R

Translational control of microglial metabolism, inflammatory responses, and survival by the eIF4E-binding proteins

Sara BermudezJung-Hyun ChoiNiaz MahmoodJacob W. VogelHashem AlmousaZiying HuangSung-Hoon KimVivian Yuchan ZhuLinqiao ZhouDiana Casas-AntillonMoein YaqubiDanielle CozachencoJo Anne StrattonOskar HanssonLuke HealyArgel Aguilar-Valles Nahum Sonenbergahttps://ror.org/01pxwe438Department of Biochemistry and Goodman Cancer Institute, McGill University, Montreal, QC H3A 1A3, Canadabhttps://ror.org/02wnxgj78Department of Biochemistry, Chungbuk National University, Cheongju 28644, South Koreachttps://ror.org/012a77v79Department of Clinical Sciences, Lund University, Lund 221 84, SwedendSciLifeLab, Lund 221 00, Swedenehttps://ror.org/0227as991Department of Microbiology and Molecular Biology, Chungnam National University, Daejeon 34134, South Koreafhttps://ror.org/03rmrcq20Department of Pathology and Laboratory Medicine, Faculty of Medicine, University of British Columbia, Vancouver, BC V5Z 1M9, CanadagThe Hospital for Sick Children, Toronto, ON M5G 1E8, Canadahhttps://ror.org/01pxwe438Department of Neurology and Neurosurgery, Montreal Neurological Institute, McGill University, Montreal, QC H3A 1A1, Canadaihttps://ror.org/03490as77Institute of Biophysics Carlos Chagas Filho, Federal University of Rio de Janeiro, Rio de Janeiro 21941-170, Braziljhttps://ror.org/02qtvee93Department of Neuroscience, Carleton University, Ottawa, ON K1S 5B6, Canada

📅 2026-10-21 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceThe link between Alzheimer’s disease (AD) risk and microglial function is well-known, yet the mechanisms governing microglial responses remain unclear. Postmortem and experimental studies demonstra...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2610766123?af=R

Naturalistic spatial education engages distinct neural processes for mental rotation, cognitive load, and linguistic demands

Eloise B. GacettaMarissa L. LawsLauren S. AuletTeoman S. OzaydinJonathan S. TsayConrad ZapantaJessica F. Cantlonahttps://ror.org/05x2bcf33Department of Psychology, Carnegie Mellon University, Pittsburgh, PA 15213bhttps://ror.org/01an3r305Center for the Neural Basis of Cognition, Carnegie Mellon University and University of Pittsburgh, Pittsburgh, PA 15213chttps://ror.org/0072zz521Department of Psychological and Brain Sciences, University of Massachusetts Amherst, Amherst, MA 01003dhttps://ror.org/05x2bcf33Neuroscience Institute, Carnegie Mellon University, Pittsburgh, PA 15213ehttps://ror.org/05x2bcf33Department of Biomedical Engineering, Carnegie Mellon University, Pittsburgh, PA 15213

📅 2026-10-21 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceSpatial visualization skill is a foundational component of human intelligence and a powerful lever for success in science, technology, engineering, and mathematics (STEM). Yet little is known about...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2601206123?af=R

C1ql2 gates stress vulnerability by regulating microglia-mediated synaptic pruning in the BLA-vHPC circuit

Chun-Yan WangWei-Zhu LiuShuang-Mei LiaoJing NiuHan-Qing PanWen-Jie YouKe ZhaoBao MaMing-Fen LuJia-Xin ZouPing HuWen-Hua ZhangBing-Xing Panahttps://ror.org/042v6xz23Jiangxi Institute of Respiratory Diseases and Department of Pathology, The First Affiliated Hospital, School of Basic Medical Sciences and Institute of Biomedical Innovation, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinabhttps://ror.org/042v6xz23Jiangxi Provincial Key Laboratory of Brain Science and Brain Health, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinachttps://ror.org/042v6xz23Postdoctoral Research Station, The First Affiliated Hospital, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinadhttps://ror.org/042v6xz23Institute of Translational Medicine, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinaehttps://ror.org/042v6xz23Department of Neurology, The Second Affiliated Hospital, Jiangxi Medical College, Nanchang University, Nanchang 330006, China

📅 2026-10-21 📖 1 min 📝 43 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceStress-related emotional disorders involve maladaptive synaptic remodeling in the basolateral amygdala (BLA), which is highly heterogeneous across neurons projecting to distinct downstream targets....

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2600854123?af=R

BVLab-analyzer: a unified visual toolbox for terabyte-scale whole-brain cell detection, neuron reconstruction, and vascular network extraction

Tingwei Quan 2026-10-07 1 min 139 words FRONTIERS NEUROINFORMATICS

Summary: BVLab-Analyzer is a standalone graphical software tool for end-to-end analysis of terabyte-scale whole-brain microscopy images. Operating through a zero-code interface, it handles data import (TIFF, BioImageVision, OME-Zarr), deep learning-based semantic segmentation, 3D skeleton reconstruction, and...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1925112

Transformers for fMRI analysis: from spatiotemporal modeling to brain foundation models

Bocheng Wang 2026-10-07 1 min 217 words FRONTIERS NEUROINFORMATICS

Summary: Functional magnetic resonance imaging (fMRI) serves as a pivotal tool for investigating the functional organization and dynamic connectivity of the human brain, indirectly reflecting neural activity through blood oxygen level-dependent (BOLD) signals. However, the high-dimensional spatiotemporal str...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1869687

Natural language processing for suicidal behavior detection in short Spanish clinical notes from first-contact interviews

👤 Ansel Y. Rodríguez-González
📅 2026-10-07 📖 1 min 📝 267 words

FRONTIERS NEUROINFORMATICS

Summary: IntroductionSuicide is a multifactorial phenomenon influenced by social, cultural, biological, and psychological factors, whose early detection remains challenging. According to the World Health Organization, suicide is the third leading cause of death among individuals aged 15–29 years, representin...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1919962

Mechanonociceptive tactile feedback enhances prosthetic grasp control during repeated human-in-the-loop interaction

👤 Elif Hocaoglu
📅 2026-10-15 📖 1 min 📝 215 words

FRONTIERS NEUROROBOTICS

Summary: Upper-limb prosthetic systems continue to face a fundamental challenge in providing sensory feedback that remains functionally effective during repeated interaction. This study investigates innocuous painful feedback (IPF), a controlled mechanonociceptive tactile modality intended to provide salient...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1910158

Brain-inspired predictive visual attention for spatiotemporal semantic understanding of complex film and television videos

Peng Han 2026-10-22 1 min 275 words FRONTIERS NEUROROBOTICS

Summary: Due to the unique characteristics of film and television, such as montage editing and narrative structure jumps, semantically salient targets frequently experience temporal feature drift, making it difficult for existing models to maintain stable semantic understanding output when keyframes are lost...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1934032

Ordered onset of regional desynchronization and a transition in movement-class discriminability during cue-paced motor imagery

Seho Lee 2026-10-21 1 min 298 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Motor imagery recruits a distributed network whose components and large-scale organization are well characterized. The temporal order in which those components are engaged is not: the latency at which each region begins to desynchronize within a trial has not been compared statistically i...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb161

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028002347&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39870674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028002347&v=2.20.1

Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028002347&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028002347&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

Robert Oostenveld 2025-08-13 1 min 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028002347&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

Suzy J Styles 2025-12-08 1 min 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41358599/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028002347&v=2.20.1

Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41676644/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028002347&v=2.20.1

Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41678848/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028002347&v=2.20.1

The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41703344/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028002347&v=2.20.1

EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42538927/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028002347&v=2.20.1

Intelligent Sensing and Adaptive Control for Low-Loss Soybean Harvesting: A Review of Multi-Scenario Technologies

👤 Zhong Tang 📅 2026-10-27 📖 1 min 📝 60 words LOW VISION

Summary: Soybean harvesting is strongly affected by low pod position, lodging, pod shattering, moisture variation, and changing material flow, making stable low-loss operation difficult across diverse field conditions. This review synthesizes 179 core publications identified through structured literature sea...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944440/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028002344&v=2.20.1

A 3D Vision-Based Localization Method for Automated Loading of Large-Sized Heat Exchangers

👤 Qing Jiang 📅 2026-10-27 📖 1 min 📝 62 words LOW VISION

Summary: Manual loading of large-sized finned-tube heat exchangers has low productivity and imposes substantial physical demands on workers. In addition, industrial 3D vision systems often produce large regions of missing depth data when imaging highly reflective metallic surfaces, which complicates grasp-po...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944478/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028002344&v=2.20.1

Hybrid Vision Perception for Low-Cost Autonomous Vehicle Navigation

Abdelkader El Kamel 2026-10-27 1 min 69 words LOW VISION

Summary: High-performance perception systems for autonomous vehicles (AVs) commonly use LiDAR, RADAR, and multiple cameras. These sensors provide useful information, but their cost and computational requirements can be difficult to justify for small prototypes and low-cost platforms. This work therefore uses...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944489/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028002344&v=2.20.1

Fidelity-oriented pelvic CBCT-to-CT synthesis using an anatomical information-attention diffusion model

Kunyuan Xu 2026-10-27 1 min 63 words LOW VISION

Summary: Cone-beam computed tomography (CBCT) is widely used in image-guided radiotherapy and adaptive radiotherapy, but its direct quantitative use is limited by scatter artifacts, low soft-tissue contrast, and inaccurate Hounsfield unit (HU) values. Diffusion models provide stable image generation capabili...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944823/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028002344&v=2.20.1

Poros: Perception-Enabled Interactive Assistance for Appliance Use by Blind and Low Vision Users

Wei Liang 2026-10-27 1 min 62 words LOW VISION

Summary: Modern home appliances pose significant accessibility barriers for individuals with visual impairments, driven by increasing functional complexity and minimalist design trends. Drawing on formative interviews with blind and low vision (BLV) participants, we propose Poros, a novel system providing pe...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945092/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028002344&v=2.20.1

DELUGE: Decomposed Entropy-coded Live Unstructured Geometry Exchange for Real-time Particle Streaming

Takefumi Hiraki 2026-10-27 1 min 74 words LOW VISION

Summary: Particle-based physics simulations, including fluids, smoke, and granular media, are fundamental to visual realism in immersive VR and AR. With the growing adoption of social VR and digital twins, demand is increasing for shared experiences in which multiple users interact with the same simulation i...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945105/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028002344&v=2.20.1

A multi-attention mechanism-improved ResNet18 approach for lithology identification and classification

Ruifeng Li 2026-10-27 1 min 61 words LOW VISION

Summary: Lithology identification is vital for improving geological exploration accuracy, optimizing petroleum exploration schemes, and ensuring construction engineering safety. Traditional identification methods suffer from low efficiency, high cost, and heavy dependence on domain expertise. This study prop...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945382/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028002344&v=2.20.1

An open-source platform for measuring optomotor responses in mice

Shogo Soma 2026-10-27 1 min 64 words LOW VISION

Summary: The optomotor response (OMR) is a valuable behavioral index for assessing visual function in basic research; however, commercially available OMR evaluation equipment is expensive and lacks flexibility, as proprietary software and fixed hardware geometry restrict custom stimulus protocols and system ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945590/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028002344&v=2.20.1

Communication patterns in home medication review for individuals with vision impairment: A qualitative study

👤 Woojin Park
📅 2026-10-27 📖 1 min 📝 37 words

LOW VISION

Summary: CONCLUSIONS: In these consultations, professional dialogue supported verification and reassurance, while direct regimen review revealed unverified beliefs and gaps in medication knowledge. The findings suggest a role for accessible counseling that combines information provision with professional ver...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945707/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028002344&v=2.20.1

Global, regional, and national burden of unintentional poisoning, 1990-2023: results from the Global Burden of Disease Study 2023

👤 GBD 2023 Unintentional Poisoning Collaborators

📅 2026-10-27 📖 1 min 📝 65 words

LOW VISION

Summary: BACKGROUND: Unintentional poisoning is a preventable cause of injury-related morbidity and mortality, yet global patterns, trends, and inequalities remain incompletely characterised. Within the Global Burden of Disease (GBD) framework, unintentional poisoning includes poisoning by carbon monoxide an...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945982/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261028002344&v=2.20.1

Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux
2026-07-18 1 min 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42470115/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028002341&v=2.20.1

A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42532383/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028002341&v=2.20.1

Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42590984/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028002341&v=2.20.1

Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42596634/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028002341&v=2.20.1

Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42614895/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028002341&v=2.20.1

A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42787568/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028002341&v=2.20.1

Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42788316/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028002341&v=2.20.1

Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42817759/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028002341&v=2.20.1

Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42923236/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028002341&v=2.20.1

Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42926384/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261028002341&v=2.20.1

Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42711906/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028002339&v=2.20.1

Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42768901/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028002339&v=2.20.1

Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42775591/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028002339&v=2.20.1

Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42789611/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028002339&v=2.20.1

Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42845092/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028002339&v=2.20.1

Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42848198/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028002339&v=2.20.1

Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42881664/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028002339&v=2.20.1

Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885551/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028002339&v=2.20.1

An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885828/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028002339&v=2.20.1

Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42941752/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261028002339&v=2.20.1

Combining Cognitive Training and Brain Stimulation: A Novel Approach to Address Disordered Eating and Antipsychotic-Induced Weight Gain in People With Severe Mental Illness

👤 Ulrike Schmidt
📅 2026-10-26 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: The initiation of antipsychotic therapy is a critical clinical window for early intervention to mitigate the long-term metabolic and psychiatric consequences of antipsychotic-induced weight gain. Combining ABMT + tDCS is acceptable and may help reduce binge eating in individuals on antip...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42929065/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028002337&v=2.20.1

High-Intensity Transcranial Alternating Current Stimulation for Post-Stroke Cognitive Impairment: A Triple-Blind Randomized Trial

Xiquan Hu 2026-10-26 1 min 60 words TDCS TACS TRNS

Summary: CONCLUSION: Hi-tACS improved global cognition and working memory in patients with PSCI and was generally well tolerated. Exploratory fNIRS findings suggested possible accompanying changes in cortical activation and network organization but did not establish a neurophysiological mechanism. Importantl...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42933772/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028002337&v=2.20.1

Comparative analysis of mitochondrial genomes in Zantedeschia parental lines and their hybrid progeny

Hongzhi Wu 2026-10-27 1 min 31 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study expands current knowledge of mt genome diversity in the Araceae and provides valuable genomic resources for future studies of Zantedeschia mitogenome evolution, genetic resource conservation, and molecular breeding.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42934513/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028002337&v=2.20.1

Combined Transcranial Direct Current Stimulation and Transcutaneous Cervical Vagus Nerve Stimulation for Anxiety and Depression: An Exploratory Pilot Study

👤 Maria-Manuela Apostol
📅 2026-10-27 📖 1 min 📝 60 words

TDCS TACS TRNS

Summary: Anxiety and depressive disorders are among the leading causes of disability worldwide, yet current treatments remain limited by modest efficacy and restricted accessibility; non-invasive neuromodulation techniques such as transcranial direct current stimulation (tDCS) and transcutaneous cervical vag...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939258/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028002337&v=2.20.1

Non-Invasive Brain Stimulation Techniques and Effects on Neuromodulation and Athletic Performance in Combat Sports: A Systematic Review

Bianca Miarka 2026-10-27 1 min 59 words TDCS TACS TRNS

Summary: CONCLUSIONS: tDCS may influence selected cognitive, neuromuscular, and fatigue-related outcomes in combat sport athletes; however, the direction and magnitude of these effects are inconsistent across studies. Given the substantial methodological heterogeneity and the low-to-very-low certainty of evi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939305/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028002337&v=2.20.1

Non-Invasive Brain Stimulation for Central Neuropathic Pain: An Exploratory Randomized Sham-Controlled Crossover Trial

Eunhee Sohn 2026-10-27 1 min 64 words TDCS TACS TRNS

Summary: Background/Objectives: Central neuropathic pain (CNP) is difficult to manage, and comparative evidence for non-invasive brain stimulation is limited. We explored the short-term efficacy and safety of repetitive transcranial magnetic stimulation (rTMS) and transcranial direct current stimulation (tDC...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939331/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028002337&v=2.20.1

Noninvasive Neuromodulation for Cognitive Dysfunction in Parkinson's Disease: A Scoping Review

Reginald B O'Hara 2026-10-27 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: Evidence specific to PD-MCI is limited and mixed. Several studies pairing stimulation with cognitive training reported favorable outcomes, but this pattern was inconsistent and does not establish superiority over stimulation alone. No VNS trial enrolled participants with confirmed PD-MC...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939339/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028002337&v=2.20.1

Comparative Plastid Phylogenomics of 29 Newly Sequenced Bougainvillea Cultivars Resolves the Phylogenetic Placement of Double-Bracted Cultivars

Huiqi Zhao 2026-10-27 1 min 59 words TDCS TACS TRNS

Summary: Bougainvillea (Nyctaginaceae) cultivars are horticulturally prized worldwide for their vivid bracts; however, repeated interspecific hybridization and somatic mutations have obscured their maternal origins, particularly for double-bracted forms. Here, we sequenced and assembled the complete plastome...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939474/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028002337&v=2.20.1

Anodal Transcranial Direct Current Stimulation Reduces Neuroinflammation and Cholinergic Dysregulation After Young Adult Traumatic Brain Injury

Guven Akcay
2026-10-27 1 min 61 words TDCS TACS TRNS

Summary: Traumatic brain injury (TBI) during childhood and adolescence can result in persistent motor and cognitive dysfunction due to secondary injury cascades involving neuroinflammation and disruption of neurotransmitter homeostasis. Although transcranial Direct Current Stimulation (tDCS) is a promising n...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939682/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028002337&v=2.20.1

Assessing brain-behaviour coupling in non-invasive brain stimulation: Insights from reliable change indices and Monte Carlo simulations

👤 Alexander D Tang
📅 2026-10-27 📖 1 min 📝 64 words

TDCS TACS TRNS

Summary: Non-invasive brain stimulation (NiBS) studies frequently report exploratory brain-behaviour correlations following the application of NiBS. However, these analyses are typically underpowered and rely on ratio-based change scores with known statistical limitations. We addressed these limitations by p...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945749/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261028002337&v=2.20.1

Neural-Discursive Dynamics in Collaborative Spatial Problem Solving: Evidence from fNIRS Hyperscanning and Accountable Talk

👤 Xingguang Guo
📅 2026-10-27 📖 1 min 📝 65 words FNIRS

Summary: High-quality collaboration cannot be inferred simply from how much students talk or how strongly their brain activity is synchronized during joint work. This study examined relations among accountable talk, inter-brain synchrony (IBS), and group task performance in fifth-grade students' collaborativ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42938505/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028002334&v=2.20.1

Probing Executive Function in the Toddler Data Desert: A Touchscreen fNIRS Study of Working Memory and ADHD-Related Symptoms

👤 Caitie Busch 📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Attention-deficit/hyperactivity disorder (ADHD) is associated with early differences in attentional control and working memory (WM), yet relatively few paradigms are developmentally feasible across the toddler-to-preschool period while also distinguishing spatial from featural information use. Addre...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42938612/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028002334&v=2.20.1

Mapping the Global Landscape and Thematic Evolution of fNIRS-Gait Research: A Systematic Bibliometric and Network Visualization Analysis

👤 Naime Uluğ 📅 2026-10-27 📖 1 min 📝 20 words FNIRS

Summary: CONCLUSIONS: Establishing standardized preprocessing pipelines and short-separation regressions may prove pivotal for transitioning mobile neuroimaging into objective, closed-loop clinical neurorehabilitation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939053/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028002334&v=2.20.1

Self-Selected Versus Assigned Emotion Regulation in Pain Expectancy: Behavioral and fNIRS Evidence

👤 Ping Shi 📅 2026-10-27 📖 1 min 📝 63 words FNIRS

Summary: Background: Expectation is a powerful cognitive modulator of pain, but how different types of emotion regulation approaches interact with expectancy to influence pain processing remains unclear. The present study compared self-selected habitual strategies (cognitive reappraisal or deep breathing) wi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939283/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028002334&v=2.20.1

Comparative Effects of Acute HIIE and MICE on Executive Function, Prefrontal Oxyhaemoglobin Responses, and Related Biomarkers in Sedentary Young Adults: A Randomized Crossover Trial

👤 Xinyue Yang
📅 2026-10-27 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSIONS: Both exercise sessions were followed by faster reaction times on selected tasks. The small sample, multiple analyses, and potential systemic contributions to fNIRS signals preclude firm conclusions about exercise-mode superiority or biological mediation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939307/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028002334&v=2.20.1

Cerebral Hemodynamic Patterns in People with Subjective Cognitive Decline: A Systematic Review Based on Functional Near-Infrared Spectroscopy

👤 Israel Contador
📅 2026-10-27 📖 1 min 📝 62 words FNIRS

Summary: Background/Objectives: Subjective cognitive decline (SCD) has been proposed as an early symptomatic manifestation of dementia/Alzheimer's disease. However, its associated brain patterns are not yet fully characterized. In recent years, functional near-infrared spectroscopy (fNIRS) has emerged as a c...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939380/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028002334&v=2.20.1

Resting-State Functional Brain Connectivity and Neurodevelopmental Profiles in 3-Year-Olds with Congenital Heart Disease

👤 Anne Gallagher
📅 2026-10-27 📖 1 min 📝 67 words FNIRS

Summary: Background/Objectives: Congenital heart disease (CHD) is associated with altered brain development and an increased risk of neurodevelopmental difficulties, particularly in more severe forms of the condition. Although alterations in functional brain connectivity have been reported before and after t...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42940032/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028002334&v=2.20.1

Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan
📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944288/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028002334&v=2.20.1

Multimodal Sensing and Digital Biomarkers in Virtual Reality-Based Cognitive Rehabilitation: Clinical Validation, Adaptive Systems, and Real-World Translation

👤 Raffaele Ferri
📅 2026-10-27 📖 1 min 📝 58 words FNIRS

Summary: Virtual reality (VR) is evolving from a platform for cognitive exercises into an instrumented rehabilitation environment capable of simultaneously delivering treatment and quantifying behavior. This critical narrative review examines how behavioral, ocular, kinematic, physiological, and neurophysiol...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944342/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028002334&v=2.20.1

Unpredictable human screams elicit anticipatory and phasic cortical responses that affect subsequent interpersonal space regulation

Alessandro Grecucci 2026-10-27 1 min 59 words FNIRS

Summary: Unpredictable threats can sustain vigilance states that bias perception, cognition, and behaviour beyond immediate stimulus processing. Human screams, as evolutionarily salient alarm signals, are especially effective in eliciting defensive responses, yet their cortical dynamics and consequences for ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42946138/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261028002334&v=2.20.1

Human Factors in Brain-Computer Interfaces: Boundaries, Key Issues, and Methodological Approaches

Yunfa Fu 2026-10-27 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: The large-scale, real-world deployment of brain-computer interface (BCI) is confronted with human factors bottlenecks, including reliability and usability. While modern human factors engineering has encompassed domains such as cognitive ergonomics, workload, adaptive automation, and human-in-the-loo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939319/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028002332&v=2.20.1

EOG Artifact Removal from EEG Using a Gated WGAN with Multi-Scale and Multi-Domain Feature Extraction

👤 Hao Tang 📅 2026-10-27 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: Background: Electroencephalogram (EEG) signals provide important information for brain-computer interfaces (BCIs) and the diagnosis of neurological disorders but are susceptible to electrooculographic (EOG) artifacts. Traditional methods often rely on multichannel data or relatively long windows, li...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939376/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028002332&v=2.20.1

EEG Features of Cortical Intent for Urination and Defecation in Healthy Aging and Spinal Cord Injury: A Descriptive Feasibility Study for BCI Neurorehabilitation

👤 Jie Jia 📅 2026-10-27 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: Background/Objectives: Systematically elucidating the neural mechanisms of bowel and bladder control is crucial for developing closed-loop brain-computer interface (BCI) neurorehabilitation for patients with spinal cord injury (SCI). This study aims to characterize the spatiotemporal Electroencephal...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939378/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028002332&v=2.20.1

The Development and Evolution of Substrate Materials for Brain-Computer Interfaces

👤 Yen Wei 📅 2026-10-27 📖 1 min 📝 58 words BRAIN COMPUTER INTERFACE

Summary: Brain-Computer Interfaces (BCIs) hold tremendous potential for neurological therapeutics and neuroprosthetics; however, their long-term clinical viability is fundamentally limited by chronic interfacial failure. This failure originates from the profound biomechanical mismatch between synthetic micro...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42942690/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028002332&v=2.20.1

Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan 📅 2026-10-27 📖 1 min 📝 60 words BRAIN COMPUTER INTERFACE

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944288/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028002332&v=2.20.1

Agentic EEG Pipeline Synthesis Using Large Language Models and the Model Context Protocol

Katy Tarrit 2026-10-27 1 min 56 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography analysis typically requires multistage processing pipelines for which configuration depends on recording characteristics and the intended analytical objective. Although automated pipelines improve reproducibility, they generally implement predefined processing sequences. We pr...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944335/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028002332&v=2.20.1

Overt and Covert Speech-State Classification from Tripolar Concentric Ring Electrode EEG: A 4 mm and 10 mm Configuration Study

Shoaib Memon 2026-10-27 1 min 76 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) speech-state classification is a noninvasive method to identify the brain's overt and covert behavior. The EEG achieved through the scalp is contaminated by artifacts and volume conduction and suffers from poor spatial resolution. We present the first application of smal...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944450/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028002332&v=2.20.1

The compositional encoding of hand-eye coordinated movements for single neurons in the posterior parietal cortex

Richard A Andersen
2026-10-27 1 min 77 words BRAIN COMPUTER INTERFACE

Summary: Human posterior parietal cortex (PPC) is known to play a role in hand-eye (H-E) coordination, yet the underlying encoding mechanisms remain uncertain. We recorded 412 single neurons across 11 sessions from motor cortex (MC; n = 251) and PPC (n = 161) in a single human participant performing an H-E c...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945213/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028002332&v=2.20.1

NCEL: Convolution-Enhanced Lightweight Neural Architecture for Robust EEG Signal Decoding in Noninvasive Brain-Computer Interfaces

Liangliang Hu
2026-10-27 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) is a cornerstone of brain-computer interfaces (BCIs) and clinical neuro-diagnostics due to its non-invasiveness and high temporal resolution. However, robust EEG decoding remains challenging, particularly for resource-constrained deployments. While lightweight architectu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945585/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028002332&v=2.20.1

DSGTANet: A semi-supervised dual-stream graph-temporal adaptation network for cross-subject continuous EEG emotion recognition

Wang-Yang Xu 2026-10-27 1 min 61 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: Continuous emotion recognition using electroencephalogram (EEG) is a critical topic in brain-computer inter-faces (BCIs), with applications in mental disorder regulation and human-computer interaction. However, its practical appli-cation is significantly constrained by several key issues,...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945587/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261028002332&v=2.20.1

PyPI Removing Reserver

/u/sadrasabouri 2026-10-28 1 min 147 words REDDIT PYTHON

Summary: <!-- SC_OFF --> <p>It's difficult to tell whether a name is available in PyPI, where searching for package name doesn't guarantee that it's available. Imagine how stressful it is to build a project, setting up its website, repository, and so on, only to find when you’re ready to publi...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xhbtxv/pypi_removing_reserver/

Python jettisons the native JIT, at least for now

👤 /u/nicholashairs 📅 2026-10-28 📖 1 min 📝 105 words REDDIT PYTHON

Summary: <!-- SC_OFF --> High level summary article More detailed article <blockquote> In June, ...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xhadxq/python_jettisons_the_native_jit_at_least_for_now/

Code-execution flaw in UEFI firmware threatens most servers on planet Earth

👤 DamnInteresting 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

Summary: Article URL: https://arstechnica.com/security/2026/10/code-execution-flaw-in-uefi-firmware-threatens-most-servers-on-planet-earth/ Comments URL: <a href="http...

🔗 **Read full article:**
https://arstechnica.com/security/2026/10/code-execution-flaw-in-uefi-firmware-threatens-most-servers-on-planet-earth/

Expanding our partnership with GSA to bring Claude to all levels of government

👤 matthieu_bl 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.anthropic.com/news/expanding-partnership-gsa Comments URL: https://news.ycombinator.com/item?id=50266701 Points: 31 # Co...

🔗 **Read full article:**
https://www.anthropic.com/news/expanding-partnership-gsa

Moxnox 1.0: NT-Like, POSIX-Compatible OS Running Atop Btrfs

👤 SockThief 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/maharmstone/moxnox Comments URL: https://news.ycombinator.com/item?id=50267569 Points: 27 # Comments: 3

🔗 **Read full article:**
https://github.com/maharmstone/moxnox

Git for US Legislation

👤 alansaber 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://lexifina.com/blog/git-for-the-us-code Comments URL: https://news.ycombinator.com/item?id=50268380 Points: 86 # Comments: 40

🔗 **Read full article:**
https://lexifina.com/blog/git-for-the-us-code

New software package: ERPLAB Studio

👤 Steve Luck 📅 2024-06-12 📖 2 min 📝 449 words ERP BOOT CAMP

> **Summary:** We are excited to announce the release of a new EEG/ERP analysis package, ERPLAB Studio. We think it’s a huge improvement over the classic EEGLAB user interface. See our cheesy <a href="https://www.youtube.com/watch?v=lIaKVQ9DD6E">...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/11/erplab-studio

Brain Age Gap Associations with Externalising Behaviours in Youth: A Longitudinal Investigation in the Adolescent Brain Cognitive Development (ABCD) Cohort

👤 Asikin, A., Visontay, R., Devine, E. K., Squeglia, L. M., Byrne, H., Mewton, L.

📅 2026-10-27 📖 1 min

📝 301 words

BIORXIV NEUROSCIENCE

Summary: Background. Early adolescence is a period of heightened risk for the emergence of externalising psychopathology, making it an important window for identifying neurodevelopmental risk markers. Differences between predicted brain age and chronological age - termed the brain age gap - reflect variation...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.761283v1?rss=1

Scalable hardware and software for high-density, multi-regional electrophysiology in non-human primates

👤 Dubey, A., Xu, Q., Khazali, M. F., Choi, J., Wingel, K., Haggerty, J., Stefanowicz, J., Pesaran, B.

📅 2026-10-27 📖 1 min 📝 238 words BIORXIV NEUROSCIENCE

Summary: Understanding how distributed neural circuits generate complex behaviors requires long-duration recordings from populations of individual neurons across multiple brain regions. High-density Neuropixels probes have revolutionized rodent systems neuroscience. Extending these capabilities to non-human ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.761135v1?rss=1

Enhancing working memories using repetitive TMS

👤 Pagnotta, M. F., D'Esposito, M.

📅 2026-10-27 📖 1 min 📝 244 words BIORXIV NEUROSCIENCE

Summary: Working memory, the ability to retain and manipulate information no longer present in the environment, is fundamental for goal-directed behavior (Baddeley, 2012; Baddeley & Hitch, 1974; D'Esposito, 2007; D'Esposito & Postle, 2015). Alpha oscillations (~8-12 Hz) have two apparently contradictory role...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.761198v1?rss=1

Manifolds for the neural representation of 3D spatial experience

Doudlah, R., Xiao, M., Postlewaite, S., Kim, B., Cowan, N., Rosenberg, A.

2026-10-27 1 min

182 words

BIORXIV NEUROSCIENCE

Summary: The precision with which animals navigate and interact with objects depends in part on how their brains represent position and orientation in three dimensions. A central challenge is to explain within a common framework how neural systems ranging from insects to primates represent three-dimensional ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.760836v1?rss=1

Dissociable episodic memory performance and subjective experience across the aphantasia imagery spectrum

Duckett, W., Simons, J.

2026-10-27 1 min 263 words

BIORXIV NEUROSCIENCE

Summary: Mental imagery is often assumed to support episodic memory, yet whether this relationship depends on conscious access to imagery, and how reduced imagery differs from total aphantasia, remains unclear. The present study compared objective episodic memory performance, subjective trial-by-trial vividn...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.761078v1?rss=1

Daylight exposure improves sleep and subsequent sleepiness

Yang, M., van Bergeyk, T., Hut, R. A., Lok, R.

2026-10-27 1 min 261 words

BIORXIV NEUROSCIENCE

Summary: Study Objectives: The causal effects of daylight exposure on sleep remain poorly understood, particularly under real-world conditions. This study aimed to address this gap by examining the effects of an afternoon daylight intervention on subsequent sleep. Methods: Ten healthy participants (4 males; ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.761062v1?rss=1

A vagal-to-sympathetic reflex controls lung vascular leak

Robinson, E., Ibanez, E. M., Santibanez Ocampo, B., Li, J., Ballal, N., Miller, J. M., Shivdikar, K., Umans, B. D., Prescott, S. L.

2026-10-27 1 min 194 words

BIORXIV NEUROSCIENCE

Summary: Dynamic regulation of the vascular barrier is required for an effective inflammatory response, allowing delivery of plasma proteins and immune cells to sites of insult. Although normally protective, sustained leak becomes costly in organs like the lung, where swelling stiffens tissue and impairs bre...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.761052v1?rss=1

Lateral-eyed tree shrews combine panoramic vision with an expanded oculomotor repertoire

Pal, R., Kempkes, E., Aeleti, R., Tubbs, E., Biswas, A., Sedigh-Sarvestani, M.

2026-10-27 1 min

243 words

BIORXIV NEUROSCIENCE

Summary: Mammals differ in how they redirect gaze. Primates shift and hold gaze with rapid, precise eye movements that can be decoupled from the head. In lateral-eyed mammals studied to date, gaze is organized around the head: saccades are coupled to head rotation in freely moving animals and typically accom...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.23.761344v1?rss=1

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt

2025-01-09 1 min 65 words

OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027193345&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39870674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027193345&v=2.20.1

Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027193345&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027193345&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027193345&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41358599/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027193345&v=2.20.1

Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41676644/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027193345&v=2.20.1

Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41678848/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027193345&v=2.20.1

The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41703344/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027193345&v=2.20.1

EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42538927/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027193345&v=2.20.1

Surface-Token-Guided Query Refinement for Weakly Supervised RGB Apple Defect Sensing and Visible Surface-Condition Scoring

👤 Yasenjiang Musha 📅 2026-10-27 📖 1 min 📝 69 words LOW VISION

Summary: Low-cost RGB imaging is attractive for postharvest apple grading, yet an image-level defect label does not reveal where damage occurs or how much of the visible surface it affects. This study examines whether folder-labeled images can support defect recognition, weak localization, binary health asse...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944398/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027193343&v=2.20.1

An Energy-Efficient Edge Vision System for Front-End Quality Monitoring in Sustainable Cold Chains: Coffee Fruit Ripeness Detection and Embedded Validation

👤 Shirong Guo 📅 2026-10-27 📖 1 min 📝 64 words LOW VISION

Summary: Reliable quality monitoring at the front end of sustainable cold chains requires direct observations of product condition in addition to environmental measurements. This study developed an energy-efficient edge vision system for six coffee fruit ripeness and visible-condition classes on an RK3576 ne...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944399/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027193343&v=2.20.1

Text-Guided Feature Completion in the CLIP Embedding Space for Occlusion-Robust Facial Expression Recognition

👤 Minsam Ko 📅 2026-10-27 📖 1 min 📝 73 words LOW VISION

Summary: Camera-based affective sensing degrades sharply when the face is partially occluded by hands, masks, eyeglasses or headwear. Existing occlusion-robust facial expression recognition (FER) methods either discard the information carried by occluded regions through attention re-weighting, hallucinate it...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944415/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027193343&v=2.20.1

Intelligent Sensing and Adaptive Control for Low-Loss Soybean Harvesting: A Review of Multi-Scenario Technologies

👤 Zhong Tang 📅 2026-10-27 📖 1 min 📝 60 words LOW VISION

Summary: Soybean harvesting is strongly affected by low pod position, lodging, pod shattering, moisture variation, and changing material flow, making stable low-loss operation difficult across diverse field conditions. This review synthesizes 179 core publications identified through structured literature sea...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944440/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027193343&v=2.20.1

A 3D Vision-Based Localization Method for Automated Loading of Large-Sized Heat Exchangers

👤 Qing Jiang 📅 2026-10-27 📖 1 min 📝 62 words LOW VISION

Summary: Manual loading of large-sized finned-tube heat exchangers has low productivity and imposes substantial physical demands on workers. In addition, industrial 3D vision systems often produce large regions of missing depth data when imaging highly reflective metallic surfaces, which complicates grasp-po...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944478/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027193343&v=2.20.1

Hybrid Vision Perception for Low-Cost Autonomous Vehicle Navigation

Abdelkader El Kamel 2026-10-27 1 min 69 words LOW VISION

Summary: High-performance perception systems for autonomous vehicles (AVs) commonly use LiDAR, RADAR, and multiple cameras. These sensors provide useful information, but their cost and computational requirements can be difficult to justify for small prototypes and low-cost platforms. This work therefore uses...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944489/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027193343&v=2.20.1

Fidelity-oriented pelvic CBCT-to-CT synthesis using an anatomical information-attention diffusion model

Kunyuan Xu 2026-10-27 1 min 63 words LOW VISION

Summary: Cone-beam computed tomography (CBCT) is widely used in image-guided radiotherapy and adaptive radiotherapy, but its direct quantitative use is limited by scatter artifacts, low soft-tissue contrast, and inaccurate Hounsfield unit (HU) values. Diffusion models provide stable image generation capabili...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944823/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027193343&v=2.20.1

Poros: Perception-Enabled Interactive Assistance for Appliance Use by Blind and Low Vision Users

Wei Liang 2026-10-27 1 min 62 words LOW VISION

Summary: Modern home appliances pose significant accessibility barriers for individuals with visual impairments, driven by increasing functional complexity and minimalist design trends. Drawing on formative interviews with blind and low vision (BLV) participants, we propose Poros, a novel system providing pe...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945092/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027193343&v=2.20.1

DELUGE: Decomposed Entropy-coded Live Unstructured Geometry Exchange for Real-time Particle Streaming

Takefumi Hiraki 2026-10-27 1 min 74 words LOW VISION

Summary: Particle-based physics simulations, including fluids, smoke, and granular media, are fundamental to visual realism in immersive VR and AR. With the growing adoption of social VR and digital twins, demand is increasing for shared experiences in which multiple users interact with the same simulation i...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945105/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027193343&v=2.20.1

A multi-attention mechanism-improved ResNet18 approach for lithology identification and classification

👤 Ruifeng Li 📅 2026-10-27 📖 1 min 📝 61 words LOW VISION

Summary: Lithology identification is vital for improving geological exploration accuracy, optimizing petroleum exploration schemes, and ensuring construction engineering safety. Traditional identification methods suffer from low efficiency, high cost, and heavy dependence on domain expertise. This study prop...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945382/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027193343&v=2.20.1

Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux 📅 2026-07-18 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42470115/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027193340&v=2.20.1

A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42532383/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027193340&v=2.20.1

Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42590984/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027193340&v=2.20.1

Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42596634/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027193340&v=2.20.1

Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42614895/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027193340&v=2.20.1

A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42787568/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027193340&v=2.20.1

Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42788316/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027193340&v=2.20.1

Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42817759/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027193340&v=2.20.1

Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

Zhifei Xie
2026-10-24 1 min 66 words TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42923236/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027193340&v=2.20.1

Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42926384/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027193340&v=2.20.1

In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42694607/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027193338&v=2.20.1

Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan
📅 2026-09-09 📖 1 min 📝 57 words

BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42711906/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027193338&v=2.20.1

Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42768901/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027193338&v=2.20.1

Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42775591/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027193338&v=2.20.1

Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42789611/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027193338&v=2.20.1

Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42845092/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027193338&v=2.20.1

Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42848198/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027193338&v=2.20.1

Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42881664/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027193338&v=2.20.1

Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885551/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027193338&v=2.20.1

An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885828/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027193338&v=2.20.1

Multimodal premotor stimulation during action observation transiently boosts corticospinal excitability and motor performance

Ishan Mahajan 2026-10-26 1 min 2 words TDCS TACS TRNS

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42929054/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027193336&v=2.20.1

Combining Cognitive Training and Brain Stimulation: A Novel Approach to Address Disordered Eating and Antipsychotic-Induced Weight Gain in People With Severe Mental Illness

👤 Ulrike Schmidt
📅 2026-10-26 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: The initiation of antipsychotic therapy is a critical clinical window for early intervention to mitigate the long-term metabolic and psychiatric consequences of antipsychotic-induced weight gain. Combining ABMT + tDCS is acceptable and may help reduce binge eating in individuals on antip...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42929065/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027193336&v=2.20.1

High-Intensity Transcranial Alternating Current Stimulation for Post-Stroke Cognitive Impairment: A Triple-Blind Randomized Trial

Xiquan Hu 2026-10-26 1 min 60 words TDCS TACS TRNS

Summary: CONCLUSION: Hi-tACS improved global cognition and working memory in patients with PSCI and was generally well tolerated. Exploratory fNIRS findings suggested possible accompanying changes in cortical activation and network organization but did not establish a neurophysiological mechanism. Importantl...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42933772/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027193336&v=2.20.1

Comparative analysis of mitochondrial genomes in Zantedeschia parental lines and their hybrid progeny

Hongzhi Wu 2026-10-27 1 min 31 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study expands current knowledge of mt genome diversity in the Araceae and provides valuable genomic resources for future studies of Zantedeschia mitogenome evolution, genetic resource conservation, and molecular breeding.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42934513/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027193336&v=2.20.1

Combined Transcranial Direct Current Stimulation and Transcutaneous Cervical Vagus Nerve Stimulation for Anxiety and Depression: An Exploratory Pilot Study

👤 Maria-Manuela Apostol
📅 2026-10-27 📖 1 min 📝 60 words

TDCS TACS TRNS

Summary: Anxiety and depressive disorders are among the leading causes of disability worldwide, yet current treatments remain limited by modest efficacy and restricted accessibility; non-invasive neuromodulation techniques such as transcranial direct current stimulation (tDCS) and transcutaneous cervical vag...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939258/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027193336&v=2.20.1

Non-Invasive Brain Stimulation Techniques and Effects on Neuromodulation and Athletic Performance in Combat Sports: A Systematic Review

Bianca Miarka 2026-10-27 1 min 59 words TDCS TACS TRNS

Summary: CONCLUSIONS: tDCS may influence selected cognitive, neuromuscular, and fatigue-related outcomes in combat sport athletes; however, the direction and magnitude of these effects are inconsistent across studies. Given the substantial methodological heterogeneity and the low-to-very-low certainty of evi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939305/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027193336&v=2.20.1

Non-Invasive Brain Stimulation for Central Neuropathic Pain: An Exploratory Randomized Sham-Controlled Crossover Trial

Eunhee Sohn 2026-10-27 1 min 64 words TDCS TACS TRNS

Summary: Background/Objectives: Central neuropathic pain (CNP) is difficult to manage, and comparative evidence for non-invasive brain stimulation is limited. We explored the short-term efficacy and safety of repetitive transcranial magnetic stimulation (rTMS) and transcranial direct current stimulation (tDC...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939331/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027193336&v=2.20.1

Noninvasive Neuromodulation for Cognitive Dysfunction in Parkinson's Disease: A Scoping Review

Reginald B O'Hara 2026-10-27 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: Evidence specific to PD-MCI is limited and mixed. Several studies pairing stimulation with cognitive training reported favorable outcomes, but this pattern was inconsistent and does not establish superiority over stimulation alone. No VNS trial enrolled participants with confirmed PD-MC...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939339/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027193336&v=2.20.1

Comparative Plastid Phylogenomics of 29 Newly Sequenced Bougainvillea Cultivars Resolves the Phylogenetic Placement of Double-Bracted Cultivars

Huiqi Zhao 2026-10-27 1 min 59 words TDCS TACS TRNS

Summary: Bougainvillea (Nyctaginaceae) cultivars are horticulturally prized worldwide for their vivid bracts; however, repeated interspecific hybridization and somatic mutations have obscured their maternal origins, particularly for double-bracted forms. Here, we sequenced and assembled the complete plastome...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939474/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027193336&v=2.20.1

Anodal Transcranial Direct Current Stimulation Reduces Neuroinflammation and Cholinergic Dysregulation After Young Adult Traumatic Brain Injury

👤 Guven Akcay
📅 2026-10-27 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Traumatic brain injury (TBI) during childhood and adolescence can result in persistent motor and cognitive dysfunction due to secondary injury cascades involving neuroinflammation and disruption of neurotransmitter homeostasis. Although transcranial Direct Current Stimulation (tDCS) is a promising n...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939682/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027193336&v=2.20.1

Social relationship is associated with distinct patterns of inter-brain synchrony and directed connectivity during interpersonal coordination

Xinhong Jin 2026-10-27 1 min 59 words FNIRS

Summary: How social relationship and effector pairing relate to behavioral coordination and inter-brain interaction during joint timing remains unclear. Using functional near-infrared spectroscopy hyperscanning during a dyadic synchronization-continuation task, we examined 21 romantic couples and 23 stranger...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42937180/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027193333&v=2.20.1

Neural-Discursive Dynamics in Collaborative Spatial Problem Solving: Evidence from fNIRS Hyperscanning and Accountable Talk

👤 Xingguang Guo
📅 2026-10-27 📖 1 min 📝 65 words FNIRS

Summary: High-quality collaboration cannot be inferred simply from how much students talk or how strongly their brain activity is synchronized during joint work. This study examined relations among accountable talk, inter-brain synchrony (IBS), and group task performance in fifth-grade students' collaborativ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42938505/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027193333&v=2.20.1

Probing Executive Function in the Toddler Data Desert: A Touchscreen fNIRS Study of Working Memory and ADHD-Related Symptoms

👤 Caitie Busch 📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Attention-deficit/hyperactivity disorder (ADHD) is associated with early differences in attentional control and working memory (WM), yet relatively few paradigms are developmentally feasible across the toddler-to-preschool period while also distinguishing spatial from featural information use. Addre...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42938612/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027193333&v=2.20.1

Mapping the Global Landscape and Thematic Evolution of fNIRS-Gait Research: A Systematic Bibliometric and Network Visualization Analysis

👤 Naime Uluğ 📅 2026-10-27 📖 1 min 📝 20 words FNIRS

Summary: CONCLUSIONS: Establishing standardized preprocessing pipelines and short-separation regressions may prove pivotal for transitioning mobile neuroimaging into objective, closed-loop clinical neurorehabilitation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939053/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027193333&v=2.20.1

Self-Selected Versus Assigned Emotion Regulation in Pain Expectancy: Behavioral and fNIRS Evidence

👤 Ping Shi 📅 2026-10-27 📖 1 min 📝 63 words FNIRS

Summary: Background: Expectation is a powerful cognitive modulator of pain, but how different types of emotion regulation approaches interact with expectancy to influence pain processing remains unclear. The present study compared self-selected habitual strategies (cognitive reappraisal or deep breathing) wi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939283/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027193333&v=2.20.1

Comparative Effects of Acute HIIE and MICE on Executive Function, Prefrontal Oxyhaemoglobin Responses, and Related Biomarkers in Sedentary Young Adults: A Randomized Crossover Trial

👤 Xinyue Yang
📅 2026-10-27 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSIONS: Both exercise sessions were followed by faster reaction times on selected tasks. The small sample, multiple analyses, and potential systemic contributions to fNIRS signals preclude firm conclusions about exercise-mode superiority or biological mediation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939307/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027193333&v=2.20.1

Cerebral Hemodynamic Patterns in People with Subjective Cognitive Decline: A Systematic Review Based on Functional Near-Infrared Spectroscopy

👤 Israel Contador
📅 2026-10-27 📖 1 min 📝 62 words FNIRS

Summary: Background/Objectives: Subjective cognitive decline (SCD) has been proposed as an early symptomatic manifestation of dementia/Alzheimer's disease. However, its associated brain patterns are not yet fully characterized. In recent years, functional near-infrared spectroscopy (fNIRS) has emerged as a c...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939380/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027193333&v=2.20.1

Resting-State Functional Brain Connectivity and Neurodevelopmental Profiles in 3-Year-Olds with Congenital Heart Disease

👤 Anne Gallagher
📅 2026-10-27 📖 1 min 📝 67 words FNIRS

Summary: Background/Objectives: Congenital heart disease (CHD) is associated with altered brain development and an increased risk of neurodevelopmental difficulties, particularly in more severe forms of the condition. Although alterations in functional brain connectivity have been reported before and after t...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42940032/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027193333&v=2.20.1

Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan
📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944288/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027193333&v=2.20.1

Multimodal Sensing and Digital Biomarkers in Virtual Reality-Based Cognitive Rehabilitation: Clinical Validation, Adaptive Systems, and Real-World Translation

👤 Raffaele Ferri 📅 2026-10-27 📖 1 min 📝 58 words FNIRS

Summary: Virtual reality (VR) is evolving from a platform for cognitive exercises into an instrumented rehabilitation environment capable of simultaneously delivering treatment and quantifying behavior. This critical narrative review examines how behavioral, ocular, kinematic, physiological, and neurophysiol...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944342/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027193333&v=2.20.1

Locomotor-State Dependence of Neuromodulation During Avian Free Flight

👤 Zhigang Shang 📅 2026-10-27 📖 1 min 📝 63 words BRAIN COMPUTER INTERFACE

Summary: Complex avian locomotor behaviors depend on the coordinated operation of neural processes involved in sensory perception, decision-making, and motor execution. Exogenous neural stimulation can alter motor output by intervening in neural processes, but its effects may depend on the current locomotor ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939144/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027193331&v=2.20.1

STDR-Net: A Spatio-Temporal Dependency Refinement Network for Motor Imagery EEG Decoding

Yongkang Yu 2026-10-27 1 min 56 words BRAIN COMPUTER INTERFACE

Summary: Background/Objectives: Motor imagery electroencephalography (MI-EEG) contains weak, non-stationary patterns that evolve across channels and time. Effective decoding requires local sensorimotor structure, trial-wide context, and ordered temporal evolution to be represented without discarding informat...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939304/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027193331&v=2.20.1

Human Factors in Brain-Computer Interfaces: Boundaries, Key Issues, and Methodological Approaches

Yunfa Fu 2026-10-27 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: The large-scale, real-world deployment of brain-computer interface (BCI) is confronted with human factors bottlenecks, including reliability and usability. While modern human factors engineering has encompassed domains such as cognitive ergonomics, workload, adaptive automation, and human-in-the-loo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939319/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027193331&v=2.20.1

EOG Artifact Removal from EEG Using a Gated WGAN with Multi-Scale and Multi-Domain Feature Extraction

👤 Hao Tang 📅 2026-10-27 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: Background: Electroencephalogram (EEG) signals provide important information for brain-computer interfaces (BCIs) and the diagnosis of neurological disorders but are susceptible to electrooculographic (EOG) artifacts. Traditional methods often rely on multichannel data or relatively long windows, li...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939376/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027193331&v=2.20.1

EEG Features of Cortical Intent for Urination and Defecation in Healthy Aging and Spinal Cord Injury: A Descriptive Feasibility Study for BCI Neurorehabilitation

👤 Jie Jia 📅 2026-10-27 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: Background/Objectives: Systematically elucidating the neural mechanisms of bowel and bladder control is crucial for developing closed-loop brain-computer interface (BCI) neurorehabilitation for patients with spinal cord injury (SCI). This study aims to characterize the spatiotemporal Electroencephal...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939378/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027193331&v=2.20.1

The Development and Evolution of Substrate Materials for Brain-Computer Interfaces

👤 Yen Wei 📅 2026-10-27 📖 1 min 📝 58 words BRAIN COMPUTER INTERFACE

Summary: Brain-Computer Interfaces (BCIs) hold tremendous potential for neurological therapeutics and neuroprosthetics; however, their long-term clinical viability is fundamentally limited by chronic interfacial failure. This failure originates from the profound biomechanical mismatch between synthetic micro...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42942690/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027193331&v=2.20.1

Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan 📅 2026-10-27 📖 1 min 📝 60 words BRAIN COMPUTER INTERFACE

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944288/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027193331&v=2.20.1

Agentic EEG Pipeline Synthesis Using Large Language Models and the Model Context Protocol

Katy Tarrit 2026-10-27 1 min 56 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography analysis typically requires multistage processing pipelines for which configuration depends on recording characteristics and the intended analytical objective. Although automated pipelines improve reproducibility, they generally implement predefined processing sequences. We pr...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944335/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027193331&v=2.20.1

Overt and Covert Speech-State Classification from Tripolar Concentric Ring Electrode EEG: A 4 mm and 10 mm Configuration Study

Shoaib Memon 2026-10-27 1 min 76 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) speech-state classification is a noninvasive method to identify the brain's overt and covert behavior. The EEG achieved through the scalp is contaminated by artifacts and volume conduction and suffers from poor spatial resolution. We present the first application of smal...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944450/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027193331&v=2.20.1

The compositional encoding of hand-eye coordinated movements for single neurons in the posterior parietal cortex

👤 Richard A Andersen 📅 2026-10-27 📖 1 min 📝 77 words BRAIN COMPUTER INTERFACE

Summary: Human posterior parietal cortex (PPC) is known to play a role in hand-eye (H-E) coordination, yet the underlying encoding mechanisms remain uncertain. We recorded 412 single neurons across 11 sessions from motor cortex (MC; n = 251) and PPC (n = 161) in a single human participant performing an H-E c...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42945213/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027193331&v=2.20.1

Survival LLM: AI assistant and emergency library that runs on your own computer

👤 Luc 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

Summary: Article URL: https://www.brangus.io/survival-llm Comments URL: https://news.ycombinator.com/item?id=50264853 Points: 126 # Comments: 80

🔗 **Read full article:**
https://www.brangus.io/survival-llm

Tailscale SSH: Investigation and remediation update

👤 figmert 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://tailscale.com/ssh-investigation-update Comments URL: https://news.ycombinator.com/item?id=50264857 Points: 146 # Comments: 89

🔗 **Read full article:**
https://tailscale.com/ssh-investigation-update

Four Months of Zig

👤 ibobev 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://gracefulliberty.com/articles/four-months-of-zig/ Comments URL: https://news.ycombinator.com/item?id=50265128 Points: 48 # Co...

🔗 **Read full article:**
https://gracefulliberty.com/articles/four-months-of-zig/

Hex Raises $100M

👤 tomwphillips 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://hex.tech/blog/hex-new-era/ Comments URL: https://news.ycombinator.com/item?id=50265413 Points: 21 # Comments: 10

🔗 **Read full article:**
https://hex.tech/blog/hex-new-era/

Raspberry Pi Zero 3W and Pi 4 Model A spotted in device trees

👤 sthlmb 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://bret.dk/raspberry-pi-zero-3-w-pi-4-model-a/ Comments URL: https://news.ycombinator.com/item?id=50265594 Points: 61 # Comments: 18...

🔗 **Read full article:**
https://bret.dk/raspberry-pi-zero-3-w-pi-4-model-a/

Early progress on accelerating drug development with Claude

👤 dataking 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.anthropic.com/research/early-progress-drug-development Comments URL: https://news.ycombinator.com/item?id=50265609 Po...

🔗 **Read full article:**
https://www.anthropic.com/research/early-progress-drug-development

One year of Dioxus fullstack in production: the good, the painful

👤 hauju 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://oxidt.com/blog/one-year-of-dioxus-fullstack-in-production-the-good-the-painful Comments URL: <a href="https://news.ycombinator.com/item?id=50265960">https://news.ycombi...

🔗 **Read full article:**
https://oxidt.com/blog/one-year-of-dioxus-fullstack-in-production-the-good-the-painful

Reviewers rarely talk about reliability. Owners bring it up twice as often

👤 tsxng 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.bettascore.ai/research/reliability-reviewers-vs-owners Comments URL: https://news.ycombinator.com/item?id=50266780 Po...

🔗 **Read full article:**
https://www.bettascore.ai/research/reliability-reviewers-vs-owners

A Hierarchical-Dynamic integration model for the formation of interpersonal neural synchrony during spoken communication

📖 1 min 📝 13 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Yating Bai, Qingfang Zhang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004567?dgcid=rss_sd_all

Neurotransmitter receptor architecture links local neurochemistry to semantic memory

Jung, J., Gim, J.-A., Lambon Ralph, M.
2026-10-27 1 min 244 words

BIORXIV NEUROSCIENCE

Summary: A fundamental challenge in neuroscience is understanding how the molecular and neurochemical architecture of the human cortex gives rise to sophisticated higher cognitive functions. The anterior temporal lobe (ATL) is a key hub for semantic memory, and its underlying GABAergic and glutamatergic neur...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.762019v1?rss=1

Heterogeneous plasticity rules generate diverse structures in recurrent networks

Cusseddu, C., Festa, D., Miehl, C., Gjorgjieva, J.
2026-10-27 1 min 154 words

BIORXIV NEUROSCIENCE

Summary: How do local changes at individual synapses give rise to structured recurrent connectivity? Experiments have revealed diverse forms of spike-timing-dependent plasticity (STDP) across the brain, making it difficult to predict which circuit structures different rules will generate. We developed an ana...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.760407v1?rss=1

Cerebello-basal ganglia networks and behavior across adulthood and aging

Delaney, C., Herrejon, I., Magalhaes, T. N. C., Jackson, T. B., Hicks, T. H., Bernard, J. A.

2026-10-27 1 min 293 words BIORXIV NEUROSCIENCE

Summary: The cerebellum contributes to aging, cognition, and reward related processing, but has been relatively understudied in these domains. The cerebello-basal ganglia (CB-BG) circuit may be key to better understanding how the brain ages. Prior work has implicated this circuit in aging in a small sample, ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.760492v1?rss=1

Ms4a4a loss preferentially remodels microglia with limited, compartment-specific effects across central and peripheral immune compartments in 5xFAD mice

Danhash, E., Verbeck, A., Iyer, A., Karch, C. M.

2026-10-27 1 min 319 words BIORXIV NEUROSCIENCE

Summary: Background: Genome-wide association studies have identified the MS4A locus as a major modifier of Alzheimers disease (AD) risk and resilience. MS4A4A is enriched in microglia, and we previously demonstrated that loss of Ms4a4a reduces amyloid plaque pathology and attenuates amyloid-associated tau sp...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.762005v1?rss=1

Information coding strategies in biological reservoir computing for multiclass time-series classification

👤 Nishi, Y., Yamamoto, H., Katori, Y., Sato, S.
📅 2026-10-27 📖 1 min 📝 139 words

BIORXIV NEUROSCIENCE

Summary: Physical reservoir computing (RC) using biological neural networks (BNNs) offers a framework for probing how high-dimensional dynamics arising from multicellular systems support complex information processing in the brain. However, how electrical inputs and extracellular readouts should be designed ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.755221v1?rss=1

Focal neuronal loss and cell replacement in a connectome model of fly steering

👤 Habib, P., Petr, M., Rust, R.
📅 2026-10-27 📖 1 min 📝 203 words

BIORXIV NEUROSCIENCE

Summary: Focal neuronal loss after stroke, traumatic brain injury or spinal cord injury can disrupt circuits and result in persistent motor deficits. Which cells and connections must be restored to recover a specific behavior is difficult to establish experimentally. Here we use a connectome-constrained spik...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.762043v1?rss=1

A canonical input to the head direction compass carries predominantly eye movement signals

Stott, J. J., Chang, H. H. V., Bao, C., Zhang, K., Cullen, K. E., van der Meer, M. A.

2026-10-27

1 min

171 words

BIORXIV NEUROSCIENCE

Summary: The mammalian head direction (HD) system is thought to update its internal compass using angular head velocity (AHV) signals relayed from the brainstem. We tested this canonical interpretation by recording from the nucleus prepositus hypoglossi (NPH), a proposed AHV relay known to be important for a...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.26.761874v1?rss=1

High-content imaging of live microglia-containing Alzheimer's disease brain organoids

Quang, D., Summers, R., Dooling, B. R., Esquer, H., Zhou, Q., LaBarbera, D. V., Vergara, M. N., Chial, H. J., Potter, H., Johnson, N. R.

2026-10-27

1 min

148 words

BIORXIV NEUROSCIENCE

Summary: Alzheimer's disease (AD) is the most common progressive neurodegenerative disease for which there is currently no safe and effective treatment. AD is characterized by the pathological aggregation of amyloid-{beta} (A{beta}) in the brain resulting in neuronal dysfunction/death and cognitive decline. ...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.26.761970v1?rss=1

Scalp EEG detects motor sequence replay signatures during wakeful rest

Dash, D., Sugata, H.
2026-10-27 · 1 min · 146 words · BIORXIV NEUROSCIENCE

Summary: Brief wakeful rest may support motor-memory processing through replay of recently practiced neural sequences, but human motor replay has been demonstrated mainly with invasive recordings or magnetoencephalography. Here we tested whether ordered motor-sequence replay can be detected from scalp EEG ac...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.760953v1?rss=1

Auditory tracking magnitude, temporal alignment, and anatomical distribution during wakefulness and propofol-based anesthesia in human sEEG

Zhu, H., Yao, C., Han, Z., Zhou, W., Wang, Y., Li, S., Wong, P. C. M., Tian, X., Teng, X.
2026-10-27 · 1 min · 155 words · BIORXIV NEUROSCIENCE

Summary: General anesthesia attenuates auditory responses, but response magnitude alone does not describe how residual activity aligns with sound over time or varies across the brain. We jointly characterized these aspects of auditory tracking using stereoelectroencephalography (sEEG) in 13 patients with dru...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.23.761298v1?rss=1

Feasibility assessment of novel binocular eye alignment monitor device for intermittent exotropia

📅 2026-10-27 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-76743-y

[18F]FDG brain PET-MR versus PET/CT in a clinical cohort: intra-subject quantitative comparisons

📅 2026-10-27 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-72030-y

A NEST-based framework unlocks massively parallel simulation of networks of multi-compartment neurons with customizable subcellular dynamics

📅 2026-10-27 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11157-0

Making ‘mind pictures’ takes mental effort

📅 2026-10-27 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/d41586-026-03276-9

White matter fiber bundles enable robust conduction and time-critical signaling despite axon caliber variation

📅 2026-10-27 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11206-8

Dynorphin counteracts excitatory co-transmission in a lateral hypothalamus–locus coeruleus circuit underlying stress-induced anxiety

📅 2026-10-27 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-78989-6

Ventral CA1 perineuronal nets modulate morphine-associated contextual memory via vCA1–NAcC synaptic remodeling in male mice

📅 2026-10-27 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-79300-3

Behavioral genomics identifies Creld1 as a cross-modal regulator of somatosensation

👤 Bailong Xiao 📅 2026-10-22 📖 1 min 📝 42 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 22 October 2026; doi:10.1038/s41593-026-02484-w Jiang et al. developed a robust postnatal knockout platform for somatosensory genomics and behavioral screening, identifying Creld1 as a cross...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02484-w

Network geometry shapes the transformation of task representations across the human cortex

Michael W. Cole 2026-10-22 1 min 40 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 22 October 2026; doi:10.1038/s41593-026-02486-8 The dimensionality of intrinsic brain connectivity systematically shapes how task representations compress and expand across the cortex, and a...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02486-8

Different modes of stimulation at the Neiguan acupoint (PC6) regulate different EEG microstates under VR immersion conditions

Guangjun Wang 2026-10-22 1 min 302 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAcupuncture at the Neiguan acupoint (PC6) alleviates postoperative and chemotherapy-induced nausea and vomiting. Nevertheless, the central regulatory effects of PC6 stimulation on virtual reality motion sickness (VRMS), a form of physiological vertigo triggered by visual-vestibular senso...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1938329

Retraction: Intervention effect of rehabilitation robotic bed under machine learning combined with intensive motor training on stroke patients with hemiplegia

📅 2026-10-14 📖 1 min 📝 0 words FRONTIERS NEUROROBOTICS

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.2023207

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027155202&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39870674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027155202&v=2.20.1

Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027155202&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027155202&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027155202&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41358599/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027155202&v=2.20.1

Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41676644/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027155202&v=2.20.1

Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41678848/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027155202&v=2.20.1

The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41703344/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027155202&v=2.20.1

EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

> **Summary:** Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42538927/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027155202&v=2.20.1

Safety and Efficacy of a Single Injection of Liposomal Latanoprost in Open-Angle and Angle-Closure Eyes

👤 Tina T Wong 📅 2026-10-27 📖 1 min 📝 67 words LOW VISION

> **Summary:** Background/Objectives: Glaucoma is a leading cause of irreversible global vision loss, with intraocular pressure (IOP) as the sole modifiable risk factor to stall progression. First-line treatment in the form of eye drops suffers from poor bioavailability and low patient adherence, necessitating sus...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42942755/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027155200&v=2.20.1

A Lightweight Real-Time Vision Framework for Low-Disturbance Veterinary Management and Behavioral Welfare Assessment in Captive Forest Musk Deer

👤 Ferrante Neri
📅 2026-10-27 📖 1 min 📝 61 words LOW VISION

Summary: Non-contact behavioral monitoring is important for low-disturbance husbandry of captive forest musk deer, but variable illumination, shadows, partial occlusion, and low target-background contrast can reduce recognition reliability. This study developed HGS-YOLO26n, a lightweight real-time detector f...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944102/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027155200&v=2.20.1

Line-Scan Defect Detection for Printed Laminated Films Using Multi-Template Matching and Residual-Driven Column-Band Compensation

Wen Ren 2026-10-27 1 min 58 words LOW VISION

Summary: Machine vision enables efficient, non-contact industrial quality inspection. Online inspection of printed laminated films is challenging because complex graphics and transport-induced stretching cause appearance variations and spatially nonuniform mismatches. Global registration cannot fully correct...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944236/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027155200&v=2.20.1

A Low-Cost Portable Machine Vision System for Non-Destructive Density Estimation of Dunaliella sp. Cultures

Han Wang 2026-10-27 1 min 61 words LOW VISION

Summary: Routine microalgal cell counting requires sampling and microscopy. We developed a low-cost portable machine-vision system (LPMVS) for non-destructive density estimation of visibly green Dunaliella sp. cultures. The compact benchtop LPMVS combines motorized flask loading, a light-isolated dark box wi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944350/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027155200&v=2.20.1

Surface-Token-Guided Query Refinement for Weakly Supervised RGB Apple Defect Sensing and Visible Surface-Condition Scoring

👤 Yasenjiang Musha
📅 2026-10-27 📖 1 min 📝 69 words

LOW VISION

Summary: Low-cost RGB imaging is attractive for postharvest apple grading, yet an image-level defect label does not reveal where damage occurs or how much of the visible surface it affects. This study examines whether folder-labeled images can support defect recognition, weak localization, binary health asse...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944398/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027155200&v=2.20.1

An Energy-Efficient Edge Vision System for Front-End Quality Monitoring in Sustainable Cold Chains: Coffee Fruit Ripeness Detection and Embedded Validation

👤 Shirong Guo 📅 2026-10-27 📖 1 min 📝 64 words LOW VISION

Summary: Reliable quality monitoring at the front end of sustainable cold chains requires direct observations of product condition in addition to environmental measurements. This study developed an energy-efficient edge vision system for six coffee fruit ripeness and visible-condition classes on an RK3576 ne...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944399/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027155200&v=2.20.1

Text-Guided Feature Completion in the CLIP Embedding Space for Occlusion-Robust Facial Expression Recognition

👤 Minsam Ko 📅 2026-10-27 📖 1 min 📝 73 words LOW VISION

Summary: Camera-based affective sensing degrades sharply when the face is partially occluded by hands, masks, eyeglasses or headwear. Existing occlusion-robust facial expression recognition (FER) methods either discard the information carried by occluded regions through attention re-weighting, hallucinate it...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944415/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027155200&v=2.20.1

Intelligent Sensing and Adaptive Control for Low-Loss Soybean Harvesting: A Review of Multi-Scenario Technologies

👤 Zhong Tang 📅 2026-10-27 📖 1 min 📝 60 words LOW VISION

Summary: Soybean harvesting is strongly affected by low pod position, lodging, pod shattering, moisture variation, and changing material flow, making stable low-loss operation difficult across diverse field conditions. This review synthesizes 179 core publications identified through structured literature sea...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944440/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027155200&v=2.20.1

A 3D Vision-Based Localization Method for Automated Loading of Large-Sized Heat Exchangers

👤 Qing Jiang 📅 2026-10-27 📖 1 min 📝 62 words LOW VISION

Summary: Manual loading of large-sized finned-tube heat exchangers has low productivity and imposes substantial physical demands on workers. In addition, industrial 3D vision systems often produce large regions of missing depth data when imaging highly reflective metallic surfaces, which complicates grasp-po...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944478/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027155200&v=2.20.1

Hybrid Vision Perception for Low-Cost Autonomous Vehicle Navigation

Abdelkader El Kamel 2026-10-27 1 min 69 words LOW VISION

Summary: High-performance perception systems for autonomous vehicles (AVs) commonly use LiDAR, RADAR, and multiple cameras. These sensors provide useful information, but their cost and computational requirements can be difficult to justify for small prototypes and low-cost platforms. This work therefore uses...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944489/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027155200&v=2.20.1

Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42470115/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027155158&v=2.20.1

A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42532383/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027155158&v=2.20.1

Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42590984/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027155158&v=2.20.1

Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42596634/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027155158&v=2.20.1

Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42614895/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027155158&v=2.20.1

A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42787568/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027155158&v=2.20.1

Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42788316/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027155158&v=2.20.1

Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42817759/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027155158&v=2.20.1

Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

Zhifei Xie
2026-10-24 1 min 66 words TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42923236/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027155158&v=2.20.1

Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42926384/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027155158&v=2.20.1

In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42694607/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027155155&v=2.20.1

Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan
📅 2026-09-09 📖 1 min 📝 57 words

BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42711906/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027155155&v=2.20.1

Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42768901/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027155155&v=2.20.1

Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42775591/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027155155&v=2.20.1

Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42789611/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027155155&v=2.20.1

Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42845092/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027155155&v=2.20.1

Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42848198/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027155155&v=2.20.1

Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42881664/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027155155&v=2.20.1

Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885551/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027155155&v=2.20.1

An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885828/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027155155&v=2.20.1

Multimodal premotor stimulation during action observation transiently boosts corticospinal excitability and motor performance

👤 Ishan Mahajan 📅 2026-10-26 📖 1 min 📝 2 words TDCS TACS TRNS

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42929054/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027155153&v=2.20.1

Combining Cognitive Training and Brain Stimulation: A Novel Approach to Address Disordered Eating and Antipsychotic-Induced Weight Gain in People With Severe Mental Illness

👤 Ulrike Schmidt
📅 2026-10-26 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: The initiation of antipsychotic therapy is a critical clinical window for early intervention to mitigate the long-term metabolic and psychiatric consequences of antipsychotic-induced weight gain. Combining ABMT + tDCS is acceptable and may help reduce binge eating in individuals on antip...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42929065/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027155153&v=2.20.1

High-Intensity Transcranial Alternating Current Stimulation for Post-Stroke Cognitive Impairment: A Triple-Blind Randomized Trial

Xiquan Hu 2026-10-26 1 min 60 words TDCS TACS TRNS

Summary: CONCLUSION: Hi-tACS improved global cognition and working memory in patients with PSCI and was generally well tolerated. Exploratory fNIRS findings suggested possible accompanying changes in cortical activation and network organization but did not establish a neurophysiological mechanism. Importantl...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42933772/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027155153&v=2.20.1

Comparative analysis of mitochondrial genomes in Zantedeschia parental lines and their hybrid progeny

Hongzhi Wu 2026-10-27 1 min 31 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study expands current knowledge of mt genome diversity in the Araceae and provides valuable genomic resources for future studies of Zantedeschia mitogenome evolution, genetic resource conservation, and molecular breeding.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42934513/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027155153&v=2.20.1

Combined Transcranial Direct Current Stimulation and Transcutaneous Cervical Vagus Nerve Stimulation for Anxiety and Depression: An Exploratory Pilot Study

👤 Maria-Manuela Apostol
📅 2026-10-27 📖 1 min 📝 60 words

TDCS TACS TRNS

Summary: Anxiety and depressive disorders are among the leading causes of disability worldwide, yet current treatments remain limited by modest efficacy and restricted accessibility; non-invasive neuromodulation techniques such as transcranial direct current stimulation (tDCS) and transcutaneous cervical vag...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939258/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027155153&v=2.20.1

Non-Invasive Brain Stimulation Techniques and Effects on Neuromodulation and Athletic Performance in Combat Sports: A Systematic Review

👤 Bianca Miarka 📅 2026-10-27 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSIONS: tDCS may influence selected cognitive, neuromuscular, and fatigue-related outcomes in combat sport athletes; however, the direction and magnitude of these effects are inconsistent across studies. Given the substantial methodological heterogeneity and the low-to-very-low certainty of evi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939305/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027155153&v=2.20.1

Non-Invasive Brain Stimulation for Central Neuropathic Pain: An Exploratory Randomized Sham-Controlled Crossover Trial

👤 Eunhee Sohn 📅 2026-10-27 📖 1 min 📝 64 words TDCS TACS TRNS

Summary: Background/Objectives: Central neuropathic pain (CNP) is difficult to manage, and comparative evidence for non-invasive brain stimulation is limited. We explored the short-term efficacy and safety of repetitive transcranial magnetic stimulation (rTMS) and transcranial direct current stimulation (tDC...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939331/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027155153&v=2.20.1

Noninvasive Neuromodulation for Cognitive Dysfunction in Parkinson's Disease: A Scoping Review

Reginald B O'Hara 2026-10-27 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: Evidence specific to PD-MCI is limited and mixed. Several studies pairing stimulation with cognitive training reported favorable outcomes, but this pattern was inconsistent and does not establish superiority over stimulation alone. No VNS trial enrolled participants with confirmed PD-MC...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939339/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027155153&v=2.20.1

Comparative Plastid Phylogenomics of 29 Newly Sequenced Bougainvillea Cultivars Resolves the Phylogenetic Placement of Double-Bracted Cultivars

Huiqi Zhao 2026-10-27 1 min 59 words TDCS TACS TRNS

Summary: Bougainvillea (Nyctaginaceae) cultivars are horticulturally prized worldwide for their vivid bracts; however, repeated interspecific hybridization and somatic mutations have obscured their maternal origins, particularly for double-bracted forms. Here, we sequenced and assembled the complete plastome...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939474/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027155153&v=2.20.1

Anodal Transcranial Direct Current Stimulation Reduces Neuroinflammation and Cholinergic Dysregulation After Young Adult Traumatic Brain Injury

👤 Guven Akcay
📅 2026-10-27 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Traumatic brain injury (TBI) during childhood and adolescence can result in persistent motor and cognitive dysfunction due to secondary injury cascades involving neuroinflammation and disruption of neurotransmitter homeostasis. Although transcranial Direct Current Stimulation (tDCS) is a promising n...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939682/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027155153&v=2.20.1

Social relationship is associated with distinct patterns of inter-brain synchrony and directed connectivity during interpersonal coordination

Xinhong Jin 2026-10-27 1 min 59 words FNIRS

Summary: How social relationship and effector pairing relate to behavioral coordination and inter-brain interaction during joint timing remains unclear. Using functional near-infrared spectroscopy hyperscanning during a dyadic synchronization-continuation task, we examined 21 romantic couples and 23 stranger...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42937180/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027155151&v=2.20.1

Neural-Discursive Dynamics in Collaborative Spatial Problem Solving: Evidence from fNIRS Hyperscanning and Accountable Talk

👤 Xingguang Guo
📅 2026-10-27 📖 1 min 📝 65 words FNIRS

Summary: High-quality collaboration cannot be inferred simply from how much students talk or how strongly their brain activity is synchronized during joint work. This study examined relations among accountable talk, inter-brain synchrony (IBS), and group task performance in fifth-grade students' collaborativ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42938505/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027155151&v=2.20.1

Probing Executive Function in the Toddler Data Desert: A Touchscreen fNIRS Study of Working Memory and ADHD-Related Symptoms

Caitie Busch 2026-10-27 1 min 60 words FNIRS

Summary: Attention-deficit/hyperactivity disorder (ADHD) is associated with early differences in attentional control and working memory (WM), yet relatively few paradigms are developmentally feasible across the toddler-to-preschool period while also distinguishing spatial from featural information use. Addre...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42938612/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027155151&v=2.20.1

Mapping the Global Landscape and Thematic Evolution of fNIRS-Gait Research: A Systematic Bibliometric and Network Visualization Analysis

Naime Uluğ 2026-10-27 1 min 20 words FNIRS

Summary: CONCLUSIONS: Establishing standardized preprocessing pipelines and short-separation regressions may prove pivotal for transitioning mobile neuroimaging into objective, closed-loop clinical neurorehabilitation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939053/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027155151&v=2.20.1

Self-Selected Versus Assigned Emotion Regulation in Pain Expectancy: Behavioral and fNIRS Evidence

👤 Ping Shi 📅 2026-10-27 📖 1 min 📝 63 words FNIRS

Summary: Background: Expectation is a powerful cognitive modulator of pain, but how different types of emotion regulation approaches interact with expectancy to influence pain processing remains unclear. The present study compared self-selected habitual strategies (cognitive reappraisal or deep breathing) wi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939283/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027155151&v=2.20.1

Comparative Effects of Acute HIIE and MICE on Executive Function, Prefrontal Oxyhaemoglobin Responses, and Related Biomarkers in Sedentary Young Adults: A Randomized Crossover Trial

👤 Xinyue Yang
📅 2026-10-27 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSIONS: Both exercise sessions were followed by faster reaction times on selected tasks. The small sample, multiple analyses, and potential systemic contributions to fNIRS signals preclude firm conclusions about exercise-mode superiority or biological mediation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939307/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027155151&v=2.20.1

Cerebral Hemodynamic Patterns in People with Subjective Cognitive Decline: A Systematic Review Based on Functional Near-Infrared Spectroscopy

👤 Israel Contador
📅 2026-10-27 📖 1 min 📝 62 words FNIRS

Summary: Background/Objectives: Subjective cognitive decline (SCD) has been proposed as an early symptomatic manifestation of dementia/Alzheimer's disease. However, its associated brain patterns are not yet fully characterized. In recent years, functional near-infrared spectroscopy (fNIRS) has emerged as a c...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939380/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027155151&v=2.20.1

Resting-State Functional Brain Connectivity and Neurodevelopmental Profiles in 3-Year-Olds with Congenital Heart Disease

👤 Anne Gallagher
📅 2026-10-27 📖 1 min 📝 67 words FNIRS

Summary: Background/Objectives: Congenital heart disease (CHD) is associated with altered brain development and an increased risk of neurodevelopmental difficulties, particularly in more severe forms of the condition. Although alterations in functional brain connectivity have been reported before and after t...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42940032/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027155151&v=2.20.1

Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan
📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944288/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027155151&v=2.20.1

Multimodal Sensing and Digital Biomarkers in Virtual Reality-Based Cognitive Rehabilitation: Clinical Validation, Adaptive Systems, and Real-World Translation

Raffaele Ferri 2026-10-27 1 min 58 words FNIRS

Summary: Virtual reality (VR) is evolving from a platform for cognitive exercises into an instrumented rehabilitation environment capable of simultaneously delivering treatment and quantifying behavior. This critical narrative review examines how behavioral, ocular, kinematic, physiological, and neurophysiol...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944342/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027155151&v=2.20.1

Brain-Computer Interfaces for Cognitive Rehabilitation: A Narrative Review

Ruben I Carino-Escobar 2026-10-27 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Cognitive diseases/disorders are a major cause of disability affecting millions of people worldwide due to pathologies or conditions such as Alzheimer's disease, Parkinson's disease, vascular cognitive impairment, and attention deficit/hyperactivity disorder. The effects of these diseases/disorders ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939079/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027155148&v=2.20.1

Locomotor-State Dependence of Neuromodulation During Avian Free Flight

Zhigang Shang 2026-10-27 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Complex avian locomotor behaviors depend on the coordinated operation of neural processes involved in sensory perception, decision-making, and motor execution. Exogenous neural stimulation can alter motor output by intervening in neural processes, but its effects may depend on the current locomotor ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939144/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027155148&v=2.20.1

STDR-Net: A Spatio-Temporal Dependency Refinement Network for Motor Imagery EEG Decoding

Yongkang Yu 2026-10-27 1 min 56 words BRAIN COMPUTER INTERFACE

Summary: Background/Objectives: Motor imagery electroencephalography (MI-EEG) contains weak, non-stationary patterns that evolve across channels and time. Effective decoding requires local sensorimotor structure, trial-wide context, and ordered temporal evolution to be represented without discarding informat...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939304/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027155148&v=2.20.1

Human Factors in Brain-Computer Interfaces: Boundaries, Key Issues, and Methodological Approaches

👤 Yunfa Fu 📅 2026-10-27 📖 1 min 📝 62 words BRAIN COMPUTER INTERFACE

Summary: The large-scale, real-world deployment of brain-computer interface (BCI) is confronted with human factors bottlenecks, including reliability and usability. While modern human factors engineering has encompassed domains such as cognitive ergonomics, workload, adaptive automation, and human-in-the-loo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939319/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027155148&v=2.20.1

EOG Artifact Removal from EEG Using a Gated WGAN with Multi-Scale and Multi-Domain Feature Extraction

👤 Hao Tang 📅 2026-10-27 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: Background: Electroencephalogram (EEG) signals provide important information for brain-computer interfaces (BCIs) and the diagnosis of neurological disorders but are susceptible to electrooculographic (EOG) artifacts. Traditional methods often rely on multichannel data or relatively long windows, li...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939376/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027155148&v=2.20.1

EEG Features of Cortical Intent for Urination and Defecation in Healthy Aging and Spinal Cord Injury: A Descriptive Feasibility Study for BCI Neurorehabilitation

Jie Jia 2026-10-27 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: Background/Objectives: Systematically elucidating the neural mechanisms of bowel and bladder control is crucial for developing closed-loop brain-computer interface (BCI) neurorehabilitation for patients with spinal cord injury (SCI). This study aims to characterize the spatiotemporal Electroencephal...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42939378/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027155148&v=2.20.1

The Development and Evolution of Substrate Materials for Brain-Computer Interfaces

Yen Wei 2026-10-27 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Brain-Computer Interfaces (BCIs) hold tremendous potential for neurological therapeutics and neuroprosthetics; however, their long-term clinical viability is fundamentally limited by chronic interfacial failure. This failure originates from the profound biomechanical mismatch between synthetic micro...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42942690/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027155148&v=2.20.1

Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan 📅 2026-10-27 📖 1 min 📝 60 words BRAIN COMPUTER INTERFACE

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944288/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027155148&v=2.20.1

Agentic EEG Pipeline Synthesis Using Large Language Models and the Model Context Protocol

👤 Katy Tarrit 📅 2026-10-27 📖 1 min 📝 56 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography analysis typically requires multistage processing pipelines for which configuration depends on recording characteristics and the intended analytical objective. Although automated pipelines improve reproducibility, they generally implement predefined processing sequences. We pr...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944335/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027155148&v=2.20.1

Overt and Covert Speech-State Classification from Tripolar Concentric Ring Electrode EEG: A 4 mm and 10 mm Configuration Study

Shoaib Memon 2026-10-27 1 min 76 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) speech-state classification is a noninvasive method to identify the brain's overt and covert behavior. The EEG achieved through the scalp is contaminated by artifacts and volume conduction and suffers from poor spatial resolution. We present the first application of smal...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42944450/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027155148&v=2.20.1

A research on the typical images of animals in the dreams of Naxi.

2026-09-10 1 min 157 words DREAMING

Summary: Animal images in dreams have long been studied, yet existing research is not systematic or in-depth. The Naxi, a small Chinese ethnic group (about 320,000), possess a rich cultural heritage. In their belief, animals mediate between humans, deities, and nature, while dreams are key communication chan...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000353

Philosophical and medical analysis of dreams in avicenna’s perspective: Nature, mechanism, and functions.

📅 2026-06-15 📖 1 min 📝 218 words DREAMING

Summary: Avicenna was an eminent Islamic philosopher and physician. Drawing primarily on The Book of Healing and The Canon of Medicine, this article presents a systematic account of his theory of dreaming within the framework of his psychology and natural philosophy. Avicenna conceives drea...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000341

Threatening events and other themes in COVID-19-related dreams: Examining pandemic dreams from the perspective of current dream simulation theories.

📅 2026-06-25 📖 1 min 📝 218 words DREAMING

Summary: In this study on pandemic dream experiences in rural Nepal, 54 interviews were collected 1 year into the COVID-19 pandemic and compared with 62 interviews collected some years earlier. As in comparable studies, the dream narratives were categorized and analyzed according to their content, employing ...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000348

Correction to “Big-five personality traits and dream experiences: A systematic review and meta-analysis” by Marin et al. (2026).

📅 2026-09-10 📖 1 min 📝 399 words DREAMING

Summary: Reports an error in "Big-Five personality traits and dream experiences: A systematic review and meta-analysis" by Jai Marin, Noah Thomas, Lan Nguyen and Jared Walters (Dreaming, Advanced Online Publication, Mar 26, 2026, np; see record 2027-47158-001). The article is being made available op...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000349

Monthly Updates [March]

📅 2026-03-01 📖 2 min 📝 571 words FMHY

Summary: INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/mar-2026

Monthly Updates [June]

📅 2026-06-01 📖 4 min 📝 954 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/june-2026

Monthly Updates [August]

📅 2026-08-01 📖 3 min 📝 700 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/aug-2026

Monthly Updates [Sept]

📅 2026-08-31 📖 5 min 📝 1079 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/sept-2026

Towards closing the gap between declaring and verifying Python dependencies

👤 /u/s5s11 📅 2026-10-27 📖 1 min 📝 129 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> ⁉️ Why does <code>pydantic>=2.11</code> sit in your <code>pyproject.toml</code>? Is it the lowest version that actually works, or just a tool default? And what evidence backs that claim? 💡 Dependency declarations are claims about package versions, yet they...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xgwbul/towards_closing_the_gap_between_declaring_and/

Pulling numbers from 10-K and 10-Q filings: XBRL or parse the document?

👤 /u/Early-Sir-932 📅 2026-10-27 📖 1 min 📝 260 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> for income statement, balance sheet and cash flow numbers from 10-k and 10-q filings there are two routes and theyre good at different things. xbrl uses the tagged data that comes with the filing. the secs companyfacts api returns every tagged fact for a comp...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xguct1/pulling_numbers_from_10k_and_10q_filings_xbrl_or/

PHP2Shell, LiteLLM, GitLab: Find What You Need to Patch

👤 bigfish2026 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://cloudsleuth.io/blog/php2shell-litellm-gitlab-find-what-you-run/ Comments URL: https://news.ycombinator.com/item?id=50261583...

🔗 **Read full article:**
https://cloudsleuth.io/blog/php2shell-litellm-gitlab-find-what-you-run/

Amazon Is Gutting Its AI Division After Sustained Failure

👤 dpflan 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://futurism.com/artificial-intelligence/amazon-ai-division-failure-nova Comments URL: <a href="https://news.ycombinator.com/item?id=50261589">https://news.ycombinator.com/item?id=50...

🔗 **Read full article:**
https://futurism.com/artificial-intelligence/amazon-ai-division-failure-nova

Vicious Cycles: Theses on a philosophy of news

👤 eustoria 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://harpers.org/archive/2020/01/vicious-cycles-theses-on-a-philosophy-of-news/ Comments URL: <a href="https://news.ycombinator.com/item?id=50261623">https://news.ycombinator.co...

🔗 **Read full article:**
https://harpers.org/archive/2020/01/vicious-cycles-theses-on-a-philosophy-of-news/

Classification of Finite Simple Groups Formalized in Lean

👤 pfdietz 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://cfsg.axiommath.ai/ Comments URL: https://news.ycombinator.com/item?id=50261647 Points: 87 # Comments: 24

🔗 **Read full article:**
https://cfsg.axiommath.ai/

Police are using AI to pull over drivers for 'suspicious' behavior

👤 gumby 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://gizmodo.com/police-are-using-ai-to-pull-over-drivers-for-suspicious-behavior-2000831220 Comments URL: <a href="https://news.ycombinator.com/item?id=50261819">h...

🔗 **Read full article:**
https://gizmodo.com/police-are-using-ai-to-pull-over-drivers-for-suspicious-behavior-2000831220

Ghostty 1.4.0

👤 deaux 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://ghostty.org/docs/install/release-notes/1-4-0 Comments URL: https://news.ycombinator.com/item?id=50261998 Points: 335 # Comments:...

🔗 **Read full article:**
https://ghostty.org/docs/install/release-notes/1-4-0

Introducing LFM3: Every Modality for Specialized Intelligence

👤 Philpax 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.liquid.ai/blog/lfm3 Comments URL: https://news.ycombinator.com/item?id=50262229 Points: 54 # Comments: 15

🔗 **Read full article:**
https://www.liquid.ai/blog/lfm3

Three TLS Bugs in GoAhead

👤 owenshen24 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://foxio.io/blog/choose-your-backend-three-tls-bugs-in-goahead Comments URL: https://news.ycombinator.com/item?id=50263701 ...

🔗 **Read full article:**
https://foxio.io/blog/choose-your-backend-three-tls-bugs-in-goahead

Discord will soon let anyone find public server messages through Google

👤 DeepLogin 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://discord.com/blog/making-content-from-discoverable-servers-easier-to-discover Comments URL: <a href="https://news.ycombinator.com/item?id=50263854">https://news.ycombinato...

🔗 **Read full article:**
https://discord.com/blog/making-content-from-discoverable-servers-easier-to-discover

US President says he will ban the term 'artificial intelligence'

👤 anigbrowl 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://thehill.com/homenews/administration/6175633-trump-bans-artificial-intelligence-term/ Comments URL: <a href="https://news.ycombinator.com/item?id=50264228">https:/...

🔗 **Read full article:**
https://thehill.com/homenews/administration/6175633-trump-bans-artificial-intelligence-term/

Recording and slides now available for ERPLAB Studio webinar

👤 Steve Luck 📅 2024-06-28 📖 1 min 📝 30 words ERP BOOT CAMP

> **Summary:** We held a webinar to demonstration ERPLAB Studio on 28 June 2024. Click here to access a recording. Click here to access a PDF of the slides.<...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/28/recording-and-slides-now-available-for-erplab-studio-webinar

New Paper: Does the P3b component reflect working memory updating?

👤 Steve Luck 📅 2025-03-21 📖 7 min 📝 1540 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Simmons, A. M., Kiat, J. E., & Luck, S. J. (in press). Enhanced working memory representations for rare events. Psychophysiology. https://doi.org/10.1111/psyp.70038 [<a href="https://doi.org/10.1101/2024.03.20...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/3/20/new-paper-oddball

10-Day ERP Boot Camp to be held June 15-24, 2026 in Davis, California

👤 Steve Luck 📅 2025-08-20 📖 1 min 📝 125 words ERP BOOT CAMP

> **Summary:** We have received another 5 years of funding from the National Institute of Mental Health, so we plan to hold ERP Boot Camps in each of the next 5 summers. The next one will be June 15-24, 2026 in Davis, California. The application portal will open around January 1, 2026. A...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/8/20/boot-camp-summer-2026

Call for Papers: IEEE Brain Special Issue

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 36 words BRAIN

Summary: In a unique interdisciplinary collaboration with the IEEE's Society on Social Implications of Technology (SSIT) and IEEE Brain, J-FLEX is joining forces to explore both the technology of the Internet-of-Medical-Things (IoMT) solutions and medical wearables/ implantables.

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-journal-on-flexible-electronics/

IEEE Brain Joins the American Brain Coalition

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 61 words BRAIN

Summary: IEEE Brain is pleased to announce its acceptance as a nonprofit member of the American Brain Coalition (ABC), a prestigious alliance of over 150 organizations dedicated to advancing brain research, advocacy, and improving treatments for individuals affected by brain conditions. The ABC Board has ent...

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-brain-joins-the-american-brain-coalition-as-a-nonprofit-member/

Call for Papers: IEEE Transactions on Human-Machine Systems

👤 Adriel Carridice 📅 2025-06-18 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/call-for-papers-ieee-transactions-on-human-machine-systems/

Q&A with Dr. Alain Dagher, Neurologist and Professor at McGill University and the Montreal Neurological Institute

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-alain-dagher-neurologist-and-professor-at-mcgill-university-and-the-montreal-neurological-institute/

IEEE Advances Responsible Neurotechnology Standard

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/news-and-insights/ieee-brain-news/ieee-advances-responsible-neurotechnology-standard/

Q&A with Dhruv Vaish, PhD student at UC Berkeley and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dhruv-vaish-phd-student-at-uc-berkeley-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Technical Parameters and Feedback Control for Blood-Brain Barrier Permeability Enhancement by Focused Ultrasound

📅 2025-12-02 📖 1 min 📝 166 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Focused ultrasound combined with intravenously infused microbubbles has been shown to effectively enhance the permeability of the blood-brain barrier, facilitating drug delivery to the brain. A wide range of technical parameters has been evaluated through preclinical studies and clinical trials. Gen...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11272135

Decoding Spikes From Multiunit Data

📅 2026-01-22 📖 1 min 📝 250 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Communication and control in biological systems is mediated by the timing of discharges –spikes– from excitable cells such as neurons and muscle fibers. Each spike is associated to a characteristic waveform that can be captured by sensors. The waveform's characteristics depend on the cell's biophysi...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11361153

Table of Contents

📅 2026-01-29 📖 1 min 📝 1 words REVIEWS BIOMEDICAL ENGINEERING

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11368645

IEEE Engineering in Medicine and Biology Society

📅 2026-01-29 📖 1 min 📝 1 words REVIEWS BIOMEDICAL ENGINEERING

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11368668

Front Cover

📅 2026-01-29 📖 1 min 📝 1 words REVIEWS BIOMEDICAL ENGINEERING

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11368638

Welcoming the Newly Elected EMBS AdCom Members for 2026–2028

👤 dziura 📅 2025-12-10 📖 1 min 📝 18 words EMBS

Summary: The post Welcoming the Newly Elected EMBS AdCom Members for 2026–2028 appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/uncategorized/welcoming-the-newly-elected-embs-adcom-members-for-2026-2028/

Welcoming & Congratulating Our Newly Elected EMBS ExCom Members

👤 Nancy Zimmerman 📅 2025-12-14 📖 1 min 📝 18 words EMBS

Summary: The post Welcoming & Congratulating Our Newly Elected EMBS ExCom Members appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/uncategorized/welcoming-congratulating-our-newly-elected-embs-excom-members/

EMBS AdCom Nominations Open

👤 Nancy Zimmerman 📅 2026-05-04 📖 1 min 📝 51 words EMBS

Summary: Nominate volunteers for terms starting Jan 1, 2027: Asia Pacific Rep (3 yr), Technical Rep (3 yr), Student Rep (2 yr), Young Professional Rep (2 yr). Help shape the future… <a class="continue" href="https://www.embs.org/uncategorized/adcom-nominations-2027/">Continue Reading EMBS AdCo...

🔗 **Read full article:**
https://www.embs.org/uncategorized/adcom-nominations-2027/

IEEE Pulse Editor-in-Chief Search – Submit an Application!

👤 Deidre Artis 📅 2026-08-28 📖 1 min 📝 17 words EMBS

> **Summary:** The post IEEE Pulse Editor-in-Chief Search – Submit an Application! appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/ieee-pulse-editor-in-chief-search-submit-and-application/

Metformin alleviates neuronal injury in MK-801-Induced Schizophrenia-Like mice by inhibiting neuronal ferroptosis

📖 1 min 📝 25 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Yanwen Lin, Gaoxia Wen, Xiaoyue Hu, Ran Hu, Yahong Duan, Na Wu, Peng Xiong, Xianlai Jin

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004610?dgcid=rss_sd_all

Distinctive patterns of right-hemisphere vulnerability in cerebral small vessel disease: a gray matter asymmetry study

📖 1 min 📝 15 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Jingdian Lu, Yanli Wu, Lei Gao

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004646?dgcid=rss_sd_all

N-methyl-D-aspartate receptors in the prelimbic cortex modulate chronic neuropathic pain and comorbid generalised anxiety in a rainbow boa snake threat paradigm

📖 1 min 📝 38 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Priscila Medeiros, Ana Carolina Medeiros, Ohana Turcato Macacare, Weverton Castro Coelho-Silva, Renata Moreira Acunha, Fernando René Bendaña-Córdoba, Farhad Ullah, Sérgio Henrique Ferreira, Renato L...

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004609?dgcid=rss_sd_all

Exploratory bone–brain transcriptomic associations and an ODE-based digital-twin framework for fracture healing

📖 1 min 📝 15 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Chao Ma, Wei Lu, Ruipeng Guo

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004592?dgcid=rss_sd_all

Zona incerta circuits in cognition and the regulation of sleep and wakefulness: potential relevance to Alzheimer’s disease

📖 1 min 📝 14 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Jun Kang, Yang Li, Wenjun Yan

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007001?dgcid=rss_sd_all

State transitions in amyotrophic lateral sclerosis: mechanistic convergence and biomarker-guided stratification

📖 1 min 📝 23 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Mo Nasb, Dan Yang, Tayyab Hamid Malik, Dandan Xu, Lamis Dayoub, Jing Zhou, Yan Zhao

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007062?dgcid=rss_sd_all

Dynamic multi-scale topology learning and cross-modal graph fusion for autism spectrum disorder diagnosis

📖 1 min 📝 22 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Huadong Zhang, Junyong Zhao, Shaoqi Wu, Krisztián Szigeti, Kellermayer Miklós, Jiande Sun, Jianping Qiao

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007050?dgcid=rss_sd_all

Diagnostic classification of ASD improves with positively sparsified DTI inter-regional correlation and graph-theoretical artificial intelligence

📖 1 min 📝 22 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Ravi Ratnaik, Sriram Kumar P., Lavanya Selvaganesh, Sandeep Singh Sengar, Jac Fredo Agastinose Ronickom

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226006937?dgcid=rss_sd_all

A shape deformation-based framework for constructing individualized subcortical morphological networks

📖 1 min 📝 16 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Shutong Lu, Jiashun Zhang, Junle Li, Jinhui Wang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006208?dgcid=rss_sd_all

Digital twin control framework uncovers individualized cross-scale links between functional connectivity deviations, communication impairment, and molecular mechanisms in autism

📖 1 min 📝 12 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Guanlu Liu, Jianfeng Qiu

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006269?dgcid=rss_sd_all

Decoding individual differences in mental information from human brain responses predicted by deep neural networks

📖 1 min 📝 20 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Kiichi Kawahata, Jiaxin Wang, Antoine Blanc, Naoya Maeda, Shinji Nishimoto, Satoshi Nishida

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006270?dgcid=rss_sd_all

Insula functional connectivity development follows a canonical antero-posterior axis of organization

👤 Flook, E. A., Shafiei, G., Cieslak, M., Horien, C., Luo, A. C., Meisler, S. L., Salo, T., Sydnor, V. J., Tong, T. T., Barch, D. M., Goldsmith, J., Gur, R. E., Somerville, L. H., Uddin, L. Q., Alexander-Bloch, A., Shinohara, R. T., Satterthwaite, T. D.

📅 2026-10-27 📖 1 min 📝 150 words BIORXIV NEUROSCIENCE

Summary: The insula is a heterogeneous region associated with a range of functional domains including sensory, cognitive, and emotional processing. Cytoarchitectural and functional studies have described an antero-posterior axis of insula organization. However, the insula's functional development remains spa...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.24.761529v1?rss=1

Loss of PEA-15 Alters ERK/MAPK Signaling and Circadian Function in the Suprachiasmatic Nucleus

👤 Wheaton, K., Halloy, N., Formanowicz, M., Ramos, J. W., Hoyt, K. R., Obrietan, K.

📅 2026-10-27

📖 1 min 📝 261 words BIORXIV NEUROSCIENCE

Summary: Circadian rhythms are a fundamental biological process that coordinates physiological functions, behavior, and metabolism with an approximately 24-hour periodicity. At the center of the mammalian circadian timing system is the suprachiasmatic nucleus (SCN) of the hypothalamus, which generates robust...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.25.761398v1?rss=1

Predictable Timing Enhances the Use of Predictable Content

Magnusson, O., Hobbelen, H., Berkhof, W., Sack, A. T., Leunissen, I., ten Oever, S.

2026-10-27

1 min

224 words

BIORXIV NEUROSCIENCE

Summary: As events unfold, they carry structure in both their timing and content, with these dimensions often interrelated. Most research has treated these dimensions separately, although processing them jointly may provide important computational benefits. It is an open question whether the brain processes ...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.23.760967v1?rss=1

Linking cross-species hemodynamics with neuronal activity in prefrontal cortex during abstract sequence viewing

Conen, K. E., Stewart, B., Rivera Nunez, M. V., Hyde, H., Yusif Rodriguez, N., Ahuja, A., Basu, D., McKim, T., Desrochers, T. M.

2026-10-27

1 min

199 words

BIORXIV NEUROSCIENCE

Summary: The ability to identify and predict abstract sequential patterns enables a wide range of complex cognition and behavior, but the mechanisms mediating it remain poorly understood. Lateral prefrontal cortex (LPFC) has been implicated in abstract sequence tracking in both humans and macaques, but it is...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.23.761183v1?rss=1

Habitual sleep duration is associated with the cortical organisation of brain network controllability

Liu, X., Lu, R., Wang, S., Yu, Q., Zhang, M., Cui, R., Xiao, J., Liu, X., Fan, T.

2026-10-27 1 min

148 words

BIORXIV NEUROSCIENCE

Summary: Sleep is fundamental to adaptive brain function, yet how habitual sleep relates to the structural network properties that support flexible transitions between brain states remains poorly understood. Combining wrist-accelerometer sleep measures with diffusion MRI connectomes from 15,799 UK Biobank ad...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.25.761633v1?rss=1

Familiarity synchronizes brain rhythms during natural viewing

Noad, K. N., Kaiser, D.

2026-10-27 1 min 200 words

BIORXIV NEUROSCIENCE

Summary: Our perception of the world is shaped not only by sensory information, but also by prior experience. However, how familiarity alters the coordination of neural representations across distributed brain networks remains unclear. Rhythmic neural activity provides a potential mechanism for coordinating ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.760672v1?rss=1

The human-specific lncRNA HD-LINC drives lipid droplet-associated neurodegeneration in Huntington's disease

Kwon, J.-S., Wang, L., Ganti, N. R., Li, J., Xia, S., Al-Dalahmah, O., Yoo, A. S.

2026-10-27

1 min

230 words

BIORXIV NEUROSCIENCE

Summary: Huntington's disease (HD) is characterized by degeneration of striatal medium spiny neurons (MSNs), yet the human-specific regulatory mechanisms that contribute to neuronal vulnerability remain poorly understood. Here, we identify HD-LINC (LINC01139), a human-specific long intergenic noncoding RNA s...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.23.761349v1?rss=1

Neuroscience in Conversation: Tracking Scientific Change Through an Open Seminar Archive

Bellin, I., Tulner, L. K., Beyh, A., Wearn, A., Forkel, S. J.

2026-10-27

1 min

262 words

BIORXIV NEUROSCIENCE

Summary: Scientific talks capture how researchers frame questions, debate ideas, and communicate emerging methods, providing a complementary record of scientific change to the published literature. We investigated whether an open neuroscience seminar archive could reveal shifting scientific attention and con...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.761073v1?rss=1

Early life adversity enhances prosocial helping towards a cagemate and reshapes brain-wide neural activity and network organization

Breton, J. M., Lynch, A., Chiang, J., Wijethunga, S., Abdallah, I., Hickman-Maynard, J., Poon, Z., Ben-Ami Bartal, I., Brenhouse, H.

2026-10-27 1 min 238 words BIORXIV NEUROSCIENCE

Summary: Early life adversity (ELA) is linked to changes in social behavior and emotional regulation and to increased risk for developing mood and conduct disorders. However, the effects of ELA on empathy and its associated behaviors, including prosocial helping, remain poorly understood. Animal models of EL...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.23.761382v1?rss=1

Collateral-defined prefrontal-striatal circuits partition behavioral vigor during appetitive goal pursuit

Ma, X., Shao, M., Ren, B., Lei, J., Liu, Y., Hong, F., Huang, D., He, M., Zhang, S., Tai, Y., Lu, J.

2026-10-27 1 min 215 words BIORXIV NEUROSCIENCE

Summary: Motivated behavior unfolds through a sequence of appetitive processes that transform an intention to act into pursuit of a goal. Yet how behavioral vigor is regulated across distinct components of appetitive goal pursuit, and the neural mechanisms underlying this regulation, remain unclear. Here we ...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.24.761528v1?rss=1

Model selection in linear mixed-effects analysis of craving outcomes in a randomized controlled trial of medications for opioid use disorders

2026-10-27 1 min 0 words NATURE NEUROSCIENCE SUBJECTS

Read full article:
https://www.nature.com/articles/s41598-026-77858-y

Circuit mechanisms for disentangled representations and generalization in the hippocampus

Antonio Fernandez-Ruiz 2026-10-27 1 min 39 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 27 October 2026; doi:10.1038/s41593-026-02491-x The hippocampus encodes different aspects of experience as disentangled variables, enabling flexible recombination and generalization to new s...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02491-x

Stimulus-specific recruitment of human amygdala neurons predicts episodic memory encoding success

Ueli Rutishauser 2026-10-27 1 min 40 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 27 October 2026; doi:10.1038/s41593-026-02493-9 The study shows that stimulus-specific neurons in the human amygdala determine which experiences become lasting memory, revealing a mechanism ...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02493-9

Multiple sources contribute to predictive tactile attenuation in the mouse whisker system

Kalpana GuptaShubhodeep ChakrabartiCornelius Schwarzahttps://ror.org/03a1kwz48Systems Neurophysiology, Werner Reichardt Center for Integrative Neuroscience, Eberhard Karls University Tübingen, Tübingen 72076, Germanybhttps://ror.org/04zzwzx41Systems Neurophysiology, Hertie Institute for Clinical Brain Research, Eberhard Karls University Tübingen, Tübingen 72076, Germanychttps://ror.org/03a1kwz48Graduate Training Center for Neuroscience, Eberhard Karls University Tübingen, Tübingen 72076, Germany

2026-10-20 1 min 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceThe brain has evolved multiple interacting neuronal circuits that predict varying aspects of the consequences of one’s own actions. The predicted consequences range from sensory consequences of mov...

Read full article:
https://www.pnas.org/doi/abs/10.1073/pnas.2617276123?af=R

Connexins are essential for the contribution of latent progenitors to healing after spinal cord injury

María Victoria FalcoGabriela FabbianiDaniel PrietoMateo VidalMilagros BenítezMaría Inés RehermannMaría Constanza SilveraRenata SimeoneFederico F. TrigoRaúl E. Russoahttps://ror.org/05b50ej63Departamento de Neurofisiología Celular y Molecular, División Neurociencias, Instituto de Investigaciones Biológicas Clemente Estable, Montevideo CP11600, Uruguaybhttps://ror.org/05b50ej63Departamento de Señalización Neuronal y Glial, División Neurociencias, Instituto de Investigaciones Biológicas Clemente Estable, Montevideo CP11600, Uruguay

2026-10-20 1 min 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceEpendymal cells of the spinal cord are latent progenitors whose manipulation is regarded as a potential strategy to improve healing after injury. Therefore, a better understanding of the mechanisms...

Read full article:
https://www.pnas.org/doi/abs/10.1073/pnas.2605647123?af=R

CX3CL1 (fractalkine)–CX3CR1 interface drives CD8⁺ T cell–induced neurodegeneration in human brain organoids

👤 Woo Sub YangMin Sun ShinMu Seog ChoeFerdi Ridvan KiralJumee KimHong-Jai ParkMario FelixLais OsmaniIn-Hyun ParkInsoo KangaDepartment of Genetics, Interdepartmental Neuroscience Program, Yale Stem Cell Center, Yale Child Study Center, Wu Tsai Institute, Yale School of Medicine, New Haven, CT 06520bDepartment of Internal Medicine, Section of Rheumatology, Allergy & Immunology, Yale School of Medicine, New Haven, CT 06520chttps://ror.org/00vvvt117College of Pharmacy, Sookmyung Women’s University, Seoul 04310, Republic of KoreadDepartment of Pathology, Yale School of Medicine, New Haven, CT 06520

📅 2026-10-20 📖 1 min 📝 45 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceNeurodegenerative diseases involve not only intrinsic neuronal pathology but also dysregulated immune responses. However, how peripheral adaptive immune cells directly influence human neurons under...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2605209123?af=R

FMRI-guided focal transcranial magnetic stimulation of anterior cingulate cortex reduces opioid seeking and restores connectivity

👤 Zilu MaYing DuanHieu NguyenSabrina LinMd Mohaiminul HaqueDanni WangSamantha HoffmanAidan CarneyTaylor ScottEleni VarlasElliot SteinZheng-Xiong XiYavin ShahamHanbing LuYihong YangaNeuroimaging Research Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224bSchool of Electrical Engineering, International University, VNU-HCM, Ho Chi Minh City 71309, Vietnamchttps://ror.org/00waaqh38Viet Nam National University Ho Chi Minh City, Ho Chi Minh City 71309, Vietnamdhttps://ror.org/043mer456Department of Electrical and Electronic Engineering, University of Nebraska-Lincoln, Lincoln, NE 68588eMolecular Targets and Medication Discovery Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224fBehavioral Neuroscience Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224

📅 2026-10-26 📖 1 min 📝 48 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceA major challenge in developing brain stimulation therapies for opioid addiction is identifying which neural circuits to target and tracking how stimulation reshapes relapse-related circuit functi...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2619013123?af=R

Prefrontal cortical morphometric covariance and cognitive task performance in healthy young adults: an exploratory analysis of Human Connectome Project data

Roman Chivers 2026-10-22 1 min 343 words FRONTIERS HUMAN NEUROSCIENCE

Summary: The prefrontal cortex (PFC) is implicated in many cognitive tasks, including working memory, reward-based decision-making, executive control, and metacognition. This exploratory analysis investigated whether individual expression of PFC cortical thickness and surface area covariance components is as...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1976371

CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-10-22 1 min 231 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundMotor imagery-based brain-computer interface (MI-BCI) holds great potential in neurorehabilitation and motor assistance. Data quality and motor intention decoding effectiveness are closely associated with participants' engagement. However, previous MI tasks ignored active engagement regula...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1908608

Correction: The impact of sleep deprivation on cognitive function in healthy adults: insights from auditory P300 and reaction time analysis

Wei Wang 2026-10-27 1 min 0 words FRONTIERS NEUROSCIENCE

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.2023176

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027104957&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39870674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027104957&v=2.20.1

Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027104957&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027104957&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027104957&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41358599/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027104957&v=2.20.1

Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41676644/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027104957&v=2.20.1

Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41678848/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027104957&v=2.20.1

The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41703344/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027104957&v=2.20.1

EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42538927/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261027104957&v=2.20.1

Clinical significance of "complex" multimorbidity in hospitalised adult patients

👤 Gino Roberto Corazza 📅 2026-10-24 📖 1 min 📝 65 words LOW VISION

Summary: The term "complex multimorbidity" lacks universal consensus and its relevance remains unclear. We aimed to assess the prevalence of complex multimorbidity and its association with major healthcare outcomes. This was a prospective, single-centre study (internal medicine ward). We collected sociodemog...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42925459/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027104954&v=2.20.1

Outcomes of subthreshold diode micropulse laser and anti-VEGF vs anti-VEGF monotherapy for diabetic macular edema: a meta-analysis

Marko M Popovic 2026-10-26 1 min 41 words LOW VISION

Summary: CONCLUSIONS: In patients with DME, SDMLP and anti-VEGF combination therapy appears comparable, and possibly modestly superior to anti-VEGF monotherapy, with a reduced injection burden over 12 months. BCVA and CMT differences were below clinically important thresholds and supported by low-certainty e...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42928376/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027104954&v=2.20.1

Tropospheric Ozone Assessment Report II: Ozone distribution and variability in the oceanic and polar regions

Alfonso Saiz-Lopez 2026-10-26 1 min 76 words LOW VISION

Summary: Oceanic and polar regions are critical for global atmospheric processes, but an evaluation of ozone distributions and trends in these regions has been missing. In this assessment, part of the TOAR-II project, we report lower atmosphere ozone concentrations over open oceans and polar regions: with da...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42929035/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027104954&v=2.20.1

Telemedicine-Enabled Medication Optimization for Adults with Polypharmacy: A Systematic Review and Implications for Saudi Vision 2030 Healthcare Transformation

👤 Mark Sim 📅 2026-10-26 📖 1 min 📝 32 words LOW VISION

Summary: CONCLUSIONS: Telemedicine-enabled medication optimization appears feasible, but evidence of clinical effectiveness and cost savings remains uncertain. Saudi implementation should begin with monitored pilots emphasizing safety, equity, Arabic-language accessibility, governance, and prospective evalua...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42930373/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027104954&v=2.20.1

Accurate Extraction of Phenotypic Parameters for DUS Testing of Oudemansiella raphanipes Based on Deep Learning

Linyi Li 2026-10-26 1 min 65 words LOW VISION

Summary: To enable the rapid and accurate acquisition of phenotypic parameters during distinctness, uniformity, and stability (DUS) testing of edible mushroom varieties, and to address the low efficiency, strong subjectivity, and insufficient standardization of conventional manual measurement methods, this s...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42930743/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027104954&v=2.20.1

Understanding decision making factors influencing referrals to vision rehabilitation services at a tertiary medical center

Christina L Thomas-Virnig 2026-10-26 1 min 74 words LOW VISION

Summary: Functional vision loss at any stage affects quality of life even before clinically quantified to be moderate or severe. Eye care providers as the gatekeepers to vision rehabilitation services often underestimate how small variations in functional vision loss can have a significant impact in a patien...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42932291/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027104954&v=2.20.1

Comparative longitudinal evaluation of DIMS, progressive addition and single-vision lenses on myopia progression, binocular and visual performance in Spanish children

José R Jiménez 2026-10-26 1 min 58 words LOW VISION

Summary: An 18-month prospective, non-randomized comparative observational study evaluated the effects of different optical corrections on myopia progression, binocular vision, and visual performance in Spanish children. Forty-five myopic participants aged 9-14 years were allocated to single-vision lenses (S...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42932299/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027104954&v=2.20.1

Will Artificial Intelligence Augment Global Health Inequalities?

Matthew J Burton 2026-10-26 1 min 66 words LOW VISION

Summary: AbstractArtificial intelligence (AI) has been promoted as a tool to reduce global health inequities by addressing workforce shortages and expanding access to specialist care. However, if AI development and validation are concentrated in high-income countries, innovation may disproportionately benefi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42932554/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027104954&v=2.20.1

A secure and explainable hybrid CNN-transformer framework for trustworthy skin lesion classification

Annavarapu Chandra Sekhara Rao 2026-10-26 1 min 67 words LOW VISION

Summary: High diagnostic accountability and data integrity in digital health systems are a prerequisite for accurate early-stage classification of malignant skin lesions. Although deep learning models are remarkably successful in laboratory settings, their clinical deployment is limited due to their opaque "...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42933066/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027104954&v=2.20.1

Highly Linear Perovskite Synapses for Neuromorphic Computing

Yue Zhang 2026-10-27 1 min 65 words LOW VISION

Summary: Neuromorphic computing offers a compelling route to data-intensive applications with boosted computing power and energy efficiency. However, its accuracy for edge vision tasks is critically bottlenecked by the inherent non-linearity and asymmetry of conductance updates in analog memristive synapses....

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42934816/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261027104954&v=2.20.1

Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux
2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42470115/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027104951&v=2.20.1

A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42532383/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027104951&v=2.20.1

Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42590984/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027104951&v=2.20.1

Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42596634/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027104951&v=2.20.1

Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42614895/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027104951&v=2.20.1

A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42787568/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027104951&v=2.20.1

Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42788316/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027104951&v=2.20.1

Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42817759/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027104951&v=2.20.1

Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

Zhifei Xie
2026-10-24 1 min 66 words TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42923236/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027104951&v=2.20.1

Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42926384/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1no_pWrlHWS46ep2l9cVOQkZ1QsEMPlx7YY7aF6AfCIqP-RYZd&fc=None&ff=20261027104951&v=2.20.1

In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42694607/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027104949&v=2.20.1

Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan
📅 2026-09-09 📖 1 min 📝 57 words

BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42711906/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027104949&v=2.20.1

Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42768901/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027104949&v=2.20.1

Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42775591/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027104949&v=2.20.1

Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42789611/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027104949&v=2.20.1

Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42845092/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027104949&v=2.20.1

Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42848198/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027104949&v=2.20.1

Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42881664/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027104949&v=2.20.1

Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885551/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027104949&v=2.20.1

An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42885828/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=165yO28ehHLjXJb8W3JvTx2bYozdDe8IvyFRBlOfHZxFR8o1uX&fc=None&ff=20261027104949&v=2.20.1

Efficacy and Safety of Perioperative Transcranial Electrical Stimulation for Postoperative Pain and Related Outcomes: A Systematic Review and Meta-Analysis With Trial Sequential and Subgroup Analyses

👤 Qing Zhong 📅 2026-10-24 📖 1 min 📝 64 words TDCS TACS TRNS

Summary: CONCLUSIONS: Current evidence suggests that tES, as an adjuvant perioperative intervention, may alleviate postoperative pain, reduce opioid consumption, and decrease the incidence of postoperative delirium. However, the effect of tES on postoperative quality of recovery remains unclear. Considering ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42924907/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027104946&v=2.20.1

Accelerated high-capacity 6 mA transcranial direct current stimulation for major depression

Mark S George 2026-10-25 1 min 14 words TDCS TACS TRNS

Summary: CONCLUSIONS: Accelerated 6 mA HC-tDCS was feasible, well tolerated, and warrants a sham-controlled trial.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42926624/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027104946&v=2.20.1

Rapid and robust effects of high-dose intermittent Transcranial Brain Stimulation (iTBS) on depression and suicidal ideation in adolescents with acute major depression: a 10-day randomised, controlled, single-blind trial

Yutao Lan 2026-10-26 1 min 56 words TDCS TACS TRNS

Summary: CONCLUSION: Current research indicates that a 10-day course of high-dose iTBS is effective in alleviating depressive symptoms and suicidal ideation in adolescents with depression, and that the therapeutic effects persist for up to one month after the end of treatment. We also found that high-dose iT...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42928194/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027104946&v=2.20.1

Comparative Efficacy of Noninvasive Neuromodulation in Patients with Disorders of Consciousness: A Systematic Review and Network Meta-Analysis of Randomized Controlled Trials

👤 Quan Wei
📅 2026-10-26
📖 1 min
📝 64 words
TDCS TACS TRNS

Summary: CONCLUSIONS: HF-rTMS-M1 and RMNS plus HF-rTMS-DLPFC, RMNS, HF-rTMS-DLPFC, and tDCS have shown positive signals in patients with DoC. The duration of the disorder and underlying etiology may influence the effectiveness of different noninvasive neuromodulation techniques in DoC. Given the marked heter...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42928473/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027104946&v=2.20.1

Associations of Combined Transcranial Magnetic Stimulation and Transcranial Direct Current Stimulation With Psychological and Functional Outcomes After Stroke: A Retrospective Cohort Study

👤 Meng Huang
📅 2026-10-26 📖 1 min 📝 46 words

TDCS TACS TRNS

Summary: CONCLUSIONS: In this retrospective cohort, combined TMS and tDCS with standard rehabilitation was associated with lower depression and anxiety symptom scores and more favorable neurological and motor outcome trajectories than standard rehabilitation. Non-random treatment allocation and residual conf...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42928674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027104946&v=2.20.1

A tool to evaluate the impact of lived experience involvement in research: the Brain and Genomics Hub Impact Log protocol

Sophie E Legge 2026-10-26 1 min 74 words TDCS TACS TRNS

Summary: CONCLUSION: The potential value of this research is to implement a novel tool and framework for facilitating, recording and evaluating LEI in complex mental health research, which can be adapted for wider use. Strengths in design are LE leadership and cross-cutting LE research integration, incorpora...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42928742/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027104946&v=2.20.1

Multimodal premotor stimulation during action observation transiently boosts corticospinal excitability and motor performance

Ishan Mahajan 2026-10-26 1 min 2 words TDCS TACS TRNS

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42929054/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027104946&v=2.20.1

Combining Cognitive Training and Brain Stimulation: A Novel Approach to Address Disordered Eating and Antipsychotic-Induced Weight Gain in People With Severe Mental Illness

👤 Ulrike Schmidt
📅 2026-10-26 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: The initiation of antipsychotic therapy is a critical clinical window for early intervention to mitigate the long-term metabolic and psychiatric consequences of antipsychotic-induced weight gain. Combining ABMT + tDCS is acceptable and may help reduce binge eating in individuals on antip...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42929065/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027104946&v=2.20.1

High-Intensity Transcranial Alternating Current Stimulation for Post-Stroke Cognitive Impairment: A Triple-Blind Randomized Trial

Xiquan Hu 2026-10-26 1 min 60 words TDCS TACS TRNS

Summary: CONCLUSION: Hi-tACS improved global cognition and working memory in patients with PSCI and was generally well tolerated. Exploratory fNIRS findings suggested possible accompanying changes in cortical activation and network organization but did not establish a neurophysiological mechanism. Importantl...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42933772/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027104946&v=2.20.1

Comparative analysis of mitochondrial genomes in Zantedeschia parental lines and their hybrid progeny

Hongzhi Wu 2026-10-27 1 min 31 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study expands current knowledge of mt genome diversity in the Araceae and provides valuable genomic resources for future studies of Zantedeschia mitogenome evolution, genetic resource conservation, and molecular breeding.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42934513/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=143rKCPgMwbasrj66gQ1r1ebioUg42SIGRyVKSoW4m6X-ecQ00&fc=None&ff=20261027104946&v=2.20.1

The test-retest reliability of cortical activation and behavioral metrics in adolescent basketball players during dual-task postural control: an fNIRS-based assessment

👤 Lulu Song 📅 2026-10-24 📖 1 min 📝 59 words FNIRS

Summary: CONCLUSION: These findings provide preliminary test-retest reliability estimates for selected multimodal neurobehavioral measures in male adolescent basketball players although reliability alone does not establish clinical utility and estimates with wide confidence intervals require confirmation in ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42920610/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027104944&v=2.20.1

How School-Based Interventions Shape the Developing Brain during 3-12 Years: A Systematic Review and Exploratory Meta-analysis

👤 Hui Li 📅 2026-10-24 📖 1 min 📝 56 words FNIRS

Summary: The effects of school-based interventions (SBIs) on children's neurodevelopment are poorly synthesized. This systematic review integrates neuroscientific evidence on how structured educational activities influence brain development in children aged 3-12 years. Following PRISMA guidelines, we identif...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42925739/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027104944&v=2.20.1

The Walking Brain: Mapping feature-specific cortical responses to naturally unfolding sensory experience in the real world

👤 Meryem A Yücel 📅 2026-10-25 📖 1 min 📝 58 words FNIRS

Summary: Understanding how the brain processes the everyday world is fundamental, yet laboratory paradigms exclude self-generated movement, unscripted events, and a continuously evolving multisensory environment. Wearable neuroimaging enables real-world measurement, but linking neural activity to uncontrolle...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42926503/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027104944&v=2.20.1

Functional near infrared spectroscopy-based dual-stream adaptive language mapping in adults with and without post-stroke aphasia

👤 Meryem Yücel
📅 2026-10-25 📖 1 min 📝 67 words FNIRS

Summary: In aphasia, variability in lesion location and post-stroke network reorganization complicates efforts to identify the regions that support semantic versus phonological processing. In prior work, researchers have often examined these domains in separate patient cohorts, further limiting direct compar...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42926860/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027104944&v=2.20.1

How Task Load Modulates the Effect of State Anxiety on Working Memory and Its Neural Mechanisms: Evidence from Verbal and Spatial Modalities

Senqing Qi 2026-10-25 1 min 73 words FNIRS

Summary: The impact of state anxiety on working memory (WM) and its neural mechanisms, especially the role of task load, remain uncertain. To address this issue, we conducted two experiments to examine how state anxiety affects WM under different task loads and whether these effects are consistent across ver...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42927827/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027104944&v=2.20.1

Prefrontal dysfunction and impulsive decision-making in female methamphetamine use disorder: a fNIRS study

Peng Lei 2026-10-26 1 min 67 words FNIRS

Summary: CONCLUSIONS: Women with MUD showed altered intertemporal choice and reduced prefrontal hemodynamic responses in regions implicated in value evaluation and cognitive control. These findings support the relevance of prefrontal dysfunction to impulsive decision-making in female MUD while also indicatin...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42928151/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027104944&v=2.20.1

Melanin Bias in Cerebral fNIRS: A Monte Carlo Study Toward Equitable Optical Brain Sensing

👤 Murad Althobaiti
📅 2026-10-26 📖 1 min 📝 63 words FNIRS

Summary: Functional near-infrared spectroscopy (fNIRS) is an optical biosensing modality that transduces cortical hemodynamics into a measurable near-infrared signal, underpinning wearable brain monitors and brain-computer interfaces (BCIs). Because epidermal melanin is a strong near-infrared absorber, devic...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42930085/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027104944&v=2.20.1

Professional Experience Moderates Behavioral Performance and Sensorimotor-Related Cortical Activity During Scoring Decision-Making in Aerobic Gymnastics Judges: An fNIRS Study

👤 Bin Shao 📅 2026-10-26 📖 1 min 📝 58 words FNIRS

Summary: Understanding how professional experience shapes cognitive and neural resource allocation across different task demands is important for elucidating expert decision-making. This study examines whether professional experience influences behavioral performance and sensorimotor-related cortical activit...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42930959/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027104944&v=2.20.1

High-Intensity Transcranial Alternating Current Stimulation for Post-Stroke Cognitive Impairment: A Triple-Blind Randomized Trial

Xiquan Hu 2026-10-26 1 min 60 words FNIRS

Summary: CONCLUSION: Hi-tACS improved global cognition and working memory in patients with PSCI and was generally well tolerated. Exploratory fNIRS findings suggested possible accompanying changes in cortical activation and network organization but did not establish a neurophysiological mechanism. Importantl...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42933772/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027104944&v=2.20.1

fNIRS-VFT is a complementary tool to (18)F-FDG PET in presurgical evaluation of drug-resistant epilepsy: an exploratory study

Yuanxiang Lin 2026-10-27 1 min 59 words FNIRS

Summary: CONCLUSIONS: In this exploratory, descriptive analysis, interictal ^(18)F-FDG PET showed higher concordance with lateralization of surgical areas compared with fNIRS-VFT among patients with DRE. Nevertheless, fNIRS may provide complementary information for cases with inconclusive PET findings. Addit...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42934830/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261027104944&v=2.20.1

Fabrication Strategies and Performance Optimization Strategies of Flexible Brain-Computer Interface Electrodes

Jianguo Tang 2026-10-25 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) provides a direct communication pathway between the brain and external devices and has been applied in neurological disease treatment, motor rehabilitation, and human-computer interaction. Electrodes play essential roles in recording and modulating neural signals, and ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42926359/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027104941&v=2.20.1

Melanin Bias in Cerebral fNIRS: A Monte Carlo Study Toward Equitable Optical Brain Sensing

Murad Althobaiti 2026-10-26 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Functional near-infrared spectroscopy (fNIRS) is an optical biosensing modality that transduces cortical hemodynamics into a measurable near-infrared signal, underpinning wearable brain monitors and brain-computer interfaces (BCIs). Because epidermal melanin is a strong near-infrared absorber, devic...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42930085/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027104941&v=2.20.1

A Complementary Dual-Branch Receptive Field Network for Cross-Channel-Density Motor Imagery EEG Decoding

Zuguang Rao 2026-10-26 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) enable direct communication between the brain and external devices, providing a bio-inspired link between humans and artificial systems. However, electroencephalography (EEG)-based motor imagery (MI) decoding continues to pose challenges, due to limited temporal expl...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42930242/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027104941&v=2.20.1

A Biodegradable Fish-Derived Gelatin-Chitosan Sponge as a Novel Hemostatic Agent in a Rat Tooth Extraction Model

Dwi Ariawan 2026-10-26 1 min 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: These findings demonstrate the potential of gelatin-rich fish-derived gelatin-chitosan composite sponges as biodegradable hemostatic biomaterials for oral surgical applications.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42930474/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027104941&v=2.20.1

Beyond Overlap: Development and Validation of the Inclusion of AI in Self (IAIS) Scale and Its Moderating Role in the Relationship Between AI Identity and AI Threat

Xiang Yao 2026-10-26 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Artificial intelligence (AI) is becoming increasingly integrated into people's self-concepts, yet a psychometrically robust multi-item pictorial instrument for assessing the inclusion of AI in self (IAIS) is lacking. Drawing on self-expansion theory and the AI-mediated communication (AI-MC) model, w...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42930953/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027104941&v=2.20.1

REFS: Robust EEG feature selection with incomplete labels for multi-dimensional emotion recognition

Li Zhuo 2026-10-26 1 min 61 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: Compared to single-type features, multi-type electroencephalography (EEG) features provide a multi-level representation for analyzing multi-dimensional emotions. However, the high dimensionality of multi-type EEG features, combined with the relatively small number of high-quality EEG samp...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42933128/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027104941&v=2.20.1

Theoretical system of brain-computer interface for basic research

Rongzhang Luo 2026-10-26 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) has achieved remarkable progress in brain signal acquisition, decoding algorithms, and related fields. Nevertheless, its underlying theoretical system remains largely fragmented, lacking a systematic characterization of system decodability, performance boundaries, and ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42934124/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027104941&v=2.20.1

Multi-scale motor imagery electroencephalogram classification framework with dynamic weighted multi-source domain alignment

Jingxia Chen 2026-10-26 1 min 55 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) technology facilitates information exchange between the brain and external devices, offering new approaches to human-computer interaction and medical rehabilitation. However, distribution shifts caused by inter-subject variability in electroencephalogram (EEG) signals ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42934128/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027104941&v=2.20.1

Mechanisms and outcome evaluation of neurofeedback in post-traumatic stress disorder intervention

Yunfa Fu 2026-10-26 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Patients with post-traumatic stress disorder (PTSD) still face challenges such as limited treatment response in some cases and substantial inter-individual variability when receiving existing interventions. Neurofeedback (NFB), as a closed-loop neuromodulation approach based on brain-computer interf...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42934145/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027104941&v=2.20.1

Auditory Rhythm Biases Visual Speed Perception and Shows Tempo-Dependent Signatures During Motion Extrapolation

Lihan Chen 2026-10-27 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: Auditory signals can bias visual motion judgements, but how dynamic auditory structure affects variable-speed motion during perception and subsequent extrapolation remains unclear. Across three experiments, we paired changing visual motion with changes in auditory tempo or pitch. In Experiment 1, sl...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42934997/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261027104941&v=2.20.1

When the body suffers, the mind dreams: The lived experience of cancer.

2026-09-10 1 min 194 words DREAMING

Summary: Dreams have intrigued humanity across centuries, serving as portals to the unconscious, reflections of lived experiences, and sources of spiritual guidance. The study of dreaming spans neuroscience, psychology, anthropology, sociology, and religious thoughts, with each discipline offering unique ins...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000355

Big-Five personality traits and dream experiences: A systematic review and meta-analysis.

📅 2026-03-26 📖 1 min 📝 369 words DREAMING

Summary: [Correction Notice: An Erratum for this article was reported in Vol 36(3) of Dreaming (see record 2028-23196-002). The article is being made available open access under the CC-BY license following the Griffith University opt-in to the CAUL/ APA Read and Publish agreement. The correct copyrig...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000336

The dream insight framework: Harnessing the power of dreams in leadership.

📅 2026-08-20 📖 1 min 📝 80 words DREAMING

Summary: This article introduces the Dream Insight Framework, an approach for integrating dreamwork into leadership development. It explores how dreams, often dismissed in existing leadership approaches, can be powerful tools for enhancing creativity, decision-making, and problem-solving. By reviewing existi...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000354

Interpretability of dreams elicited by progressive-interval rapid-eye-movement and fixed-schedule awakening methods: Trajectories of dream affect, themes, and insight across the night.

📅 2026-07-30 📖 1 min 📝 201 words DREAMING

Summary: This study examined whether interpreting REM dream reports elicited using a REM-cycle-accommodating awakening protocol yielded more associations and greater insight than interpreting REM and non-REM dream reports obtained through a fixed-schedule awakening method. It also explored how interpretabili...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000350

🪣 **Bucket Newsletter**

Generated automatically from 35 RSS feeds

Powered by GitHub Actions • Updated every 30 minutes

Visit: yuckyman.github.io/bucket